Program overview

19-May-2025 13:10

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master CS 2024	**Master Computer Science 2024**						
Core courses 2024							
Core courses (compulsory)							
CS4500	Responsible Computer Science	5					
CS4505	Software Architecture	5					
Core courses (choose 3 out of 6)							
CESE4045	High-performance data networking	5					
CS4510	Formal Reasoning about Software	5					
CS4515	3D Computer Graphics and Animation	5					
CS4520	Security and Cryptography	5					
CS4525	Web-Scale Data Management	5					
DSAIT4005	Machine and Deep Learning	5					
Themes 2024							
Algorithmics 2024							
CS4205	Evolutionary Algorithms	5					
CS4530	Modelling and Problem Solving	5					
CS4535	Constraint Solving	5					
Computer Graphics 2024							
CS4540	Geometric Data Processing	5					
DSAIT4080	3D Visualization	5					
DSAIT4120	Applied Image Processing	5					
Computer Systems 2024							
CESE4025	Real-time Systems	5					
CESE4030	Embedded Systems Laboratory	5					
CESE4120	Smart Phone Sensing	5					
Cyber Security 2024							
CS4150	Systems Security	5					
CS4380	Privacy Enhancing Technologies	5					
CS4430	Network Security	5					
Distributed Systems Engineering 2024							
CS4160	Blockchain Engineering	5					
CS4545	Distributed Algorithms	5					
CS4550	Distributed Data Systems	5					
Information Management 2024							
CS4145	Crowd Computing	5					
DSAIT4050	Information Retrieval	5					
DSAIT4055	Web Science and Engineering	5					
Machine Learning Apllications Engineering 2024							
CS4295	Release Engineering for Machine Learning Applications	5					
DSAIT4020	Elements of Statistical Learning	5					
DSAIT4065	Conversational Agents	5					
Programming Languages 2024							
CS4555	Compiler Construction	5					
CS4560	Parallel and Concurrent Programming	5					
CS4565	Advanced Functional Programming	5					
Software Engineering 2024							
CS4570	Machine Learning for Software Engineering	5					
CS4575	Sustainable Software Engineering	5					
CS4580	Automated Software Testing and Reverse Engineering	5					
Research courses 2024							
CESE4065	Advanced Practical I.o.T. and Seminar	5					
CS4700	Literature Survey	5					
CS4705	Research Seminar Computer Graphics	5					
CS4710	Research in Cyber Security Hacking Lab	5					
CS4715	Programming Languages Research Seminar	5					
CS4720	Research in Program Analysis	5					
CS4725	Research Seminar on Scalable Learning Systems	5					
DSAIT4200	Empirical Research of Computational Solutions	5					
DSAIT4210	Research in Web Data and Information Management	5					
DSAIT4220	Research in Intelligent Decision Making	5					
DSAIT4230	Research in Social Signal Processing and Affective Computing	5					

DSAIT4235	Research Seminar on Multimedia Computing and Systems	5

Deepening/broadening Electives 2024

Interdisciplinary projects

IFEEMCS520201	Advanced Interdisciplinary Artificial Intelligence Project	15
IFM4040	Joint Interdisciplinary Project	15
TPM423A	Technology Venture Development	15

Deepening electives

CS4410	Category Theory for Programmers	5
DSAIT4300	Formal Methods for Machine Learning	5
DSAIT4305	Machine Learning for Graph Data	5
DSAIT4310	Modeling and Data Analysis and Complex Networks	5
DSAIT4315	Performance Analysis	5
DSAIT4320	Building Serious Games	5
DSAIT4325	Socio-Cognitive Engineering	5
TPM027A	Cyber Risk Management	5

Broadening electives

TPM401A	Technology Entrepreneurship and Innovation	5
TPM404B	Technology Entrepreneurship and Internationalization	5
TPM409A	Digital technology entrepreneurship and management	5
TPM411A	Idea to Start-up IT & AI	5

Thesis project 2024

CS5000	Thesis Project	45

Additional courses for cohort 2023 and earlier

AP3132	Advanced Digital Image Processing	6
CESE4025	Real-time Systems	5
CESE4030	Embedded Systems Laboratory	5
CESE4045	High-performance data networking	5
CESE4055	Ad hoc and Sensor Networks	5
CESE4060	Wireless IoT and Local Area Networks	5
CESE4120	Smart Phone Sensing	5
CS4090	Quantum Communication and Cryptography	5
CS4195	Modeling and Data Analysis in Complex Networks	5
CS4225	Educational Technologies	5
CS4235	Socio-Cognitive Engineering	5
CS4345	Seminar Formal Methods for Learned Systems	5
CS4350	Machine Learning for Graph Data	5
CS4410	Category Theory for Programmers	5
EE4715	Array Processing	5
EE4740	Data Compression: Entropy and Sparsity Perspectives	5
EE4C06	Networking	5
ET4030	Error Correcting Codes	4
IFEEMCS4070	Multivariate Data Analysis	5
IN4302TU	Building Serious Games	5
IN4306	Literature Survey	10
IN4310	Seminar Computer Graphics	5
IN4341	Performance Analysis	5
IN5000	Final Project	45
QIST4310	Fundamentals of Quantum Information	4
SEN1141	Managing Multi-actor Decision-making	5
SEN1611	I&C Architecture Design	5
SEN1622	I&C Service Design	5
TPM010A	Cyber Crime Science	5
TPM025B	User-Centred Security	5
TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5
WI4049TU	Introduction to High Performance Computing	6

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Master CS 2024

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Core courses 2024	
Introduction 1	The master's programme starts with a common core that aims to further develop your knowledge and mastery of advanced concepts and theories in Computer Science and fosters a responsible mindset preparing you for your future career. Compulsory courses: - CS4505 Software Architecture - CS4500 Responsible Computer Science In addition to the compulsory courses you can choose three courses from the following list: - DSAIT4005 Machine and Deep Learning - CS4510 Formal Reasoning about Software - CS4515 3D Computer Graphics and Animation - CS4520 Security and Cryptography - CS4525 Web-Scale Data Management - CESE4045 High-performance data networking

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Core courses (compulsory)

CS4500	Responsible Computer Science	5
Responsible Instructor	Dr.ir. C.C.S. Liem	
Responsible Instructor	Dr. M.S. Pera	
Instructor	Dr. A. Anand	
Instructor	Prof.dr.ir. D.M. van Solingen	
Instructor	L. Cavalcante Siebert	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	The goal of this course is for students to further develop a responsible attitude that will contribute to the role they play in their future careers. Students will learn to make and reflect on choices related to Responsible Computer Science Engineering with an ethical and sustainable stance and act with integrity. Students will engage with social, organizational, professional, and technical aspects of developing Computer Science, from both systems and human-centric perspectives. Topics discussed include entrepreneurial dilemmas, craftship/professionalism, safety/privacy and data governance, transparency, non-discrimination, fairness, inclusivity, accessibility, and accountability. The course will run in the first year of the master, in quarter 3. Responsible CS Engineering-related content in courses in quarter 1 and 2 will be a means for students to start reflecting on their professional roles and on their work in the different courses and projects they undertake. No deliverables are expected nor required in quarter 1 and 2 In this course, students will be presented with organizational, social and technical approaches towards designing and developing Ethical and Responsible Computer Science, integrity and ethics in Computer Science and engineering, discuss and engage with colleagues on their duties and roles in their future careers, connect different perspectives related to ethical and responsible Computer Science as they think of new problems, and prepare their final presentations.	
Study Goals	At the end of this course, the student is able to: 1. Reflect on your role, responsibilities, possible dilemmas and trade-offs in developing Computer Science methodology and technology, both as a professional and as a team member. 2. Analyze Computer Science approaches and principles towards addressing societal and ethical issues, considering the perspectives of multiple stakeholders. 3. Contribute to appropriate responsible design, development and impact assessments on Computer Science work. 4. Integrate concepts related to integrity and professional duties when presenting and justifying your choices to address matters of responsibility connected to Computer Science.	
Education Method	Most lectures with ~45min content + 45 min discussions (guided by cases). In addition, selected content and guest lectures.	
Literature and Study Materials	Slides, real-life case scenarios, scientific articles, and impact assessment methods.	
Assessment	The final grade of the course consists of the following components: - Group Report: case study analysis and presentation (P/F) - Individual Presentation: individual interview (P/F) Both components need to be passed, to receive a pass for the course. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Report: Repair opportunity - Individual Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4505	Software Architecture	5
Responsible Instructor	Prof.dr. A. van Deursen	
Responsible Instructor	Prof.dr.ir. D. Spinellis	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The software architecture course offers students a chance to learn and experience the concepts of designing, modeling, analyzing and evaluating software design and software architectures. Furthermore, the course provides students with a discussion forum in which recent articles in the area of software architecture are presented and discussed. The course also features a number of guest lectures to show the state-of-the-art of software architecture in industry. Topics covered by this course are: fundamentals of software architectures, modeling and designing software architectures, architectural patterns and styles, architecture viewpoints and perspectives, the role of the software architect, analyzing and evaluating software architectures, component and plug-in frameworks, software product lines, service oriented architectures, code quality, technical debt, refactoring, and software architecture evolution. The course includes extensive labwork in groups of four. Each group is tasked to develop the architecture for a system of choice, covering problem analysis, a developed architecture, and a proof-of-concept implementation, and to describe these in written reports.	
Study Goals	Bring students into the position that they can (1) structure a systems problem space; (2) architect a software solution that meets stakeholder needs appropriately using models, views, and perspectives; (3) manage evolving needs, keeping architecture aligned, recognizing and addressing technical debt; (4) understand and assess architectures of existing software systems; (5) Use proof-of-concepts for architectural decision making; (6) communicate architectural decisions; (7) fulfil the role of an architect;	
Education Method	Interactive lectures, lab assignment, paper presentation, and discussion.	
Literature and Study Materials	The course uses the books: Cesare Pautasso. Software Architecture: Visual Lecture Notes. Leanpub, 2020; Additional reading material will be announced in the lectures.	
Assessment	The final grade of the course consists of the following components: - Group Report: Essay (weighting 60%) - Group Assignment: coding lab assignment (weighting 20%) - Group Presentation (weighting 20%) Final grade calculation = 0.6 * Group Report + 0.2 * Group Assignment + 0.2 * Group Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Report: Repair opportunity - Group Assignment: Repair opportunity - Group Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

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Core courses (choose 3 out of 6)

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet plays a critical role in modern society, but the complexity of large-scale networks and the diversity of protocols make network management increasingly challenging. As networks become more programmable, the distinction between compute systems and network elements is blurring, leading to a paradigm shift in network operations. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network operations, programmability and management. The course will examine different networking protocols and techniques used across various organisational domains, from edge computing to Internet-scale systems. The course will cover essential topics such as Quality of Service, Internet architecture/organisation, and network measurements.	
Study Goals	At the end of the course, the student is able to: 1. Understand advanced networking concepts, such as Software-Defined Networking, Inter/Intra-domain routing, etc. 2. Understand network management and operation techniques applied in internet, edge, and cellular (5G) domains 3. Evaluate network performance via a network emulator (Mininet) and understand real-world Internet architecture via measurements. 4. Create network programs in the domain-specific language P4.	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = 1 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. N. Mohan	

CS4510	Formal Reasoning about Software	5
Responsible Instructor	Dr. B.P. Ahrens	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, we learn the fundamentals of formal reasoning about programs. The basis for reasoning is logic, specifically, propositional and predicate logic. Within logic, we formally specify programming languages, that is, their syntax and operational semantics. We also formulate specifications of programs and prove that given programs satisfy their specification. The logic we work in is itself implemented as a computer program, in a so-called computer proof assistant. We learn how to implement our logical reasoning in a computer proof assistant; the activity of "proving" thus becomes a form of programming itself.	
Study Goals	At the end of this course, the student is able to: 1. Formulate and prove statements in propositional and predicate logic 2. Formally specify a programming language 3. Specify and prove properties of programs 4. Mechanize proofs of program behaviour in a computer proof assistant	
Education Method	Lectures and problem sessions	
Literature and Study Materials	https://softwarefoundations.cis.upenn.edu/	
Assessment	The final grade of the course consists of the following components: - Individual Report (weighting 100%) Final grade calculation = 1 * Individual Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. J.G.H. Cockx	

CS4515	3D Computer Graphics and Animation	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	In this course, you will be introduced to Computer Graphics. The topic is of very high relevance for the industry and the research community and has numerous applications in different domains, such as scientific visualization, video games, simulators, special effects, animated movies and many more. Here, you will learn about basic and advanced algorithms, as well as modern implementation techniques for image, object and motion synthesis on a computer.	
Study Goals	After the course, the student is able to: 1. Evaluate properties of algorithms and data structures linked to Computer Graphics 2. Reproduce the mathematical models and algorithms associated with Computer Graphics 3. Analyze and explain advanced Computer Graphics methods 4. Apply mathematical modeling and theory of geometric computations and transformations, object representations, simulation, and encoding. 5. Create algorithms and data structures using the C++ programming language and modern OpenGL	
Education Method	Lectures, instructions, research papers, lab work	
Literature and Study Materials	Research Papers in domain of selected topics, lecture sheets, online sources, optional books (see below) Books: - Fundamentals of Computer Graphics by Shirley et al. - CRC Press - Real-time Rendering by Tomas Akenine-Möller, Eric Haines, Naty Hoffman - Peters, Wellesley - Real-Time Shadows by Elmar Eisemann, Michael Schwarz, Ulf Assarsson, Michael Wimmer - Taylor & Francis - Computer Graphics. Principles and Practice by James D. Foley, Andries VanDam, Steven K. Feiner - Addison Wesley Advanced Material: Physically Based Rendering, From Theory to Implementation by Matt Pharr, Wenzel Jakob and Greg Humphreys - MIT Press	
Assessment	The final grade of the course consists of the following components: - Individual Assignment 1 (weighting 60%) - Individual Assignment 2: final project (weighting 40%) Each individual assignment may be composed of multiple parts. The Final Project is realized in groups, but grades are individual. Final grade calculation = 0.6 * Individual Assignment 1 + 0.4 * Individual Assignment 2 A passing final grade for the course can only be earned when, for all components, at least a 5.0 is earned and the weighted final grade (rounded to 0.5) is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. The repair option for Individual Assignment 1 is in the form of a new assignment. A repair option is not offered for the final project since it spans several weeks, covers different topics, and is realized in groups. Nevertheless, feedback on progress can be given during the practical sessions and on-demand, giving the students ample opportunity to improve their work during the assignment timeframes. Note: For all assignments, it is not allowed to use Generative AI tools. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4520	Security and Cryptography	5
Responsible Instructor	Dr. Z. Erkin	
Instructor	Dr. K. Liang	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Recommended: - Probability and Statistics - Programming skills	
Course Contents	Motivation: Computers are now found in every layer of society, and information is being communicated and processed automatically on a large scale. Examples include medical and financial files, automatic banking, video-phones, pay-tv, teleshopping and global computer networks. In all these cases there is a growing need for the protection of information to safeguard economic interests, to prevent fraud and to ensure privacy. Synopsis: Security and cryptography are essential components of any digital system. In this course, the fundamentals of secure data storage and transportation of information are described. In particular, classical (e.g. Caesar, Vigenere) and modern encryption schemes (RSA, DES, AES, Elliptic curves) are described along with their mathematical background such as number theory. Methods for authentication, data integrity and digital signatures are discussed in detail, as these are the main components of many security architectures. The course also investigates more advanced topics such as zero-knowledge proofs and secret-sharing schemes. Aim: It is the aim that at the end of the course one has a survey of the state of the art of both cryptographic algorithms and protocols for security and privacy, as well as is familiar with present applications. Among other things, the following topics are covered: -Classical systems -Information-theoretic security -Definition of Security notions -Symmetric encryption (e.g. DES, AES) -Asymmetric encryption (RSA, Elliptic Curves) -Hash functions -Random number generation -Key Management -Digital Signatures.	
Study Goals	At the end of this course, the student is able to: 1. Understand the basic concepts of cryptography, including those used in classical cryptography, modern secret cryptography and public key cryptography 2. Explain the fundamental concepts of security, including its importance in cryptography. 3. Understand the societal needs for confidentiality, integrity, and authenticity of information. 4. Understand the role of cryptographic primitives in securing information, including the differences between symmetric and asymmetric cryptography, the functions of hash functions, digital signatures, and PKI (Public Key Infrastructure). 5. Grasp advanced cryptographic topics relevant to modern society, where trust in entities cannot be guaranteed.	
Education Method	Lectures, assignments and weekly exercises. Through assignments, students are expected to have the chance to work on the topics covered in the lectures.	
Literature and Study Materials	- Introduction to Modern Cryptography, Katz and Lindell, CRC Press, Second Edition - Cryptography made simple, Nigel P. Smart, 2nd Edition, Springer, 2016 (PDF Available Online) - Handouts of lectures	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 70%) - Mandatory Assignments (weighting 30%) Final grade calculation = 0.7 * Written exam + 0.3 * Mandatory Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Mandatory Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4525	Web-Scale Data Management	5
Responsible Instructor	Dr. C. Lofi	
Responsible Instructor	Dr. A. Katsifodimos	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: - BSc in Computer Science (or equivalent degree), or - Basic Knowledge of Data Management and (relational) Databases. - Programming Skills Recommended: Knowledge of Operating Systems, Networking Systems, Distributed Systems, Implementation of relational databases.	
Course Contents	This course addresses the challenges of Data Management at Web-scale. Especially, it covers the need for large-scale distributed data storage systems. The lecture therefore introduces step-by-step increasingly complex distributed storage systems, leading up to modern implementations of different NoSQL data storage systems. The challenges arising from such systems are presented and discussed, especially focusing on the CAP theorem and the resulting trade-offs with respect to data models, transactional power, query expressivity, and replication consistency. These discussions lead to different variants of NoSQL database systems, like Key-Value Stores, Document Stores, Wide-Columnar stores, and Graph Databases. The advantages, disadvantages, and general properties of these systems are discussed in more detail. The course focuses also on database transactions, and the implications those have in modern, Web-scale application development and deployment, i.e., there is special focus on distributed transactions and consistency guarantees of different data management systems and methods.	
Study Goals	At the end of this course, the student is able to: 1. Explain the trade-off between data consistency and scalability of data processing techniques. 2. Design a scalable data management architecture 3. Analyse a use case using high-level data management and use-case requirements 4. Evaluate a self-designed scalable data management architecture with experiments 5. Apply scalable data management techniques and methods in practical scenarios	
Education Method	The course comprises weekly lectures that introduce core data management concepts that are then exemplified via the presentation of real-world web-scale data management systems. From week 2, a project is announced, that the students can complete in teams of 3-5 persons. At the beginning of each lecture, the lecturer and the students discuss issues the projects, alongside summative feedback on progress as well as specific suggestions about the project. At the end of the class, the students present their project results, and the instructor also executes the project in a computing cluster (or the Cloud) infrastructure.	
Literature and Study Materials	Primarily course slides, and research papers. Recommended book: "Designing Data Intensive Applications", by Martin Klepmann.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 80%) - Group Assignments (weighting 20%) Final grade calculation = 0.8 * Written Exam + 0.2 * Group Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4005	Machine and Deep Learning	5
Responsible Instructor	Dr. D.M.J. Tax	
Responsible Instructor	Dr. J.C. van Gemert	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The range of problems that a computer scientist may encounter is very broad, from the very abstract (to determine if some problem can be solved by a computer) to very specific (to implement an electronic health record system). Due to the complexity and noisiness of the real world, a modern computer scientist has to rely more and more on the analysis and models of data. Machine Learning, and a recently successful extension of it, Deep Learning, deals with the analysis of data and the development of models. To be able to apply these algorithms, the computer scientist needs to have a good understanding of the foundations of Machine Learning. This course provides a recapitulation of (un)supervised learning, classification, decision theory and overfitting. It introduces some classical statistical learning classifiers and discusses their complexity, and ways to adapt their complexity (regularisation). The course focuses further on Deep Learning, including backpropagation, Optimization, Convolutional Neural Networks, Recurrent Neural Networks, self-attention and Transformers. Finally, the course discusses biases in datasets, and the design and analysis of Machine Learning experiments. This makes the student aware how the Machine and Deep Learning models can be applied in a responsible manner to real world problems.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and assumptions in the fields of Machine and Deep Learning, like the classical supervised learning, generalization, overfitting, model selection, and specialized architectures such as CNNs, RNNs, Transformers, etc. 2. Reflect on the strengths and limitations of classical and modern Machine and Deep Learning concepts 3. Apply the covered techniques, with reasons about their effectiveness 4. Implement key Machine Learning and Deep Learning techniques	
Education Method	In-person lectures with additional labs/practical projects	
Literature and Study Materials	- Bishop, "Pattern Recognition and Machine Learning" - Goodfellow et al., "Deep Learning"	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 100%) Final grade calculation = 1 * Digital Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

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Themes 2024	
Introduction 1	Through our challenging themes, you obtain advanced knowledge and skills in subject areas of Data Science and AI Technology through inspiring themes, allowing you to work on specific types of challenges you might encounter as a professional. With the flexibility to select two themes, you have the freedom to create a programme that is tailored to your individual strengths and area of interest. Each theme consists of 3 courses, offered in the 2nd, 3rd and 4th quarters of the 1st year.
Introduction 2	Themes: - Algorithms - Computer Graphics - Computer Systems - Cyber Security - Distributed Systems Engineering - Information Management - Machine Learning Applications Engineering - Programming Languages - Software Engineering

Algorithmics 2024	
Introduction 1	Many of the important problems in society are combinatorial (NP-hard) optimisation problems. For example, health treatment planning, optimizing logistic operations, automated production, efficient planning for the energy transition, and scheduling of train operations are immensely important since this may directly result in lower costs, increased productivity, and/or higher quality. Although such problems may seem very different, the same approaches may be applied. In this theme, students learn how to formally specify deterministic problems in an unambiguous and rigorous way, apply advanced algorithms to solve these problems, and reason about the trustworthiness of the solutions.
Introduction 2	Courses: Q2: CS4530 Modelling and Problem Solving Q3: CS4535 Constraint Solving Q4: CS4205 Evolutionary Algorithms

CS4205	Evolutionary Algorithms	5
Responsible Instructor	Prof.dr. P.A.N. Bosman	
Responsible Instructor	Dr.ir. P.A. Bouter	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we consider a specific subfield of Artificial Intelligence: Evolutionary Algorithms (EAs). These algorithms, sometimes also identified as being part of the class of bio-inspired algorithms, have as a metaphor the concept of natural evolution, i.e., the mechanisms by which, the fittest individuals in a population survive, reproduce, and in doing so, over time, change to be better equipped to thrive in their environment. Initiated in the 60s and 70s of the 20th century, research on EAs has progressed immensely. Today, EAs are being used to solve real-world problems in many areas, e.g. to optimize the layout of electrical wind farms, to automatically create radiation therapy treatment plans, and to optimize the architectures of deep neural networks. This course covers a spectrum of topics in EAs, ranging from basic concepts to advanced, recent, and state-of-the-art research, and ranging from theoretical to applied. In particular, topics include genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, optimal mixing evolutionary algorithms, multi-objective optimization, (GPU) parallelization, and real-world applications.	
Study Goals	At the end of this course, the student is able to: 1. Explain the key concepts underlying the main streams in Evolutionary Algorithm (EA) research, with in particular genetic algorithms, evolution strategies, genetic programming, estimation-of-distribution algorithms, and optimal mixing evolutionary algorithms. 2. Explain key ingredients underlying the rationale of when these algorithms work and when they do not work. In particular: schema analysis and how the match between the search bias of an EA and the fitness landscape is influenced by aspects such as variable dependencies and multi-modality. 3. Explain key research lines along which state-of-the-art research in EAs is done to achieve more robust, efficient, and effective EAs. 4. Identify good opportunities for using EAs, or hybrid versions thereof, in practice. 5. Properly (scientifically) experiment with EAs as well as program your own.	
Education Method	8-10 lectures 4 smaller (individual) projects 1 larger group project	
Literature and Study Materials	Various scientific papers, course-specific texts, and slides	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment (weighting 30%) - Individual Assignments: weekly assignments (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.3 * Group Assignments + 0.1 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4530	Modelling and Problem Solving	5
Responsible Instructor	Dr. N. Yorke-Smith	
Instructor	S. Dumani	
Instructor	Dr. E. Demirovi	
Instructor	Ir. S.E. Verwer	
Instructor	Y. Murakami	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: bachelor course on Algorithm Design, programming Recommended: bachelor course on reasoning, complexity theory	
Course Contents	The course covers major (combinatorial) solving approaches, namely constraint programming, integer programming, SAT, decision diagrams, and local search. The course focuses on modelling practical problems, expressing them in the survey paradigms; algorithms for solving such problems are covered in the follow-up "Constraint Solving" course. Lastly, the course focuses on conducting proper empirical evaluations.	
Study Goals	At the end of this course, the student is able to: 1. Comprehensively explain main modelling paradigms such as constraint programming, mixed-integer programming, SAT and decision diagrams. 2. Motivate a choice for an appropriate modelling paradigm. 3. Model real-world problems in a suitable modelling paradigm. 4. Conduct proper empirical evaluation of their models and related algorithms.	
Education Method	Lectures and labs	
Literature and Study Materials	* Russell and Norvig, Artificial Intelligence: A Modern Approach, 4th (Global) Edition * provided study material	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Group Assignment 1: lab assignment (weighting 10%) - Group Assignment 2: lab assignment (weighting 10%) - Group Assignment 3: lab assignment (weighting 10%) - Group Assignment 4: lab assignment (weighting 10%) Final grade calculation = 0.6 * Digital Exam + 0.1 * Group Assignment 1 + 0.1 * Group Assignment 2 + 0.1 * Group Assignment 3 + 0.1 * Group Assignment 4 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity - Group Assignment 4: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Special Information	For personal administrative matters only: mps-cs-ewi@tudelft.nl For general questions: please use answers.ewi.tudelf.nl	
Tags	Artificial intelligence Group work Modelling	
Co-Instructor	Prof.dr. M.M. de Weerdt	

CS4535	Constraint Solving	5
Responsible Instructor	Dr. N. Yorke-Smith	
Responsible Instructor	Dr. E. Demirovi	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Covers state-of-the-art algorithms used in practice to solve combinatorial optimisation problems, focusing on (Max)SAT and (lazy clause generation) constraint programming. The course offers an in-depth view of the algorithms including advanced techniques (e.g., propagator design, conflict analysis, decomposition approaches), alongside techniques that can certify the correctness of the output of the algorithms beyond reasonable doubt. The assignments include a notable programming component with Rust as the programming language. After the course, students will have a principled understanding of the algorithms and will be able to use that knowledge to design their own tailored approaches to solve combinatorial optimisation problems.	
Study Goals	At the end of this course, the student is able to: 1. Explain how combinatorial optimisation paradigms based on (Max)SAT, (lazy clause generation) constraint programming, and to some extent mixed-integer linear programming work. 2. Understand the main mechanisms behind ensuring the correctness of the output of algorithms (certificates of unsatisfiability / optimality). 3. Analyse the inner workings of the algorithms to determine the reason for the effectiveness of one paradigm over the other on a given problem. 4. Design and implement algorithms to solve combinatorial optimisation problems.	
Education Method	Lectures, practical assignments, labs	
Literature and Study Materials	Lecture notes, lecture videos, academic papers, package documentation, various textbooks.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Group Assignment 1: lab assignment (weighting 10%) - Group Assignment 2: lab assignment (weighting 10%) - Group Assignment 3: lab assignment (weighting 10%) - Group Assignment 4: lab assignment (weighting 10%) Final grade calculation = 0.6 * Written Exam + 0.1 * Group Assignment 1 + 0.1 * Group Assignment 2 + 0.1 * Group Assignment 3 + 0.1 * Group Assignment 4 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity - Group Assignment 4: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Computer Graphics 2024	
Introduction 1	Computer graphics is the science of generating images with a computer. This topic has gained a tremendous importance for many domains. Computational photography and image processing are key enablers to photo manipulation. Image synthesis is important for simulations, such as in an architectural context, movies, or video games but also to generate targeted data for machine learning. Data Visualization can give insights into medical datasets. Throughout the theme, real-world applications will be covered that combine theory and practice. In this theme, you will gain working knowledge in Computer Graphics. Starting with 2D captures of the world in form of photos, you will acquire skills in image processing, learn about algorithms to clean, complete, manipulate photos or extract additional information, such as 3D structures. This step links to the next focus of the theme on 3D representations. We will learn about generating images from 3D data via various methods and include physical simulations to achieve realistic and informative results, where the latter will be illustrated with examples from scientific computing. In this context, shader programming aspects will be covered as well to extend the practical skills of the students. Finally, the objects in these 3D scenes will be the focus of geometric data processing. The focus on geometry is important for simulations of physical interaction, but also for processing of acquired 3D scans, or fabrication of real-world objects from a virtual counterpart. The latter aspect brings us full circle back to the real world, where we started. The theme will provide a wide overview of the field, covering the breadth of Computer Graphics and its importance. The topics will be linked to contemporary results and research, and the courses will provide you with skills that are of high value even beyond pure image generation, as illustrated with many applications using real-world examples.
Introduction 2	Courses: Q2: DSAIT4120 Applied Image Processing Q3: DSAIT4080 3D Visualization Q4: CS4540 Geometric Data Processing

CS4540	Geometric Data Processing	5
Responsible Instructor	Dr. K.A. Hildebrandt	
Instructor	Dr. M. Skrodzki	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Geometric data processing is concerned with the representation, analysis, manipulation, and optimization of digital shapes. Thanks to the advances in 3D acquisition and manufacturing technologies (like 3D-Scanning and 3D-printing), the usage of geometric data is continuously increasing and an efficient processing of digital shapes plays an important role for a variety of applications in areas such as computer graphics, computer-aided design and engineering, medical imaging and surgery planning, architecture, and entertainment. In this course, we will study concepts and algorithms for creating, analyzing, editing and optimizing digital geometric shapes.	
Study Goals	At the end of this course, the student is able to: 1. Describe the fundamental techniques used for representing, analyzing, processing and modeling digital 3D-shapes treated in the course and to explain the mathematical and algorithmic concepts associated with them 2. Apply the learned mathematical concepts to solve basic geometric problems arising in geometry processing applications 3. Design algorithms that can solve simple geometric processing tasks and evaluate the drawbacks, benefits and limitations of the proposed algorithms 4. Implement the designed algorithms in a geometric processing software framework	
Education Method	The course combines lectures, tutorials, practical project work, and assignments.	
Literature and Study Materials	References to textbooks and recent research and survey papers are given in the lectures.	
Assessment	The final grade of the course consists of the following components: - Group Assignments: practical projects (weighting 60%) - Individual Assignment: theoretical assignment (weighting 40%) Final grade calculation = 0.6 * Group Assignment + 0.4 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignment: Repair opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr. E. Eisemann	

DSAIT4080	3D Visualization	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	T. Höllt	
Instructor	Dr. P. Kellnhofer	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Recommended: Computer Graphics (CSE2215), Programming skills in C++, Open GL (a crash course in OpenGL will be provided)	
Course Contents	Theory and practice of spatial data visualization such as medical scans or simulations. This includes the following aspects: data acquisition basics, clinical practice; image processing, e.g., filtering, segmentation and measurement; medical volume visualization; illustrative visualization; advanced visualization for complex modalities; interaction techniques for 3D; advanced applications and specific requirements for different fields, such as medical.	
Study Goals	At the end of this course, the student is able to: 1. Explain 3D visualization algorithms and their applications for example in medicine. 2. Discuss the advantages and disadvantages of 3D-oriented visualization. 3. Discuss implications with regards to ethical issues, trust, etc. with a particular focus on medical visualization systems. 4. Build a 3D visualization system for a given problem.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of data visualization, common problems and their solutions. Various algorithms and their properties will be discussed. During the lab sessions (2 hours each), you will work in groups on specific data visualization projects and you will receive formative feedback on your approach.	
Literature and Study Materials	"Visual Computing for Medicine", Second Edition: Theory, Algorithms, and Applications, Authors: Bernhard Preim and Charl P. Botha	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 40%) - Group Assignments: implementation assignments (weighting 60%) Final grade calculation = 0.4 * Written Exam + 0.6 * Group Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade (rounded to .5) is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4120	Applied Image Processing	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. M. Weinmann	
Instructor	Dr. P. Kellnhofer	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course provides an overview of image processing with a deeper focus on applications for visual data. This will include: image representations, color models, linear filters, image sampling, digital photography (including HDR and tone mapping), image retargeting, image abstraction and non-photorealism, multiscale image processing (pyramids, Fourier transform), geometrical image transformations (warping, morphing), and neural image processing (concept, texture synthesis, style transfer, super-resolution, denoising, editing).	
Study Goals	At the end of this course, the student is able to: 1. Describe image processing techniques and concepts such as image formation, multi-scale analysis, denoising, super-resolution, style-transfer, texture synthesis, image synthesis, etc. 2. Implement efficient algorithms for processing digital images in C++ or Pytorch. 3. Apply image processing algorithms to a (simple) real-world problem which may need to run, evaluate, or extend existing code. 4. Analyse algorithms for image processing based on their computational costs, qualitative performance, and application areas. 5. Reflect on an image processing/editing/manipulation assignment in the context of responsible engineering and ethics.	
Education Method	The course consists of two types of weekly contact activities. In the lectures (2 hours each), you will be presented with the theory of image processing, common problems and their solutions. Various algorithms and their properties will be discussed. During the lab sessions (2 hours each), you will be introduced to guidelines for efficient implementation of selected algorithms, and you will receive formative feedback on your approach to practical assignments. The final project will be solved in small groups, and it will focus on survey and assessment of state-of-the-art approaches to a specified image processing problem. It will be closed by a presentation and a report.	
Literature and Study Materials	Required literature: - Lecture slides with additional references. Recommended literature: - Szeliski, Richard. Computer vision: algorithms and applications. Springer Science & Business Media, 2010. - Chapters 3, 4 and 10. - Burger, Wilhelm, and Mark J. Burge. Digital image processing: an algorithmic introduction using Java. Springer, 2016. - Chapters 22-26. - Sundararajan, Duraisamy. Digital image processing: a signal processing and algorithmic approach. Springer, 2017. - Chapters 2 -7. - Goodfellow, I., Bengio, Y. and Courville, A.. Deep Learning. MIT Press, 2016. - Chapters 6-9 and 14.	
Assessment	The final grade of the course consists of the following components: - Individual Assignments (weighting 55%) - Group Assignment (weighting 45%) Final grade calculation = 0.55 * Individual Assignments + 0.45 Group Assignment A passing final grade for the course can only be earned if both components are at least a 5.0 and the weighted final grade (rounded to 0.5) is at least a 6. The group assignment can lead to individual grades and not necessarily a single grade for all group members. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Repairs are individual and the group assignment can only be repaired, if the individual assignments (without repair) was superior or equal to a 6.0. - Individual Assignments: Repair opportunity via an oral exam with potential preparation time. - Group Assignment: Repair opportunity in form of an individual project. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Computer Systems 2024	
Introduction 1	The Computer Systems theme targets students who like to understand how low-level systems software manages and operates computer hardware (CPU, storage, network, I/O, etc.), with a focus on embedded systems that monitor and control real-world machinery. - The real-time systems (CESE4025) course addresses the scheduling of tasks under time constraints. - The embedded systems lab (CESE4030) is a project-based course in which teams of students write the software to control an in-house developed quadcopter (drone). - The smart phone sensing (CESE4120) course provides basic signal processing knowledge (including Baysian reasoning and ML) suited for execution on a smart phone. All three courses involve actual hardware and require advanced programming and debugging skills to complete successfully.

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course covers the following topics: - basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and terminology of real-time systems 2. Construct task schedules using different scheduling policies under a given set of realistic system constraints 3. Analyze the timing behavior of a system for a given system model and scheduling policy 4. Discuss advantages and disadvantages of different scheduling policies for a given platform or system 5. Discuss the effect of hardware and software interferences on the timing behavior of a given system 6. Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system 7. Derive (reverse engineer) the system specification from a given implementation (in the lab) 8. Evaluate the scheduling overheads of a given implementation (in the lab) 9. Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Individual Assignment: lab work (weighting 40%) Final grade calculation = 0.6 * Written Exam + 0.4 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 6 (5.8 and higher). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Instructor	Ir. J.B. Dönszelmann	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level. At the end of this course, the student is able to: 1. Explain and analyse real-time communication, sensor data processing, actuator control, control theory in the context of an embedded system 2. Consider various multidisciplinary aspects at the system level 3. Apply real-time programming concepts in an embedded context	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding at home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	The final grade of the course consists of the following components: - Lab Project (weighting 75%) - Written Report (weighting 25%) Final grade calculation = 0.75 * Lab Project + 0.25 * Written Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Lab Project: Additional assignment - Written Report: Additional report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. At the end of this course, the student will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	The final grade of the course consists of the following components: - Written Report 1: two intermediate reports (10% each, 2 pages each) (weighting 20%) - Written Report 2: final report (4 pages) (weighting 10%) - Oral Exam: final project demonstration (weighting 70%) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. Final grade calculation = 0.2 * Written Report 1 + 0.1 * Written Report + 0.7 * Oral Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Report 1: Resit about a new related project - Written Report 2: Resit about a new related project - Oral Exam: Resit about a new related project Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Cyber Security 2024	
Introduction 1	The digital transformation of our society relies on secure computer and network systems. Global loss in revenue due to cyberattacks is estimated to be trillions of dollars annually and devastates enterprises, governments, and individuals. This theme aims to teach Computer Science Master students fundamental methods and principles to assess and evaluate security readiness, explore vulnerabilities, design architectures, and propose attack mitigations of computer and network systems. In the core mandatory course of the theme, Computer Security, students study and apply security concepts in a single computer. In the first theme course, Systems Security, students study and apply security concepts in networked computer systems. In the second theme course of the theme, Privacy-enhanced Technology, students study and apply anonymity, credentials, and identity management techniques in different areas. In third theme course, Network Security, students study and apply network concepts, methods, and best practices in network security. In the research course, lab Hacking Lab, the students work in teams of 2-4 students to reproduce and extend results in a domain of computer or network security. Topics include malware and ransomware, distributed denial of service attacks, web security and privacy, operating systems security, side channel attacks, and Internet security.
Introduction 2	Courses: Q2: CS4150 Systems Security Q3: CS4380 Privacy Enhancing technologies Q4: CS4430 Network Security

CS4150	Systems Security	5
Responsible Instructor	G. Smaragdakis	
Responsible Instructor	A. Voulimeneas	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course overviews principles, concepts, methods, and best practices in computer systems security. The students will obtain the necessary background and hands-on experience to design secure computer systems, study attacks and defense methods in computer systems, develop and test code to explore vulnerabilities, and apply practical defenses against software and hardware attacks. Topics covered in the course include: - Hardware primitives - Operating systems abstractions - System-level attacks and defenses - Secure systems engineering	
Study Goals	At the end of this course, the student is able to: 1. Study security architecture of computer systems 2. Analyse security issues of a specific operating system 3. Develop skills in exploiting vulnerabilities of computer systems 4. Develop counter measures against exploitation of computer systems 5. Design secure computer systems	
Education Method	Lectures and lab assignments	
Literature and Study Materials	Lecture slides, "Hacking: The Art of Exploitation", Jon Erickson	
Assessment	The final grade of the course consists of the following components: - Group Assignment 1: lab assignment on system call interposition (weighting 10%) - Group Assignment 2: lab assignment on code scanning (weighting 10%) - Group Assignment/Presentation: paper presentation (weighting 20%) - Written Exam (weighting 60%) Final grade calculation = 0.1 * Group Assignment 1 + 0.1 * Group Assignment 2 + 0.2 * Group Assignment/Presentation + 0.6 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment/Presentation: Repair opportunity - Written Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4380	Privacy Enhancing Technologies	5
Responsible Instructor	Dr. Z. Erkin	
Instructor	Dr. E.A. Markatou	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Concepts like the Internet of Things or Big Data inherently utilize massive amounts of data containing private information collected and stored by websites, sensors, monitoring systems, auditing systems, and so on. Examples include electronic records in health care systems and location information in ubiquitous computing applications. But how can we protect the privacy of participating users while at the same time enable effective sharing and utilization of the distributed data? There are several dimensions in the area of privacy, ranging from technical and juridical to societal and economical. While we will touch upon all these different aspects in the course, we will focus on the technical dimension. We will explore potential techniques for building new platforms, services, and tools that protect users' privacy. The study of promising component technologies ranging from advances in anonymous communication and identity management to theoretic tools like differential privacy and cryptography will be the core of this course.	
Study Goals	At the end of this course, the student is able to: 1. Explain the privacy in the Internet of Things 2. Analyse and evaluate anonymity mechanisms, both for anonymous communication and for database privacy 3. Apply and analyse the concept of secure multiparty computation to protect privacy in different application domains 4. Design solutions using privacy-enhancing technologies	
Education Method	Lectures	
Literature and Study Materials	Lecture Slides, essential reading material provided with the lectures (articles and tutorials)	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 70%) - Individual Assignments (weighting 30%) Final grade calculation = 0.7 * Digital Exam + 0.3 * Individual Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Individual Assignments: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. K. Liang	

CS4430	Network Security	5
Responsible Instructor	Dr.ir. H.J. Griffioen	
Responsible Instructor	G. Smaragdakis	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The course provides an overview of the most important concepts, methods, and best computer and network security practices. In this course, students will obtain the knowledge and hands-on experience to secure networking and communication systems. The course will primarily focus on technologies, protocols, attacks, and defenses. More precisely, starting from a review of common vulnerabilities and attack scenarios, the course will discuss the fundamentals of security engineering and their application in system design and finally review tools and methods to assess and test communication infrastructure from a security perspective. As a result, students will gain theoretical knowledge and hands-on experience in network attacks and defense methods. Knowledge activation and the transfer from conceptual understanding towards practical experience will be further facilitated by students implementing their own attack or defense tools on selected topics and conducting measurements on the effectiveness of attack and defense schemes. Topics covered in this course are: (1) physical layer security (2) link layer security (3) network layer security (4) transport layer security (5) application layer security (6) Internet security (7) cyber threats and countermeasures	
Study Goals	At the end of this course, the student is able to: 1. Design secure computer networks 2. Detect vulnerabilities in computer networks 3. Apply mitigation techniques for network attacks 4. Measure the effectiveness of network defence mechanisms 5. Explain and investigate security incidents 6. Design and implement network defence mechanisms	
Education Method	Lectures and lab assignments	
Literature and Study Materials	- Lecture slides - "Network Security Essentials: Applications and Standards Paperback", William Stallings, Pearson - "Computer Security: Principles and Practice", William Stallings, Pearson	
Assessment	The final grade of the course consists of the following components: Component 1: - Individual Assignment 1: lab assignment on Link Layer Security (weighting 10%) - Individual Assignment 2: lab assignment on Network Layer Security (weighting 10%) - Individual Assignment 3: lab assignment on Transport Layer Security (weighting 10%) - Individual Assignment 4: lab assignment on Application Layer Security (weighting 10%) Component 2: - Individual Assignment 5: lab assignment on Internet Security (weighting 25%) - Individual Assignment 6: lab assignment on Cyberthreat Intelligence (weighting 25%) Component 3: - Individual Exam (weighting 10%) Final grade calculation = 0.1 * Individual Assignment 1 + 0.1 * Individual Assignment 2 + 0.1 * Individual Assignment 3 + 0.1 * Individual Assignment 4 + 0.25 * Individual Assignment 5 + 0.25 * Individual Assignment 6 + 0.1 * Individual Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignment 1, 2, 3, 4: Repair opportunity - Individual Assignment 5, 6: Repair opportunity - Individual Exam: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Distributed Systems Engineering 2024	
Introduction 1	This theme aims to develop the system engineering skills needed to design scalable computing systems. The focus of this theme is on such systems for distributed computation, data processing, and distributed machine learning. Engineering real-world systems is a core part of this theme, student must be able to turn theory into reality. Distributed systems engineering teaches students about the principles, technologies, and practices used in the design, implementation, deployment and management of distributed computing systems. These systems are composed of multiple interconnected computers that work together to achieve a common goal. Examples of such goals can be scalable data processing, distributed machine learning, blockchain systems, and the Cloud. At its core, systems engineering utilizes systems thinking principles. Systems thinking is a way of making sense of the complexity of the digital world by looking at it in terms of wholes and relationships rather than by splitting it down into its constituting parts. This theme develops the skills to develop complex distributed systems which are used daily in the real world. Throughout the courses, the complete spectrum of systems engineering will be covered: from foundational theory on distributed systems (consensus, distribution, parallelization, etc.) to scalable data storage & processing architectures, distributed machine learning architectures (GPUs, model-/data-parallel machine learning), and the design of decentralized systems such as blockchain. Famous applications of distributed systems within this theme are the Google search engine and Spanner, the Netflix platform, Bitcoin payment systems, and systems like Apache Flink or Amazons DynamoDB. A recent master thesis within this theme is titled: Web3Recommend: Decentralised recommendations with trust and relevance (available online) where a master student crafted a decentralised alternative to Spotify.
Introduction 2	Courses: Q2: CS4545 Distributed Algorithms Q3: CS4550 Distributed Data Systems Q4: CS4160 Blockchain Engineering

CS4160	Blockchain Engineering	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	

Course Contents

In this course you will learn all aspects of blockchain technology, including tamper-proof data structures, digital identities, transitive trust, fault tolerance, distributed consensus, smart contracts and applications. Ledgers and blockchains are an emerging technology with the potential to radically improve financial transactions, supply-chain flows, transactions in general, and distributed databases. The first three weeks of the course will provide a fast-paced introduction to Bitcoin, Ethereum, and TrustChain developed by TUDelft itself.

The main component in this course is a team-based complex engineering project. This course is designed for computer scientists to understand blockchain technology and to produce significant hands-on experience. To provide a deep understanding of blockchain technology and understand why it is special you need to experience first-hand how it operates at a detailed technical level. Students design, implement, and test their own independent project in teams of 3-5 students. Students can choose from a pool of possible project ideas. This course requires you to like software engineering.

Topics covered:
-Blockchain basics and evolution Bitcoin 1st generation, smart contract generation, future 3rd generation (trust or trust in math)
-identity and transitive trust Authentication and security primitives, tamper-proof identities, trust models, MITM attacks, Sybil attacks, and TrustChain by TUDelft
-Consensus models Proof-of-work, permissioned, Proof-of-stake, Corda no-global-consensus, TUDelft bottom-up fast consensus model
-Smart Contract pro/con encrypted data, Bitcoin scripts, Etherium execution model, Hyperledger + Docker issues, Corda Jar file approach, Tezos difficult to use, powerful technology, vision of the future: trusted verified execution
-Markets and exchanges Disruption by open markets, winner-takes-all, and multi-sided market platforms, Uber, Airbnb, 22 years of eBay, Silk Road, honesty among drug dealers, the role of trust in markets, P2P exchange markets
-Decentralized Autonomous Organization, novel method to collaborate and organise any economic activity
-Web3: how the Internet can be re-decentralised using DAOs

After this course you will have a firm grasp on the current operational blockchain-based systems, realistic view of real-world applications that may be built on top of ledger technology. You will be able to reason and discuss the open challenges and questions that still need to be resolved. This course is a key course for distributed systems students.

Study Goals

At the end of this course, the student is able to:
1. Engineer systems with core distributed ledger concepts (replication, consensus)
2. Transform the current state-of-the-art operational blockchain-based systems
3. Create real-world applications on top of ledger technology.
4. Ability to reason with the open challenges (scalability, ethics, and trust) in distributed ledgers

Education Method

Lecture, instruction, and group work

Literature and Study Materials

Online course textbook: Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction by Narayanan, Bonneau, Felten, Miller and Goldfeder.

Assessment

The final grade of the course consists of the following components:
- Group Assignments: class project (weighting 100%)

Final grade calculation = 1 * Group Assignments
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:
- Group Assignments: Repair opportunity

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

CS4545	Distributed Algorithms	5
Responsible Instructor	Dr.ir. J.A. Pouwelse	
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Knowledge of Computer Networks (CSE1405) and Operating Systems (CSE2430) is useful as background for understanding this course.	
Course Contents	Introduction to distributed algorithms; notions of time and ordering of events; distributed algorithms for network overlays, message ordering, detecting global states, termination detection, deadlock detection, mutual exclusion, election, minimum-weight spanning trees, fault tolerance, broadcast, consensus, and agreement; blockchain technology and its relation with consensus.	
Study Goals	At the end of this course, the student is able to: 1. Understand the fundamental problems that distributed algorithms face 2. Understand the most important distributed algorithms that solve these problems 3. Reason about the execution of distributed algorithms 4. Program distributed algorithms	
Education Method	Lectures and lab. Lab work executed in groups of two students	
Literature and Study Materials	Additional resources are not necessary to attend and succeed in the course, which is entirely self-contained. However, the following list contains additional resources that might be helpful. - Distributed algorithms for message-passing Systems' book by Michel Raynal. - 'Introduction to secure and reliable distributed Programming' book by Cachin et al. - 'Distributed algorithms' book by Wan Fokkink. - 'Principles of Distributed Computing' lecture notes by Roger Wattenhoffer. - 'Introduction to Reliable and Secure Distributed Programming' book by Christian Cachin, Rachid Guerraoui, Luís Rodrigues. Springer. - 'Distributed Systems' book by Maarten van Steen, Andrew S. Tanenbaum.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) - Group Assignment: lab (Pass/Fail) Final grade calculation = Written Exam and need to pass the lab A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4550	Distributed Data Systems	5
Responsible Instructor	Dr. A. Katsifodimos	
Responsible Instructor	K. Atasu	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Starting in the mid-1990s, computing is undergoing a revolution, in which collections of independent computers appear to users as a single, albeit distributed, computing system. Motivated by the increase in the computation capacity of consumer computers, by the commoditization of server-grade machines, and by the advent of the Internet, the distributed computing paradigm has permeated all fields using computers. Current distributed computing applications range from the consumer social networks, peer-to-peer file-sharing, and massively multiplayer online games; to scientific computing using Big Data and stream processing; and to engineering fields and industrial control systems. This course focuses on the systems aspects of distributed computing.	
Study Goals	At the end of this course, the student is able to: 1. Explain the objectives and functions of distributed computing systems. 2. Describe the architecture and operation of distributed computing systems. 3. Explain how distributed computing systems can process data. 4. Implement complex operations of modern distributed computing systems in real-world use-cases such as distributed learning and graph processing. 5. Analyse the trade-offs inherent in the design of distributed computing systems (performance, efficiency, scalability, reliability, availability, fault-tolerance.)	
Education Method	Lectures and labs	
Literature and Study Materials	Designing Data-Intensive Applications, Martin Kleppman, O'Reilly Media, Inc. 2007.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 30%) - Group Assignment (weighting 70%) Final grade calculation: 0.3 * Written Exam + 0.7 * Group Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Information Management 2024	
Introduction 1	Information management is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, structured, and unstructured data is becoming more and more challenging. In the context of AI, solid data and information management is one of the core prerequisites of most AI and Machine Learning applications; it is often accepted that up to 80% of the efforts to deploy and effective AI system typically are spent on managing and preparing the required data. This theme introduces the principles and techniques of Web Science and Engineering (WSE), Information Retrieval (IR), Crowd Computing (CC). The Web is now the primary source of data and information for leisure and work activities. The real value of the Web can only be unlocked if the huge amount of available data can be found, stored, analysed, and exploited so that the needs and desires of users can be satisfied in various contexts. The courses WSE and IR focus on these aspects. However, most tasks on the Web (and in data management and data-driven AI in general) are focused on task using human-created data or which are designed to bring utility to other humans. Therefore, humans are needed in the data processing pipeline to assist with human semantics and needs. The challenges stemming from this is the focus of the Crowd Computing course. The three courses provide the fundamental knowledge related to each topic, including the concepts, methods, and techniques, and highlight the research and practical challenges in the design, development, and usage of data and information management systems in contexts. Through the lectures, assignments, and the group project, students will gain an understanding of scientific and practical challenges in information management, and learn the skills to design, develop, and evaluate their own systems, from both a technological and human point of view.
Introduction 2	Courses: Q2: CS4145 Crowd Computing Q3: DSAIT4050 Information Retrieval Q4: DSAIT4055 Web Science and Engineering

CS4145	Crowd Computing	5
Responsible Instructor	Prof.dr.ir. A. Bozzon	
Responsible Instructor	Dr.ir. U.K. Gadiraju	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Crowd Computing is a research field at the intersection of computer science and data science. Crowd computing studies how large groups of people can solve complex tasks that are currently beyond the capabilities of artificial intelligence algorithms, and that cannot be solved by a single person alone. Crowd Computing involves the algorithmic engagement and coordination of people by means of Web-enabled platforms. These complex tasks are mainly focused on the creation, enrichment, and interpretation of data, making crowd computing a building block of data science. Examples of such tasks include the coordinated creation of data about real-world events when electronic sensors are not available; the annotation of existing data sets to create ground truth data for the training of machine learning algorithms; and the analysis and interpretation of Web data to identify or moderate inappropriate content (e.g., hate speech, or fake news). Crowd computing is an essential tool for any data-driven company: from Facebook to Microsoft, from Google to IBM, from Spotify to Pandora, all major companies employ crowd computing to fulfill their data needs, both by involving employees, and by reaching out to anonymous crowds through online marketplaces like Amazon Mechanical Turk, Appen, Prolific, and Toloka. The objective of the CS4145 Crowd Computing course is to introduce the scientific and technical underpinnings of crowd computing, and to investigate how it can be used for computer science applications (e.g., information retrieval, machine learning, next-generation interfaces, and data mining) and for real-world applications (e.g., cultural heritage preservation, online knowledge creation, smart cities, etc.)	
Study Goals	At the end of this course, the student is able to: 1. Identify the requirements for a Crowd Computing system 2. Design and develop Crowd Computing systems. Support and defend the relevance and correctness of his/her choices 3. Describe and compare several Crowd Computing techniques. 4. Describe and compare design decisions in the context of Crowd Computing interaction paradigms 5. Determine which Crowd Computing technique(s) is most appropriate for being used in a certain problem domain 6. Apply the appropriate Crowd Computing technique to an application domain and evaluate the obtained results. 7. Analyse the performance of a Crowd Computing system by applying the proper evaluation measures.	
Education Method	Lectures, Group Assignments, Individual Assignments, Discussions	
Literature and Study Materials	"Books: - Human Computation. Author(s): Edith Law and Luis von Ahn. Synthesis Lectures on Artificial Intelligence and Machine Learning, June 2011, Vol. 5, No. 3. http://www.morganclaypool.com/doi/abs/10.2200/S00371ED1V01Y201107AIM013 - A. Marcus and A. Parameswaran. Crowdsourced Data Management: Industry and Academic Perspectives. Foundations and TrendsR in Databases, vol. 6, no. 1-2, pp. 1161, 2013. DOI: 10.1561/1900000044. https://people.eecs.berkeley.edu/~adityagp/papers/crowd-book.pdf - An Introduction to Hybrid Human-Machine Information Systems. Demartini, G., Difallah, D.E., Gadiraju, U. and Catasta, M., 2017. Foundations and Trends in Web Science, 7(1), pp.1-87. https://edu.nl/np4th Slides: available on Brightspace Articles: available on Brightspace Recommended reading: - Interaction Design: Beyond Human-Computer Interaction (4th Ed, 2015). Authors: Jenny Preece, Helen Sharp, Yvonne Rogers"	
Assessment	The final grade of the course consists of the following components: - Individual Assignment 1 (weighting 15%) - Group Assignment (weighting 55%) - Individual Assignment 2 (weighting 30%) Final grade calculation = 0.15 * Individual Assignment 1 + 0.55 * Group Assignment + 0.3 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignment 1: Repair opportunity - Group Assignment: Repair opportunity - Individual Assignment 2: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4050	Information Retrieval	5
Responsible Instructor	Dr. A. Anand	
Responsible Instructor	Dr. M.S. Pera	
Instructor	J. Yang	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: - BSc in Computer Science (or equivalent degree), or - Programming Skills and - Linear Algebra Recommended: - Natural Language Processing Course	
Course Contents	Retrieving relevant information is one of the central activities in modern knowledge-driven societies. As the amount and variety of data increase at an unprecedented rate, access to relevant, possibly unstructured information is becoming more and more challenging. Information Retrieval (IR) is the discipline that deals with the representation, storage, organisation of, and access to information items, and it is concerned with providing efficient access to large amounts of unstructured contents, such as text, images, videos etc. The objective of the IN4325 - Information Retrieval course is to introduce the scientific underpinnings of the fields of Information Retrieval. The course aims at providing students basic information retrieval concepts and more advanced techniques for efficient data processing, storage, and querying. Students are also provided with a rich and comprehensive catalogue of information search tools that can be exploited in the design and implementation if IR-related solutions. Covered topics include: - Information Retrieval Models - Indexing techniques - Web Search - Evaluation of information retrieval systems - Recommender Systems	
Study Goals	At the end of this course, the student is able to: 1. Relate to basic concepts and terminology in IR 2. Distinguish different IR tasks that can be applied to solve problems in different information seeking scenarios 3. Create and execute a strategy for designing solutions to IR specific problems and challenges	
Education Method	Lectures, project-based learning	
Literature and Study Materials	Lectures, lecture notes, and supplementary reading material provided in class	
Assessment	The final grade of the course consists of the following components: - Individual Assignment: foundational IR tasks - text processing and matching (weighting 10%) - Individual Report (weighting 20%) - Group Report (weighting 70%) Final grade calculation = 0.1 * Individual Assignments + 0.2 * Individual Report + 0.7 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Assignments: Repair opportunity - Individual Report: Repair opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4055	Web Science and Engineering	5
Responsible Instructor	J. Yang	
Responsible Instructor	Dr. C. Lofi	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: - BSc in Computer Science (or equivalent degree), or - Programming Skills Recommended: - Scientific Writing	
Course Contents	The main subject of the course is the Web and in particular Web Data. The course considers developments in the Web and the (big) data management challenges associated to it. In particular, the course considers the relationship between people and technology that come with the Web and Web-based information systems. The course considers the Web both from an engineering perspective as well as from an analytical perspective. The course explains the concept of Web-based Information System and thus concentrates on a large class of modern information systems that use the web and web data in one way or another. The course gives an insight into the research area of Web Engineering, where methods and techniques for the design and development of web-based information systems are investigated. The course outlines the developments related to Web Data, and its management, processing and retrieval. The course gives an overview of the research and practice concerning the Semantic Web, with its main languages, theory and applications and tools for describing semantics in machine-processable manner. It also considers the concepts behind Linked Open Data and the data processing pipelines to create and analyse Linked Open Data. With the social-technical nature of the Web and its systems, the course pays attention to the interplay between people and systems. The course gives an overview of the research area of User Modeling, with its main approaches and techniques to represent and capture properties of users that provide a basis for user-adaptation and personalisation in web-based information systems. In relation to user modeling, the Social Web plays a major role, for example because data from the social web creates a great source of knowledge for user modelling. Therefore, the course also considers research in social web data analytics and data science techniques to extract user knowledge from social web data. The course also considers recent developments in the research area of Human Computation concerning the role of humans in the processing of (human-related) web data, for example using crowdsourcing to create or annotate web content. As the web and its data are mirroring the world and the people in it, the course also takes a look at Web Science, as a branch of data science that considers the largest human-made artefact ever, i.e. the Web, and how that analytical research is addressing a whole new range of challenges. These challenges include studying how data analytics can be done by means of Web data, as well as studying how new systems can be created and engineered to make use of the Web and its properties.	
Study Goals	At the end of this course, the student is able to: 1. Apply well-reasoned selected principles and concepts of Web-based Information systems, including e.g., Web-based analytics 2. Apply well-reasoned main methods, techniques and languages used for Data Management in the area of web-based information systems, in particular concerning the Semantic Web and Linked Open Data. 3. Apply well-reasoned methods and concepts for User Modelling and adaptation, and of the role of Social Web data and Human Computation for user modeling. 4. Explain the major challenges and principles from the research in the field of Web Science, and the role of web data for Web Science 5. Create a project plan to address a provided problem, which includes a project goal, and impact plan, using suitable methodologies and technologies	
Education Method	Lectures and supervised group-based project	
Literature and Study Materials	Research Papers and custom slide-deck	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 30%) - Group Report (weighting 70%) Final grade calculation = 0.3 * Written Exam + 0.7 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Machine Learning Apllications Engineering 2024	
Introduction 1	This theme delves into the intricate realm of machine learning algorithms, techniques, and applications with a focus on engineering robust and scalable solutions. The central thread running through this theme is the fusion of theoretical understanding of ML with practical implementation ML-enabled software systems. It explores the art of designing, developing, and deploying machine learning models in a real-world context.
Introduction 2	Courses: Q2: DSAIT4020 Elements of Statistical Learning Q3: DSAIT4065 Conversational Agents Q4: CS4295 Release Engineering for Machine Learning Applications

CS4295	Release Engineering for Machine Learning Applications	5
Responsible Instructor	L. Miranda da Cruz	
Responsible Instructor	Dr.ing. S. Proksch	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	While the course does not have formal pre-requisites, some skills are expected: - You are able to use basic Git commands - You have a basic understanding of networking concepts - You are not afraid of using a command line interface The course will provide introductory material for self-study that can serve as a refresher or provide pointers for students to fill knowledge gaps.	
Course Contents	The world of Software Engineering has been revolutionized in the last decade. Instead of releasing software updates yearly, companies can now release multiple times per week, sometimes even per day, to their customers. This allows quick reactions to market demands, software failures, and it is crucial for increasing the business value of software. These improvements have been mostly enabled by advances in release engineering and, in this course, we will learn about the techniques and technologies that build the foundation for modern release engineering. This course is addressing two main target audiences: 1) software engineers that would like to learn how to systematically release their applications and make them scalable in an orchestrated environment and 2) machine learning experts that would like to learn how to make their proof-of-concept applications production ready. Together, we will go on a journey that starts at continuous integration and then moves on to continuous delivery, continuous deployment, and continuous experimentation. We will discuss the theory and the current research on various related subjects like containerization, testing, or monitoring and will put the learned theory into practice. As a running example, we will build a pipeline for a machine learning application, which -compared to traditional release engineering- poses additional challenges, like data versioning or model deployment. *Important:* Make sure to register for a group in Brightspace AT THE LATEST during the first lecture. *Important:* The course introduces several state-of-the-art tools that are used in industry. While -in theory- all major operating systems are supported, experience has shown that Windows users will face challenges. To reduce the friction points, we recommend affected students to either dual-boot a Linux system or to make use of a Linux Live system via USB when working on the exercises and assignments.	
Study Goals	After following this course, you will be able to... 1) Explain the concepts of DevOps and MLOps 2) Automate the development and operation process ... Automate packaging, versioning, and releasing in modern build environments like GitLab or GitHub ... Release artifacts in public registries like the GitHub Maven Registry. ... Provision servers with Vagrant and Ansible ... Automate the model training pipeline using DVC 3) Use modern containerization technology to build a scalable micro-service architecture ... Use Docker to isolate a complex, web-based application in a container ... Use Kubernetes to orchestrate a non-trivial, distributed application ... Create an Istio service mesh in Kubernetes that enables custom routing strategies ... Use Prometheus and Grafana to monitor the state of a Kubernetes-based application ... Find and understand documentation of open-source tools 4) Deploy and maintain machine learning models in production ... Embed a typical ML model in an HTTP micro-service ... Apply version-control techniques to data and models in a machine learning project ... Implement quality-control techniques in a machine learning application ... Design a testing strategy that covers data, model, infrastructure, and monitoring aspects 5) Document the state and extensions of release engineering pipeline ... Find related work and inspiration for release engineering tasks ... Identify extension opportunities in existing release pipelines	
Education Method	- Follow interactive lectures and pre-recorded material for self-study - Execute a programming project in small teams - Transfer lecture contents into a project through practical assignments - Assess the work of yourself and others through rubric-supported peer-feedback - Actively participate in Q&A sessions - Document your release pipeline in a report - Present your project results and answer questions about the lecture contents.	
Literature and Study Materials	- Lecture slides and recordings - In-class exercises - Tool documentation - External articles - Research papers	
Assessment	--- Formative Assessment --- You can check your individual learning progress through weekly rubrics ... You will receive feedback from other groups as a neutral outsider ... You can check your own progress via self-assessment Joint Q&A sessions will discuss common points of the peer feedback and concrete questions --- Individual Knock-Out Elements --- The course enforces several criteria to protect the active members of a team. Students not reaching the following criteria will be removed from the course:	

Upholding a professional attitude when interacting with peers and staff.
Actively participating in the team process and contributing to all team deadlines and deliverables
Constructive participation in all peer reviews for assignments (one for another team and one for your own)
Meaningful and visible project contribution throughout each of the seven project weeks (W2-W4, W6-W9)
... Create 1+ mergeable PR with meaningful* contribution
... Approve 1+ PR that is not your own
... Failing to satisfy the criteria in two subsequent weeks will lead to a course fail
At the end of the couse, you must meet the following criteria
... 7+ meaningful* PRs of you were merged to main
... You approved 7+ meaningful* PRs of others
... You did not miss peer reviews for more than one assignment.
Ability to explain lecture contents and defend all aspects of the team project in the oral examination. The inability to explain or justify the project in the oral examination can be repaired in a second oral examination.
Availability of team results for grading (repositories, report, artifacts)

(*) A "meaningful" contribution adds source code, configuration options, or report content. We do not count comments or whitespace.

The full specification of the knock-out criteria will be posted on Brightspace.

--- Summative Assessment ---

You can only get a grade for the course, if you pass the following pass/fail elements:

You have passed all knock-out criteria.
You passed the Presentation & Oral Examination

Your grade then consists of the following partial grades. ALL partial grades are rounded to .1 and must at least be "almost passed" (>=5.0). Their weighted average will be rounded to .5 to calculate the final team grade.

Versioning & Releases (10%)
Containers & Orchestration (15%)
Kubernetes & Monitoring (15%)
Provisioning (15%)
Istio Service Mesh (15%)
ML Configuration Management (15%)
ML Testing (15%)

In case of large performance differences in a group, we may choose to assess the team members individually, based on the individual contributions to the deliverables and the interaction traces that we find in the repository.

--- Please Note ---

All assignment deadlines throughout the course are fixed, we do not accept late submissions. Late submissions in systems that do not enforce due dates will be treated as not submitted (e.g., Brightspace, Buddycheck).
Failing to submit a deliverable cannot be repaired.
Insufficient deliverables (>=4.0 and <6.0) can be repaired, but the grade of the deliverable is then capped at 6.0.
There is NO resit opportunity for this course.
Partial grades are not carried over to the next academic year.

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5)

DSAIT4020	Elements of Statistical Learning	5
Responsible Instructor	Dr. D.M.J. Tax	
Responsible Instructor	T.J. Viering	
Instructor	Dr. J. Sun	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2 Summer Holidays	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: Linear Algebra, Probability Theory and Statistics, Machine and Deep Learning core course	
Course Contents	This course covers all the basic concepts of Statistical Learning, focusing on the classical techniques of Machine Learning before the era of Deep Learning. These concepts include classification, (ridge) regression, (hierarchical) clustering, feature reduction and extraction, model selection and bootstrapping, fairness in ML. The emphasis is on the concepts rather than the mathematical details.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and assumptions in the field of Statistical Learning, like the classical supervised learning, but also regression, clustering and dimensionality reduction 2. Reflect on the strengths and limitations of classical and modern Statistical Learning techniques. 3. Apply the covered learning techniques to analyse a dataset and reason about their effectiveness	
Education Method	There are lectures introducing the topics, a lab course and a final project where a dataset is analysed.	
Literature and Study Materials	"Elements of Statistical Learning' Hastie et al.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 60%) - Group Report: project report on analysis of a dataset (weighting 40%) Final grade calculation = 0.6 * Digital Exam + 0.4 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4065	Conversational Agents	5
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Responsible Instructor	Z. Yue	
Instructor	S. Tan	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Chatbots, embodied and conversational virtual agents, and social robots are becoming more and more popular. Many people are owning an Alexa, Cortana or Echo or are talking to their virtual assistant on their phone. Indeed, such technologies have the potential of making our lives easier and relieve people from the more repetitive tasks. For example, it is imaginable that such systems are being used for financial applications by helping customers with frequently asked questions but also to advise them on in the long term more impactful decisions such as their pension plans. Further applications can be imagined in the area of healthcare and education, some of which are already in existence today. In this course, attention will be given to different verbal and nonverbal behavioral characteristics of conversation and embodied interaction, such as, intonation, gaze and gestures that humans show when communicating with both other people and machines. This behavior is then related to different multimodal dialogue functions, including turn-taking, addressing others, and backchanneling, that give shape to the communication process. Topics on interaction with embodied agents and how to design memory models for conversational interaction are also discussed. Students apply this knowledge through the design, development, and evaluation of an embodied conversational agent application that uses a memory model to interact.	
Study Goals	At the end of this course, the student is able to: 1. Apply relevant linguistic, speech and psychological theory to conversational agent systems 2. Design the conversational memory of your agent focusing on a perception/generational model 3. Analyse human-human conversational data to better design ML models for adding generation or perception capabilities to conversational agents 4. Develop the conversational memory in your agent 5. Quantitively evaluate a conversational agent in a user-experiment. 6. Analyse your results and the limitations of your implementation	
Education Method	There are 2 lectures scheduled per week. Students work in groups of 4 on a group project. There are 3 labs that focus on components needed for the project.	
Literature and Study Materials	"The conversational interface " by Michael McTear, Zoraida Callejas, David Griol. This book is freely available through the TU Delft library. Also, recent papers selected by instructors.	
Assessment	The final grade of the course consists of the following components: - Digital Exam (weighting 30%) - Group Report:Group report on the design, implementation, and evaluation of a conversational agent (weighting 70%) - Group Assignment 1: group lab exercise - memory (pass/fail) - Group Assignment 2: group lab exercise - perception (pass/fail) - Group Assignment 3: group lab exercise - evaluation (pass/fail) Final grade calculation = 0.3 * Digital Exam + 0.7 * Group Report + Pass/Fail * Group Assignment 1 + Pass/Fail * Group Assignment 2 + Pass/Fail * Group Assignment 3 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Digital Exam: Resit opportunity - Group Report: Repair opportunity - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Assignment 3: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Programming Languages 2024	
Introduction 1	The overall aim of the Programming Languages theme is to improve the effectiveness and reliability of programming languages and systems. Effectiveness such that programmers can express intent at the right level of abstraction and get actionable feedback that is relevant and timely. Reliability such that programmers can trust the execution and analysis of programs. These dual goals of effectiveness and reliability are achieved through the design and study of programming languages and tools for editing, analysing, transforming, and compiling programs written in those languages. In this theme you will learn everything from the mathematical foundations of programming languages to the practical engineering of programming language tools, and from latest new developments in the field of programming languages to how to design your own programming language from scratch.
Introduction 2	Courses: Q2: CS4555 Compiler Construction Q3: CS4560 Parallel and Concurrent Programming Q4: CS4565 Advanced Functional Programming

CS4555	Compiler Construction	5
Responsible Instructor	Dr. S.S. Chakraborty	
Instructor	Dr. M.A. Costea	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Compilers translate the source code of programs in a high-level programming language into executable (virtual) machine code. In this course you will study the architecture of compilers, and important concepts and techniques underlying the components of that architecture. You will study these techniques by applying them in practice to implement your own compiler for a small language. The course covers the following topics: - Syntax and parsing - Static semantics and type checking - Virtual machines, assemby code, byte code - Interpretation vs. compilation - Code generation - Memory management, garbage collection - Data-flow analysis - Optimization	
Study Goals	At the end of this course, the student is able to: 1. Explain the architecture of a compiler and the role of each component of that architecture 2. Explain the algorithms and techniques for the implementation of compiler components and apply these techniques to examples 3. Implement a code generator that translates source language abstract syntax trees to target language instructions 4. Implement simple type checkers and data-flow analyzers 5. Integrate the course-relevant components into a working compiler	
Education Method	- Attending weekly lectures - Attending weekly lab sessions (which may start with a group tutorial) - Reading lecture material and papers - Making homework assignments (building a compiler)	
Literature and Study Materials	Required reading: We will use selected chapters from the book ""Essentials of Compilation"" by Jeremy Siek, available online: https://github.com/IUCompilerCourse/Essentials-of-Compilation Lecture slides and selected papers from the literature Optional reading: - We will use Scala to implement our compilers. The following book is optional reading which may be helpful in getting up to speed with programming in Scala: https://www.artima.com/shop/programming_in_scala_5ed (Additional) reading materials will be made available on Brightspace. We will use WebLab (https://weblab.tudelft.nl/cs4200/) for the submission of homework assignments.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 80%) - Individual Homework Assignments (weighting 10%) - Groupwork Assignments (weighting 10%) Final grade calculation = 0.8 * Written Exam + 0.1 * Individual Homework Assignments + 0.1 * Groupwork Assignments A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Homework Assignments: Complete and submit assignment after course ends - Group Assignments: Complete and submit assignment after course ends Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4560	Parallel and Concurrent Programming	5
Responsible Instructor	Dr. S.S. Chakraborty	
Instructor	Dr. M.A. Costea	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Required: Knowledge in programming Recommended: Familiarity with multithreaded or parallel programming is desired	
Course Contents	Software systems are becoming highly concurrent and parallel to utilize modern multicore architectures and increasing speed and bandwidth in networks. In the multicore era, all performance-critical software employs some form of concurrent programming; typically shared memory concurrency. In this setting, parallelism and concurrency are the dominant programming paradigms for performance gains. Writing correct and efficient parallel and concurrent programs is challenging, and requires understanding the behaviors of the primitives and reason about them. This course aims to explore parallel and concurrent programming approaches along with their analysis techniques. This course aims to give students a deep understanding and hands-on experience of parallel and concurrent programming and analyzing them. Lecture plan In this course, we will cover three parts parallelism, concurrency, and automated analysis techniques for concurrency. Parallelism can bring significant performance improvement as multiple independent components execute in parallel and thereby achieve (almost) linear speedup to the number of processing units. In the second part, we will focus on concurrent programs. Concurrency is another paradigm for efficiently programming shared memory hardware. In concurrency, multiple threads execute independently with occasional communication via the shared memory. However, concurrent programs demonstrate many subtle complexities and thereby developing correct and optimal concurrency programs is a challenging problem. In the third part we will focus on the automated analysis approaches for concurrency. We will focus on concurrency bugs and analysis techniques for various properties such as data race, consistency checking, and so on. Outline of Lectures: Parallelism (3 lectures): Parallel programming Automatic parallelization Dependency analysis Concurrency (5 lectures): Concurrent programming Correctness of concurrent programs Relaxed memory concurrency Automated analysis techniques (2 lectures): Property checking e.g. data race, consistency	
Study Goals	At the end of the course, the student is able to: 1. Describe parallel and concurrent programming paradigms 2. Explain the parallel and concurrent programming principles 3. Discover bugs in parallel and concurrent programs by analysis and testing techniques 4. Evaluate the pros and cons of program analysis and testing techniques for specific parallel and concurrent programs	
Education Method	The course consists of the following education methods: - Lectures for reviewing concurrency and distribution concepts - Homeworks/assignments - Developing a course project, writing a report, and presenting it (course project) To finish the course, students (in teams) will have to: - Study several papers which will be discussed during the lectures - Deliver their assignments - Deliver and present their implementation project	
Literature and Study Materials	The Art of Multiprocessor Programming 2nd Edition, by Maurice Herlihy, Nir Shavit , Victor Luchangco, Michael Spear	
Assessment	The final grade of the course consists of the following components: - Research project: implementation (weighting 40%) - Research project: report (weighting 20%) - Research project: presentation (weighting 20%) - Individual Assignment 1: parallel & concurrent programming (weighting 10%) - Individual Assignment 2: reasoning about concurrency (weighting 10%) Final grade calculation = 0.4 * Research project implementation + 0.2 * Research project report + 0.2 * Research project presentation + 0.1 * Individual Assignment 1 + 0.1 Individual Assignment 2 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Research project implementation: Repair opportunity - Research project report: Repair opportunity - Research project presentation: Resit opportunity - Individual Assignment 1: Repair with 10% penalty - Individual Assignment 2: Repair with 10% penalty Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4565	Advanced Functional Programming	5
Responsible Instructor	C.B. Poulsen	
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: CS4510 Formal Reasoning About Software Recommended:Experience with functional programming as taught for example in CSE3100 Functional Programming	
Course Contents	The goal of functional programming is to write software that is efficient, scalable, and robust, all while maintaining a high level of abstraction. In this course we will investigate a number of techniques and language features that have been developed to apply functional programming to large-scale software projects. Concretely, we will discuss and apply the following topics: - Efficiency of functional programs: purely functional data structures, controlling lazy evaluation, persistence and amortization, parallel and concurrent functional programming - Scalability of functional programs: monad transformers, free monads, and effect systems, lenses and optics, embedded domain-specific languages, generic programming - Robustness of functional programs: type-level programming, generalized algebraic datatypes, refinement types, security types	
Study Goals	At the end of this course, the student is able to: 1. Analyse and discuss the efficiency of different purely functional data structures and algorithms. 2. Structure and organize large-scale functional programs using type classes, monad transformers, and lenses. 3. Design domain-specific languages as libraries embedded in a functional programming language. 4. Construct concurrent functional programs that are composable and deadlock-free using Software Transactional Memory. 5. Use type-level programming to statically enforce domain-specific abstractions and invariants. 6. Explain how to use advanced type systems such as refinement types and security types.	
Education Method	Interactive lectures, programming project	
Literature and Study Materials	This course uses chapters from the following books: - Purely Functional Data Structures (1998) by Chris Okasaki, Cambridge University Press - The Craft of Functional Programming (2011) by Simon Thompson (available online at https://simonjohnthompson.github.io/craft3e/craft3e.pdf) - Haskell in Depth (2021) by Vitaly Bragilevsky, Manning (ebook available via TU Delft library)	
Assessment	The final grade of the course consists of the following components: - Homework Assignments (weighting 30%) - Programming Project (weighting 70%) Final grade calculation = 0.3 * Homework Assignments + 0.7 * Programming Project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Homework Assignments: Repair opportunity - Programming Project: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Software Engineering 2024	
Introduction 1	The purpose of the software engineering theme is to equip students with solid knowledge of both the theory and practice of the conception, design, construction, evolution, and deployment of modern complex software systems. This includes principles guiding the design and evolution of software systems, as well as architectural styles and patterns to realize required quality attributes such as scalability, security, and sustainability. Furthermore, the theme investigates novel means to automate key aspects of the software development process, including the use of fuzzing for test suite generation, advanced program analysis techniques, and large language models as well as other machine learning techniques to boost software development quality and productivity.
Introduction 2	Courses: Q2: CS4570 Machine Learning for Software Engineering Q3: CS4575 Sustainable Software Engineering Q4: CS4580 Automated Software Testing and Reverse Engineering

CS4570	Machine Learning for Software Engineering	5
Responsible Instructor	M. Izadi	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Required: - Experience with programming is required (preferably Python) - Experience with machine learning and specifically deep learning techniques is a must-have - Familiarity with transformer-based language models and modern NLP is needed Recommended: - Experience with research methods is nice to have - Taking the NLP course (CS4360) before this class is recommended	
Course Contents	Software repositories archive valuable software engineering data, such as source code, execution traces, historical code changes, mailing lists, and bug reports. This data contains a wealth of information about a projects status and history. Advances in Machine Learning and AI technologies, as demonstrated by the successful application of deep neural networks in various domains did not go unnoticed in the field of Software Engineering; researchers have applied machine learning to tackle different software engineering tasks. This course aims to give students an understanding of a hands-on approach to how deep neural networks and NLP techniques are used to represent knowledge and solve existing SE problems in novel ways.	
Study Goals	At the end of this course, the student is able to: 1. Understand current literature in the area of software analytics and machine learning for software engineering. 2. Apply machine learning / deep learning algorithms to solve software engineering tasks. 3. Apply software analytics techniques to extract actionable software engineering insights. 4. Evaluate model performance on a software engineering task.	
Education Method	Lectures, Seminars and Self-study	
Literature and Study Materials	Research papers	
Assessment	The final grade of the course consists of the following components: - Group Assignments: Project LLM4code report, including code. Propose an idea for improvement of an existing language model for code and implement and assess the same model (weighting 70%) - Group Presentation 1: Project LLM4code presentation: Report and discuss results of the project (weighting 20%) - Individual Presentation: Individual questions and answers about the project LLM4code (weighting 10%) - Group Presentation 2: Present a state-of-the-art approach in the literature (pass/fail) - Group Presentation 3: Present project progress in mid-term (pass/fail) Final grade calculation = 0.7 * Group Assignments + 0.2 * Group Presentation 1 + 0.1 * Individual Presentation + Pass/Fail * Group Presentation 2 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignments: Repair opportunity - Group Presentation 1: Resit opportunity - Individual Presentation: Resit opportunity - Group Presentation 2: Resit opportunity - Group Presentation 3: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4575	Sustainable Software Engineering	5
Responsible Instructor	L. Miranda da Cruz	
Instructor	Dr. C.E. Brandt	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Sustainable Software Engineering is an overarching discipline that addresses the long-term consequences of designing, building, and releasing a software project. By definition, sustainability covers five main perspectives: environmental, social, individual, economic, technical. This course mainly focuses on the first, also known as Green Software Engineering. On top of that, we also cover fundamental aspects of social and individual sustainability of software projects. Software Engineering (SE) has long addressed sustainability by narrowing it down to economic and technical sustainability. However, our society is facing major sustainability challenges that can no longer be overlooked by software engineers and computer scientists. It is estimated that, by 2040, the ICT sector will contribute to 14% of the global carbon footprint. Hence, the environmental, social, and individual perspectives ought to be part of the equation when it comes to design, build, and release software systems. The problem is far from simple and we need expert computer scientists to bring sustainability into the core values of the next generation of tech-leading organisations.	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental sustainability principles and their relevance to software engineering. 2. Assess the energy consumption of software systems using measurement and testing techniques. 3. Create actionable strategies to address sustainability issues within software organisations. 4. Evaluate and compare different solutions for sustainable software systems based on their effectiveness in achieving sustainability goals and their potential impact on various stakeholders.	
Education Method	Lectures and Labs	
Literature and Study Materials	Lecture slides and research papers	
Assessment	The final grade of the course consists of the following components: - Group Assignment 1: write a scientific report about AI model energy testing and verification (weighting 40%) - Group Assignment 2: project code and report (weighting 40%) - Group Presentation: project presentation (weighting 10%) - Individual Presentation: individual Q&A after group presentation (weighting 10%) Final grade calculation = + 0.4 * Group Assignment 1 + 0.4 * Group Assignment 2 + 0.1 * Group Presentation 2 + 0.1 * Individual Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignment 1: Repair opportunity - Group Assignment 2: Repair opportunity - Group Presentation: Resit opportunity - Individual Presentation: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4580	Automated Software Testing and Reverse Engineering	5
Responsible Instructor	Ir. S.E. Verwer	
Responsible Instructor	Dr. A. Panichella	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	"Software is one of the most complex artifacts of mankind has ever created, but complexity is the enemy of correctness. Modern software testing and validation tools use a multitude of techniques geared toward correct computer code, most of these are base on artificial intelligence. In this course, we study these techniques in details, specifically we will understand and implement: - Execution monitoring - Branch distance computation - Hill-climbing and genetic algorithms - Concrete and symbolic (concolic) execution - Active state machine learning - Genetic programming The goal is to better understand and test software using artificial intelligence. Using the taught techniques you will be able to automatically: - Discover which code is reachable - Find (security) bugs in software - Write tests that cover all reachable code - Reverse engineer a code's functionality - Patch code to remove bugs and failing tests"	
Study Goals	At the end of this course, the student is able to: 1. Understand modern AI techniques for software testing. 2. Implement several such techniques from scratch: - smart fuzzing (probing software with input to find crashes/bugs), - symbolic execution (using logic to construct inputs that trigger specific code branches), - fault localization (given that a program fails, - find the line of code responsible for the failure), - automated program repair (using a patch library and genetic programming to improve code) 3. Apply this technology to locate bugs in real-world software implementations. 4. Evaluate and analyze the performance of the different techniques	
Education Method	lectures: theoretical part lab: implementing of the taught techniques	
Literature and Study Materials	Lecture slides, research papers presented during the lectures, the fuzzing book (https://www.fuzzingbook.org)	
Assessment	The final grade of the course consists of the following components: - Lab Assignments (Pass/Fail) - Group Report (weighting 100%) Final grade calculation: Pass/Fail * Lab Assignments + 1 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Lab Assignments: No Group Report: Additional assignment Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Research courses 2024	
Introduction 1	In the fourth quarter research course, students are prepared for conducting research in Computer Science. They gain proficiency in state-of-the-art research methodologies to critically evaluate Computer Science knowledge. Students choose one research course from the list below.
Introduction 2	Courses: CS4700 Literature Survey CS4705 Research Seminar Computer Graphics CS4710 Research in Cyber Security - Hacking Lab CS4715 Programming Languages Research Seminar CS4720 Research in Program Analysis CS4725 Research Seminar on Scalable Learning Systems CESE4065 Advanced Practical I.o.T. and Seminar DSAIT4200 Empirical Research of Computational Solutions DSAIT4210 Research in Web Data and Information Management DSAIT4220 Research in Intelligent Decision Making DSAIT4230 Research in Social Signal Processing and Affective Computing DSAIT4235 Research Seminar on Multimedia Computing and Systems

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	Dr. R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0 and 0/0/0/2	
Education Period	1 4	
Start Education	1 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands-on course for self-motivated students. Students usually work in a group of two (max 3). Students are expected to have sufficient programming and hardware development skills. The course is project-oriented thus students are expected to deliver a working model with demonstrations. -- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. -- This course is done in a group of two (max 3). The instructor/TA will help connect a potential groupmate in the first class using Brightspace/Feedback Fruits. Activities: How to in Groups? ----------------------------- Part 1: Seminar We guide the groups in selecting a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar, all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID-19-related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to the 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/Raspberry Pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki-based networks or Open testbed projects. In these cases, online group activities are possible. NOTE: In some extreme cases, the Instructor/TA may allow one person to project after assessing the situation and after discussing it with the student. Supply of Hardware: ------------------- 1. Instructor/TA is able to provide take-home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	

Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc. Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Tags	Circuits Group work Programming Project Software
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CS4700	**Literature Survey**	**5**
Responsible Instructor	L. Miranda da Cruz	
Instructor	Dr.ir. A.R. Bidarra	
Instructor	Dr. K.A. Hildebrandt	
Instructor	C.B. Poulsen	
Contact Hours / Week x/x/x/x	n/a	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The Literature Survey is an individual assignment carried out under the supervision of a Computer Science staff member, i.e. an assistant, associate or full professor. For this assignment the student reads a broad range of papers in the chosen specialisation field and writes a report in which the ideas found in the papers are discussed and compared. It is not allowed to merge this assignment with the thesis project.	
Study Goals	After the course, the student will be able to: 1. Formulate a research question that is relevant, meaningful, and scientifically sound. 2. Find high-quality contemporary scientific literature in the chosen research field. 3. Explain the scientific contribution of the reviewed papers within the context of the research question.	
Education Method	Individual assignment and individual guidance by a scientific staff member.	
Assessment	The final grade of the course consists of the following components: - Individual assignment: scientific report (weighting 100%) Writing a scientific report, individually and under supervision of a staff member. This staff member will also mark the report. Final grade calculation = 1 * Individual assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The Literature Survey may be part of an individual exam programme of a student, which has to be approved by the Board of Examiners (BoE). To apply for a literature study the student should contact a staff member of the research group of her/his chosen specialisation after having received approval of his or her individual exam programme. The staff member and the student make arrangements regarding content and scope of the survey. The abovementioned staff member will supervise the student during this course. Students need to register on the Brightspace page CS4700 Literature Survey. To complete the course the final report needs to be uploaded on the Brightspace page.	

CS4705	Research Seminar Computer Graphics	5
Responsible Instructor	Prof.dr. E. Eisemann	
Responsible Instructor	Dr. P. Kellnhofer	
Instructor	Dr. M. Skrodzki	
Instructor	Dr. M. Weinmann	
Instructor	Dr.ir. A.R. Bidarra	
Instructor	Dr. K.A. Hildebrandt	
Instructor	T. Höllt	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Recommended for students following the Computer Graphics Theme.	
Course Contents	In this research course you conduct deep investigation of recent techniques solving attractive problems in a research area of Computer Graphics including Visualization, Interaction, Image Processing, Inverse rendering and 3D data processing. You will survey research outputs in the area of your interest, present a selected paper and develop an implementation supporting the education on the topic. The goal is to strengthen your analytical and presentation skills and prepare you for research in your MSc project.	
Study Goals	At the end of this course, the student is able to: 1. Explain representative methods covering selected topics from the field of Computer Graphic, Visualization or Visual Computing in form of a public lecture. 2. Formulate insightful questions during presentations of others. 3. Implement a tool that helps to further explain the discussed technique. 4. Critically assess results of the implementation in form of a short report and a presentation.	
Education Method	This course has the format of a student seminar. You will prepare a lecture on a selected topic from Computer Graphics. Typically, you will present a detailed look at one representative method based on a scientific paper. The presentation goes in-depth and covers the strengths and weaknesses of the described technique. Depending on number of participants, we may organize the paper presentations individually or in groups. After each presentation in the plenary colloquium session, a research discussion takes place. All students participate in a scientific discussion and ask relevant critical questions. Finally, you will implement a tool that further explores the presented method. Depending on the number of participants and scheduling constraints, you will then be asked to present your work either in a form of a short public demonstration or in a pre-recorded video presentation. Additionally, you will prepare a short report summarizing your solution.	
Literature and Study Materials	Scientific journals and conference proceedings from conferences including SIGGRAPH, Eurographics, Visualization, EuroVis, CVPR, ICCV, ECCV and similar.	
Assessment	The final grade of the course consists of the following components: - Presentation of a recent scientific paper (50%) - Implementation component reproducing an aspect of the paper (40%, including short report and presentation) - Preparation of questions for the presented research papers (10%) Final grade calculation: (0.5 * Presentation + 0.4 * Implementation + 0.1 * Questions) A passing final grade for the course can only be earned if all components have at least a grade of 5.0 and the weighted average rounded on 0.5 is at least a 6. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Presentation of a recent scientific paper can be repaired by an individual video presentation of a state-of-the-art report + an oral examination. The content of each presented paper must be understood by the student. - Implementation component can be repaired by a new implementation following a new approved plan. - Preparation of each question for the assigned paper can be repaired by a short video presentation of the same paper. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4710	Research in Cyber Security Hacking Lab	5

Responsible Instructor	Dr.ir. H.J. Griffioen
Responsible Instructor	G. Smaragdakis
Instructor	Dr. G.C. Moreira Moura
Contact Hours / Week x/x/x/x	0/0/0/2
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	English
Expected prior knowledge	Necessary background differs per student project, see first lecture or contact instructors for details.
Course Contents	The security of computer and telecommunication systems is becoming an increasing concern. In this course, we will review the current state of the art on security research and gain practical experience in assessing the security and vulnerabilities of communication systems. Engineers are typically taught to focus on performance, correctness, scalability, and maintainability when building communication and information processing systems. However, an additional set of design principles are required to achieve security. In this course, we discuss security principles, common pitfalls and vulnerabilities. Lectures provide an introduction into security research, with a focus on real-world security, privacy-enhancing technology and common security pitfalls. Each student participates in a "Hacking Project", with a group of one to four students. Students can select between a wide range of available Hack Project outlines within the first week. The goal may be to evaluate the security of a real-world IT system, developing a proof-of-concept exposing a vulnerability or focussed on preserving privacy in a post-Snowden world. Students may propose their own Hack Project based on their background knowledge and skills. Such Hack Projects need to be approved and shaped together with the instructor. Example of possible outlined hardware-oriented projects are: development of a wifi tracker, programming an FPGA system to break passwords, assess the security of RFID cards, or to transparently intercept Ethernet traffic. Concrete software projects are: hacking Bitcoin, improving the TOR anonymity protocol and create Android-based tools for human rights activists in Iran, Egypt and Russia, reprogramming neural networks attacks. Each Hack Project is documented with a written report. This can be in the form of a 8 page ACM-style scientific article or a traditional more lengthy report. All results, experiences and findings are presented to the entire class in the last week of the course. Hacking Projects also report their progress several times during the course, after the weekly lectures.
Study Goals	At the end of this course, the student will be able to: 1. Demonstrate a thorough understanding of security principles in real-world systems and independently explore relevant literature. 2. Analyze the vulnerabilities present in real-world computer systems. This includes explaining common security pitfalls, why these problems persist despite known solutions, and the underlying factors contributing to the overall poor state of security.
Education Method	Lectures, student presentations, written final report and active participation. Attendance and active participation during lectures and presentations is mandatory.
Literature and Study Materials	Customize literature lists and study materials are provided per project topic
Assessment	The final grade of the course consists of the following components: - Code and Documentation (weighting 40%) - Written Report (30%) - Presentation (weighting 30%) Final grade calculation = 0.4 * Code and Documentation + 0.3 * Written Report + 0.3* Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Project Report: re-submit deliverable - Final presentation of results: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

CS4715	Programming Languages Research Seminar	5

Responsible Instructor	Dr. B.P. Ahrens
Responsible Instructor	Dr. J.G.H. Cockx
Instructor	Dr. S.S. Chakraborty
Instructor	C.B. Poulsen
Contact Hours / Week x/x/x/x	0/0/0/2
Education Period	4
Start Education	4
Exam Period	4 5
Course Language	English
Expected prior knowledge	Required: Theme Programming Languages
Course Contents	Software has revolutionized our society: most industries and aspects of our daily lives make essential use of software. It is therefore vital that we are able to develop software that is bug-free and adheres to its specification with high confidence; and that we can modify software to match changed needs by improving its efficiency, adding new features, or modifying its behavior, all while preserving its correctness. Programming languages and their associated tools are an important means to this end. In this course, you will read scientific journal and conference articles from the field of programming languages, and learn about state of the art tools and principles for language engineering. You will also do a course capstone project related to a research problem, applying methods, knowledge, and techniques from the programming language literature. Possible projects could for example be implementing a small programming language using a language workbench, creating an embedded domain-specific language with a unique feature, extending an existing compiler with a new code transformation or optimization, or using a proof assistant to formalize and prove things about a language or code transformation.
Study Goals	At the end of this course, the student is able to: 1. Analyze and discuss research papers from the broad area of programming language research. 2. Explain topics related to the development, application, and verification of state-of-the-art programming language methods and tools, such as type theory, domain-specific language design, static analysis, and compiler optimizations. 3. Apply research methodologies in the field of programming languages. 4. Apply programming language (meta-)tools, methods, and/or formalisms to replicate state of the art research results. 5. Identify research questions and objectives and relate them to the relevant programming language literature.
Education Method	Each week all students will read a scientific article in the field of programming languages, and discuss this article with other students in the discussion seminars. Each student is expected to lead one of these discussions, alone or in a group (depending on the total number of students). In parallel with these seminars, each student will work independently on a project that involves using and/or implementing tools based on state-of-the-art research in the field of programming languages. At the end of the course, each student will present their project to the other students.
Literature and Study Materials	Scientific articles in the field of programming languages and documentation of the design and implementation of languages and tools (both will be provided to students during the course).
Assessment	The final grade of the course consists of the following components: - Individual project (weighting 70%) - Oral presentation of individual project (weighting 30%) Final grade calculation = 0.7 * Project + 0.3 * Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. The grade for repair submissions is in both cases capped to 6.0. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

CS4720	Research in Program Analysis	5
Responsible Instructor	Dr. B. Özkan	
Instructor	Prof.dr. A. van Deursen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course provides a comprehensive introduction to the field of software analysis, a critical aspect of software development and maintenance. We will delve into both theoretical foundations and practical applications, learning how to analyze, understand, and improve software systems. This course explores a comprehensive set of techniques for software analysis, including practical formal methods, model checking, static and dynamic analysis, and other key approaches to evaluating and improving software systems. Through a hands-on project, you will gain experience in understanding and applying these techniques and presenting them using a scientific valid methodology.	
Study Goals	At the end of this course, you will be able to: 1. Explain different software analysis approaches, such as model checking, static, and dynamic analyses. 2. Apply the software analysis techniques to sample programs. 3. Analyze the results of software analysis techniques in terms of their precision and computational performance. 4. Evaluate the performance of different software analysis techniques and compare their pros and cons.	
Education Method	Lectures: These provide foundational knowledge in program analysis, introducing theoretical concepts and practical tools. Paper Readings and Presentations: Students read and present academic papers, enhancing their understanding of specific techniques and applications. This is coupled with peer feedback, improving research, synthesis, and presentation skills. Course Project: A hands-on project allows students to apply learned techniques to a real-world software system, fostering practical experience and professional skills like problem-solving and teamwork.	
Literature and Study Materials	More information is available at: http://cs4720.github.io/	
Assessment	The final grade of the course consists of the following components: - Group Report (weighting 80%) - Group Presentation (weighting 20%) Final grade calculation = 0.8 * Group Report + 0.2 * Group Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Report: Resit opportunity - Group Presentation: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CS4725	Research Seminar on Scalable Learning Systems	5
Responsible Instructor	Dr. Y. Chen	
Responsible Instructor	K. Atasu	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: Theme Distributed Systems Engineering	
Course Contents	This seminar course aims to teach the students how to design and build parallel and distributed machine learning (ML) and deep learning (DL) solutions. The learning activities include paper reading, presentation, discussion, and project prototyping. We will provide a broad overview of the state-of-the-art parallel and distributed ML and DL algorithms and systems, with a strong focus on the scalability, resource efficiency, data requirements, and robustness of the solutions. We will cover ways of mapping state-of-the-art ML and DL solutions to massively-parallel AI accelerators such as GPUs. We will present an array of techniques for efficiently scaling ML and DL workloads to a large number of distributed nodes in the presence of system failures and malicious attacks. Lastly, we will cover methods for improving the scalability and the efficiency of deep generative learning approaches. Course topics include - Overview of parallel and distributed ML/DL algorithms - Performance and scalability of state-of-the-art systems - Hardware-accelerated ML/DL solutions - Federated machine learning systems - Deep generative learning systems	
Study Goals	At the end of this course, the student is able to: 1. Demonstrate deep understanding of parallel and distributed machine learning and deep learning algorithms and systems by analyzing and discussing research papers 2. Design and implement scalable and efficient machine and deep learning algorithms and systems, evaluate time-space and cost-performance tradeoffs 3. Analyze and evaluate the resiliency of federated machine learning algorithms and systems against various attacks 4. Analyze and evaluate the accuracy and the scalability of generative learning systems and their applications 5. Formulate research questions and objectives and relate them to the relevant literature in the broad field of deep generative learning systems	
Education Method	The course materials will be based on a mixture of classic and recently published papers as well as reference books. For each topic, background concepts and the technology landscape will be briefly introduced and then state-of-the art of papers will be presented and discussed by students. In parallel, the students will be working on a research project to improve the speed, scalability, resilience, or accuracy of an existing ML/DL approach.	
Literature and Study Materials	Lecture slides, scientific articles, and reference books on deep learning	
Assessment	The final grade of the course consists of the following components: - Group Presentation: project (weighting 20%) - Group Report: project (weighting 40%) - Individual Report (weighting 30%) - Group Presentation: paper (weighting 10%) Final grade calculation = 0.2 * Group Presentation + 0.4 * Group Report + 0.3 * Individual Report + 0.1 * Group Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation project: Repair - Group Report: Repair opportunity - Individual Report: Repair opportunity - Group Presentation paper: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4200	Empirical Research of Computational Solutions	5
Responsible Instructor	Dr.ir. W.P. Brinkman	
Responsible Instructor	J. Urbano Merino	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Recommended: Basic programming skills (e.g., CSE1100), Probability Theory and statistics (e.g., CSE1210)	
Course Contents	In today's world, where an increasing number of computational solutions like recommender systems, social robots, and health monitoring and wearable devices are being introduced, empirical research plays a crucial role in driving innovation, uncovering insights, and making informed decisions. This course equips students with fundamental skills in conducting empirical research to evaluate these computational solutions and their underlying theories. The course begins by providing students with a solid understanding of empirical research methods. They learn to set up a research study, collect and analyse data, and draw scientifically valid conclusions. Emphasis is placed on reproducible research practices, including techniques like pre-registration and creating a data plan. To make sense of data samples, students study how to make statistical inferences about the population. They explore frequentist and Bayesian data analysis approaches, gaining the ability to make meaningful statistical inferences based on collected data. Throughout the course, students work with computational tools that aid in conducting statistical analyses of real-world data. Using these tools, students gain practical experience analysing and interpreting data to draw meaningful conclusions. By the end of this course, students will have a strong foundation in empirical research methods and the ability to evaluate computational solutions. They will be equipped with practical skills in data analysis, statistical inference, and working with real-world datasets. The course aims to empower students to become proficient researchers who can contribute to advancements in computational solutions and their underlying theories. The course's main topics: - Conceptualizing research questions and experimental design, and data planning - Frequentist and Bayesian data analysis - Generalized linear models for statistical analysis - Multilevel modelling for hierarchical and longitudinal data analysis - Measuring and sampling, validity and reliability - Principles of statistical testing	
Study Goals	At the end of this course, the student is able to: 1. Compose a scientifically sound research plan for an empirical study of a computational solution 2. Make statistical inferences with computational tools based on the Frequentist approach to data analysis 3. Make statistical inferences with computational tools based on the Bayesian approach to data analysis 4. Formulate conclusions that can be drawn from statistical inferences about computational solutions and their underlying theories 5. Address typical biases and misunderstandings in empirical research	
Education Method	In the lectures, theories, principles and methods are presented, discussed and illustrated with examples. Outside the lectures, students work in small groups on their assignments using R.	
Literature and Study Materials	- McElreath, R. (2020). Statistical rethinking: A Bayesian course with examples in R and Stan. CRC press. - Robson. C., McCartan K. (2016) Real World Research. John Wiley & Sons - Field, Z., Miles, J., & Field, A. (2012). Discovering statistics using R. Discovering Statistics Using R, 1-992.	
Assessment	The final grade of the course consists of the following components: - Group Presentation: presentation of pre-registration of empirical study (including a data plan) and the results of data analysis assignment based on the Bayesian approach (weighting 35%) - Individual Presentation: individual Q&A session about the pre-registration of empirical study (including a data plan) and the results of data analysis assignment based on the Bayesian approach (weighting 15%) - Group Report: report on the results of data analysis assignment based on a frequentist approach (weighting 50%) Final grade calculation = 0.35 * Group Presentation + 0.15 * Individual Presentation + 0.5 * Group Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation: Resit opportunity - Individual Presentation: Resit opportunity - Group Report: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 9, sub 5).	

DSAIT4210	Research in Web Data and Information Management	5
Responsible Instructor	Prof.dr.ir. G.J.P.M. Houben	
Responsible Instructor	Dr. M.S. Pera	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Required: BSc Computer Science (or equivalent)	
Course Contents	In this course we discuss recent developments in the area of web data and information management. We select topics to discuss from the areas of: - web technology (e.g., web engineering, hypertext, adaptive web), - web data management (e.g., web data interoperability, system and data integration), - web data and semantics (e.g., ontologies, semantic web, metadata), - web data analytics (e.g., user modeling, web personalization, web information filtering and retrieval), - social web (e.g., social web data analytics, social networking, human computing), - web science (e.g., crowdsourcing, trust, data science). We discuss this content while learning about the role of scientific communication and about the scientific methodologies and approaches for conducting research in the area. In this research course the students will have to prepare and give scientific presentations on the basis of research papers about selected topics - the topics are selected in the first session together with the students. The students will also have to attend the presentations and participate in discussions on the presented papers. In addition, students will have to write a short survey about a topic in the area of web data and information management of their choice.	
Study Goals	At the end of this course, the student is able to: 1. Explain the current developments in research on web information systems, the methodologies and approaches to conduct research in the area; 2. Analyse literature relevant to a given problem 3. Critically assess scientific literature, including the importance of the academic journals and conferences in the area and their review process. 4. Create a research plan 5. Write the draft of a research paper guided by the research plan	
Education Method	Seminar, with discussions, presentations, feedback sessions, and 1on1 Mentoring	
Literature and Study Materials	Is provided in the research course depending on the chosen subjects.	
Assessment	The final grade of the course consists of the following components: - Individual Presentation: presentation discussion state of the art literature (weighting 15%) - Individual Report: report outlining a research plan and research proposal (weighting 75%) - Group Presentation: group discussion in class (weighting 10%) Final grade calculation = 0.15 * Individual Presentation + 0.75 * Individual Report + 0.1 * Group Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Presentation: Repair opportunity - Individual Report: Repair opportunity - Group Presentation: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4220	Research in Intelligent Decision Making	5
Responsible Instructor	S. Dumani	
Responsible Instructor	Dr. J.W. Böhmer	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	ONLY OPEN FOR STUDENTS FROM M-CS AND M-DSAIT THAT FOLLOW THEMES: - M-CS Theme Algorithmics - M-DSAIT Theme Optimisation & Reasoning - M-DSAIT Theme Probabilistic Decision Making	
Course Contents	Intelligent decision making is a key skill of computational agents. Research on this topic focuses on building models and algorithms that enable AI systems to take appropriate decisions. Building upon theoretical knowledge gained in the Probabilistic Decision Making or Reasoning and Search themes, students collaborate in small groups on a distinct research project per group, for instance on decision-making problems in transport, logistics or smart energy grids. Purely algorithmic challenges will also be provided. The research projects provide a good opportunity to learn about topics suitable for Masters projects.	
Study Goals	At the end of this course, the student is able to: 1. Apply algorithms for decision making to problem domain for comparison and evaluation. 2. Create an extension of a decision-making algorithm. 3. Critically assess relevant topics in the research field of algorithms for intelligent decision making. 4. Apply the appropriate research methodology. 5. Communicate their findings effectively.	
Education Method	A research project in a small group	
Literature and Study Materials	Research papers	
Assessment	The final grade of the course consists of the following components: - Group Assignment: quality of work of the research project (weighting 40%) - Group Report: a scientific report of the research project (weighting 20%) - Individual Assignment: performance during the project (weighting 30%) - Group Presentation: presentation of the research project (weighting 10%) Final grade calculation = 0.4 * Group Assignment + 0.2 * Group Report + 0.3 * Individual Assignment + 0.1 * Group Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Assignment: Repair opportunity - Group Report: Repair opportunity - Individual Assignment: No - Group Presentation: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Prof.dr. M.T.J. Spaan	

DSAIT4230	Research in Social Signal Processing and Affective Computing	5
Responsible Instructor	Dr. H.S. Hung	
Responsible Instructor	Dr. C.R.M.M. Oertel Genannt Bierbach	
Instructor	Dr. B.J.W. Dudzik	
Instructor	Dr. C.A. Raman	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The core of social and emotional intelligence is our ability to understand and interpret social and affective signals of a person we are communicating with is. Social-emotional intelligence is a facet of human intelligence that has been argued to be indispensable and perhaps the most important for success in life. Social Signal Processing and Affective Computing (SSPAC), aims to develop automated approaches that can interpret human social and/or affective behaviour through machine analysis and production of nonverbal and verbal behavior. In this course you will learn how next-generation computing can make use of such social and affective signals by giving it the ability to recognize and produce human social signals and social behaviors. Think about turn taking, politeness, disagreement, emotions, rapport. You will learn about relevant findings in social psychology, and you will learn computational techniques that allow systems to make use of social and affective signals to become more effective and more efficient by being able to detect but also simulate (e.g. in virtual agents) blinks, smiles, crossed arms, laughter. Socially aware computing. These techniques can be used in robots, virtual agents, smart homes, crowd monitoring, etc.	
Study Goals	At the end of this course, the student is able to: 1. Describe what social and affective signals are. 2. Apply computational methods to detect and simulate social and/or affective signals. 3. Position the field of social signal processing and affective computing in computer science and psychology, and identify its major goals and angles of study. 4. Critique, major psychological theories of social interaction. 5. Explain and critique major social and affective signal recognition, simulation and expression techniques in computational systems. 6. Develop and motivate a research project that uses social and/or affective signals in a non-trivial manner with hypotheses, research questions, and a supporting literature survey. 7. Implement and evaluate (in groups) a research project that uses social and/or affective signals in a non-trivial manner with hypotheses, research questions, and a supporting literature survey.	
Education Method	Seminar: - Crash Course in Social Signal Processing and Affective Computing - Self-study of papers and book. - The papers will be made available at the start of each lecture. Project: Perform a piece of research (survey, research question, programming, testing) and write a paper about it. Students will work in teams of about 3-4 people	
Assessment	The final grade of the course consists of the following components: - Group Presentation: midterm presentation (weighting 10%) - Group Report 1: proposal project (weighting 30%) - Group Report 2: final project report (weighting 40%) - Individual Presentation: presentation final project (weighting 20%) Final grade calculation = 0.1 * Group Presentation + 0.3 Group Report 1 + 0.4 * Group Report 2 + 0.2 * Individual Presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Group Presentation: Resit opportunity - Group Report 1: Repair opportunity - Group Report 2: Repair opportunity - Individual Presentation: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4235	Research Seminar on Multimedia Computing and Systems	5
Responsible Instructor	P.S. Cesar Garcia	
Responsible Instructor	M. Mansoury	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Through all the exciting recent advances in digital media technology and the rapid growth of social media platforms, multimedia content is increasingly embedded in our daily lives, gaining enormous potential in improving the traditional educational, professional, business, communication and entertainment processes. To be able to use this potential for transferring these processes into user-centric interactive multimedia applications, technology is required that can help us access, deliver, enrich and share rich-media content. This course provides insight into the state-of-the-art cross-disciplinary research efforts related to the development of such technology. The topics covered by the course include, but are not limited to, multimedia engagement (emotional and social signals and recommender systems), multimedia technology (language and speech technology, telepresence and VR, mobile), and multimedia experiences (Quality of Experience, collaboration), and Multimedia in the Generative AI Era.	
Study Goals	At the end of the course, the stand is able to: 1. Understand the state-of-the-art research and development activities in the field of Multimedia Computing and systems 2. identify the "knowledge gap" (i.e., the place in which more research is needed in order to advance the state of the art). 3. write a survey paper on one specific topic 4. Present this survey paper to an audience	
Education Method	Readings, seminar discussions, presentations, survey paper	
Literature and Study Materials	Papers and articles	
Assessment	The final grade of the course consists of the following components: - Individual Report: write own paper/article (weighting 65%) - Individual Presentation 1: presentation pre-existing paper (weighting 10%) - Individual Presentation 2: present own survey (weighting 25%) Final grade calculation = 0.65 * Individual Report + 0.1 * Individual Presentation 1 + 0.25 * Individual Presentation 2 A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Individual Report: Repair opportunity - Individual Presentation 1: Resit opportunity - Individual Presentation 2: Resit opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29 sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Deepening/broadening Electives 2024	
Introduction 1	In Q5, the start of the second year, you have the opportunity to personalise your education. Within the elective space, you can deepen your Computer Science knowledge by following Computer Science electives, broaden your knowledge by following education offered by other engineering masters programmes or by doing interdisciplinary project courses.

Interdisciplinary projects

IFEEMCS520201	Advanced Interdisciplinary Artificial Intelligence Project	15
Responsible Instructor	J. Zatarain Salazar	
Responsible Instructor	Dr. P.K. Murukannaiah	
Contact Hours / Week x/x/x/x	10/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	

Expected prior knowledge

FROM SEPTEMBER 2025, THIS COURSE WILL BE PART OF IFM4040 JOINT INTERDISCIPLINARY PROJECT (JIP) AND WILL NO LONGER BE OFFERED UNDER THIS COURSE CODE. ENROLLMENT FOR THE COURSE STARTS MARCH 17TH 2025 VIA BRIGHTSPACE (IFM4040)

This course is aimed at MSc students from all TU Delft faculties who have already obtained AI-related knowledge and skills in previous courses, and who want to apply their skills to a real-world AI challenge. Prospective students, for example, will have completed an AI minor or have a good amount of AI knowledge from their bachelor (e.g. Mechanical Engineering), on topics such as data handling and machine learning OR possess demonstrated expertise in the domain relevant to a specific AI project on offer. This is an indicative guideline and we ask students that want to enroll for this course, to send a motivation letter and give an indication of their experience and knowledge on AI.

Course Contents

FROM SEPTEMBER 2025, THIS COURSE WILL BE PART OF IFM4040 JOINT INTERDISCIPLINARY PROJECT (JIP) AND WILL NO LONGER BE OFFERED UNDER THIS COURSE CODE. ENROLLMENT FOR THE COURSE STARTS MARCH 17TH 2025 VIA BRIGHTSPACE (IFM4040)

Artificial Intelligence (AI) is an interdisciplinary challenge. This Advanced Interdisciplinary Artificial Intelligence Project (AI2P) (IFEEMCS520201) is a 15EC MSc elective course aims to provide students with a unique experience to learn, develop and apply AI to a real-world challenge, working in an interdisciplinary project team. In this course you work with a group (4-6 students) on a large interdisciplinary project that is run by TU Delft AI Labs.
[link: https://www.tudelft.nl/ai/tu-delft-ai-labs]

Further, some of the projects may offer business cases from the industry. You and your fellow students will be offered to join workshops, tutorials and seminars organized by cutting edge researchers from TU Delft. You will also share kick-offs, reviews, and events with students from JIP (Joint Interdisciplinary Project). You find the projects for next academic here on Project Forum
[link: https://projectforum.tudelft.nl/course_editions/109]

Study Goals

Upon finishing this course, the student should be able to:

1. position an AI challenge in the broader literature, e.g., by performing a literature review on the AI challenge in a domain;

2. set-up a research strategy for an interdisciplinary AI project;

3. develop the AI solution according to the research strategy;

4. effectively collaborate with the team and supervisor;

5. evaluate and reflect on the research / developed AI solution;

6. assess the impact of deploying AI-based solutions and interventions on individuals, organizations, and society; and

7. report and present their research on a scientific level.

Education Method

The students may work and discuss in groups. Each student is responsible for gathering, sharing and implementing information within the overall project. This allows for collaboration and communication as well as self-directed learning.

At least once a week, the students meet with their supervisor. Peer feedback as well as feedback from the supervisor is organized on essential steps in the research project, which include:

1. Project proposal
2. Mid-term review, and
3. A scientific report, describing the problem, methods, results, and conclusion
4. A scientific presentation

Assessment

The scientific report and the project presentation are assessed by the project supervisor and an external examiner.

The assessment based on the clarity of presentation, the soundness of the methods used, and the conclusions drawn.

Resit next quarter: A repair option is allowed only for students that received a fail mark (<5.8). Overall, the resit mark will be capped to 5.8.

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman,VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (Ilo;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

	The minimum grade for the summative assessment is =/> 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

TPM423A	Technology Venture Development	15
Module Manager	Dr. A. Giga	
Instructor	J. Schmutzler	
Instructor	Dr. A. Giga	
Responsible for assignments	Dr. A. Giga	
Co-responsible for assignments	Dr. L. Hartmann	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Summary	Join us in an interdisciplinary exploration of technology startups business model and strategy. This pioneering course offers you an opportunity to work with a technology-based startup from the local ecosystem or a corporate venture in an internship-like setting and help them progress. In a group of three to four students from different faculties and study programs, you will work on two projects: 1) Validate and challenge startups business model and strategy 2) Provide consulting on a specific business challenge provided by the startup. In addition, you will be coached and taught solid business theoretical foundation and practical lessons. You will apply your learnings directly, solving pressing business challenges and driving innovation. This hands-on experience allows to you communicate directly with startups key decision-makers and see the effects of your work making a difference. This course is a real-world expedition into business model innovation, designed to inspire, challenge, and prepare you for a future where you drive change. We expect you to spend around three days a week with the startup and about one day a week in the classroom setting. Given its 15 ECTS, we expect your full weekly commitment, as in any joint interdisciplinary project. The applications opens on April 30th and the application deadline will be on June 15th. Here is the link to the application form: https://forms.office.com/e/tX9hXqNM7u	
Course Contents	What to Expect Internship-like Experience: Step into the fast-paced world of startup or corporate venture, through our internship-like experience. Here, you'll gain invaluable hands-on exposure to real-world business practices, navigating challenges, and seizing opportunities alongside industry professionals. Lectures: Immerse yourself in captivating lectures that serve as the cornerstone of your theoretical understanding. Delivered by experts in the field, these sessions will unravel the intricacies of business model innovation, offering profound insights and cutting-edge perspectives. Each week delves into critical topics such as market dynamics, design thinking, team development, financial strategies, risk management, and growth mechanisms, ensuring you're equipped with a comprehensive toolkit for the entrepreneurial world. Sessions illuminate the intersection between technical innovation and business strategy, enriching your learning experience and equipping you with versatile skills for multifaceted business environments. Coaching Sessions: These sessions help you apply your newfound knowledge and skills to your cases and tackle pressing business challenges head-on, devising innovative solutions that drive tangible impact and transformation. What You'll Experience and Learn Interdisciplinary Exploration: Navigate the complex nexus of diverse disciplinary backgrounds in engineering, basic science and architecture and design, entrepreneurship, and management, transcending traditional disciplinary boundaries to craft innovative business models that redefine the status quo. Practical Application: Move beyond theoretical concepts and immerse yourself in practical, hands-on experiences. From market analysis to financial modelling, you'll apply cutting-edge techniques and frameworks to real-world scenarios, honing your problem-solving skills and business acumen. Innovation Mindset: Cultivate an entrepreneurial mindset rooted in creativity, resilience, and adaptability. Through design thinking workshops, risk management exercises, and growth strategy simulations, you'll embrace uncertainty as an opportunity for growth and innovation. Networking Opportunities: Forge meaningful connections with industry professionals, guest lecturers, and fellow classmates, expanding your professional network and gaining valuable insights into the startup ecosystem. Personal and Professional Growth: Reflect on your journey, identify your strengths, and chart a course for future success. Whether it's sharpening your communication skills or refining your leadership abilities, this course offers a transformative experience that extends far beyond the classroom. Why Choose This Elective Cutting-Edge Curriculum: Stay ahead of the curve with a curriculum that reflects the latest trends, technologies, and best practices in business model innovation. Hands-On Experience: Gain practical, real-world experience that sets you apart in today's competitive job market. Our internship-like experience and consulting project offer unparalleled opportunities for skill development and professional growth. Inspiring Faculty: Learn from esteemed faculty members who are experts in their fields and passionate about empowering the next generation of innovators and entrepreneurs. Impactful Learning: Make a meaningful impact with a course that challenges you to think critically, solve complex problems, and drive positive change in the world.	
Study Goals	After completing this course you will have learned to: Identify and innovatively transform a business idea for a technology venture. Conduct rigorous market analysis and validation using primary and secondary research methods. Construct a detailed and financially viable business proposition. Apply niche strategies to commercialize radical high-tech products. Come up with innovative, viable, high quality recommendations that will solve real-life challenges for an existing venture, under	

	considerable time constraints. Employ relevant academic theories to support your recommendations; Effectively communicate your product and business proposition to diverse stakeholders. Emulate effective entrepreneurial behaviour in decision-making under uncertainty. Reflect on the evolution of entrepreneurial growth and business model adaptation.
Education Method	This course will entail a mix of lectures, and guest lectures. You will be provided with course materials. You will go through an onboarding process emulating a traineeship experience with the startup or corporate venture. There will be coaching sessions with experienced mentors while will offer guidelines, feedback, and tools to help you develop a solution to the consulting project.
Prerequisites	There are no prerequisites for this course.
Assessment	Your grade is determined as follows: 80% is group work and based on the quality of your group project. Quality is assessed based on your problem statement & presentation, midterm report & presentation and final report and presentation. Startup team will have a strong say on your performance. Adjustment is made through peer evaluation of your individual contribution to the team work. 20% is an individual grade based your personal learning and team reflections during the course.
Enrolment / Application	This course is open to master students across all TUD faculties starting their second year. MAXIMUM CAPACITY is contingent upon the capacity of our partners (startups and corporates) but initially limited to 30 participants. Please apply for the course no later than June 15th. If you would like to apply , fill out the application form: https://forms.office.com/e/tX9hXqNM7u If you are having trouble accessing the form, need help approving this course as your master programs elective, or have any other questions, please write to DCE@tudelft.nl If more than 30 participants apply for the course, we will select based on your application. If you apply after June 15th, you may still join if there is space available.
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.
Elective	Yes
Tags	Business Economics Finance
Targetgroup	MSc students from all faculties can apply.

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Deepening electives

Introduction 1	You can deepen your mastery of knowledge on Computer Science-related topics. The courses listed are interesting for all Computer Science students, though some courses are recommended for students who have followed specific themes.

CS4410	Category Theory for Programmers	5
Responsible Instructor	Dr. B.P. Ahrens	
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Categorical structures occur in programming languages on different levels: (1) within programming languages, providing design principles and guidance on how to write modular and correct-by-design programmes (as demonstrated in the practical programming language Haskell) and (2) in the design and study of programming languages, as a guiding meta-theory. In particular, category theory provides a mathematical justification for recursion schemes for inductive datatypes. This course aims to provide solid foundations on both (1) and (2).	
Study Goals	At the end of the course, the student is able to: 1. Use categorical constructions in the design and structuring of computer programmes 2. Specify datatypes as initial algebras 3. Prove properties of computer programmes, guided by categorical intuition 4. Use categorical fusion laws to optimize functions	
Education Method	Lectures and problem sessions	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable	

DSAIT4300	Formal Methods for Machine Learning	5
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	At the end of this course, the student is able to: 1. modeling complex systems, 2. formally defining critical properties, 3. applying verification tools to learned systems, 4. synthesizing controllers for learned systems, 5. providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: - Participation (weighting 15%) - Presenting Papers (weighting 15%) - Assignment #1 (weighting 15%) - Assignment #2 (weighting 15%) - Final Report (weighting 40%) Final grade calculation = 0.15 * Participation + 0.15 * Presenting Papers + 0.15 * Assignment #1 + 0.15 * Assignment #2 + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Presenting Papers: resit - Assignment #1: re-submit deliverable - Assignment #1: re-submit deliverable - Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4305	Machine Learning for Graph Data	5

Education Period 1

Start Education 1

Exam Period 1

Course Language English

Course Contents

PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026

Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the sate of a power or water network, and coordinate group of autonomous moving sensor, to name a few.

This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability and privacy of graph neural networks.

Study Goals

At the end of this course, the student is able to:
1. Recognise key challenges in learning from graph-structured data.
2. Compare representation learning techniques on graphs.
3. Analyse different learning techniques for graph data.
4. Design machine learning solutions for real world graph data tasks

Education Method

The course consists of:
- 12 lectures in which the main concepts are explained and discussed;
- 3 hands-on homework labs, in which practical assignments need to be solved individually;
- At least two Q&A and feedback sessions about the final project

Assessment

The final grade of the course consists of the following components:
- Oral Exam: Group presentation (weighting 40%)
- Group Report: Project report (weighting 60%)

Final grade calculation = 0.4 * Oral Exam + 0.6 * Group Report
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:
- Oral Exam: Resit opportunity
- Group Report: Repair opportunity

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

DSAIT4310	Modeling and Data Analysis and Complex Networks	5

Education Period 1

Start Education 1

Exam Period 1
2

Course Language English

Course Contents

PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026

Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other.

This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.

Study Goals

At the end of this course, students are able to:
1. Construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies.
2. Model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share.
3. Obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.

Education Method

In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).

Assessment

The final grade of the course consists of the following components:
- Assignments (weighting 20%)
- Final project (weighting 80%)

Final grade calculation = 0.2 * Assignments + 0.8 * Final project
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:
- Assignments: re-submit deliverable
- Final project: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

DSAIT4315	Performance Analysis	5
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer, electric and transportation networks as well as biological, brain and social networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). In addition, queuing theory is key to assess many kinds of scheduling problems. Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included.	
Study Goals	At the end of this course, the student is able to: 1. Understand and apply mathematical techniques, in particular probabilistic methods and graph theory. 2. To compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = 1.0 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4320	Building Serious Games	5
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026 Project-based interdisciplinary course, open to MSc students of all faculties. The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, training, health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose. Assignments for this course will be provided by real-world end-users (e.g. companies or the Science Centre Delft), to whom the group will be reporting throughout the term of the project. Project-based interdisciplinary course, open to MSc students of all faculties (of LDE universities). The main goal of the project is to take students with varying talents, backgrounds, and perspectives and put them together to do what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: 1. Identifying and valuing the soft skills necessary to work effectively in interdisciplinary teams 2. Responsibly interacting in a team, integrating its members' talents and expertise in a variety of team roles 3. Adapting with flexibility to the dynamic requirements of a complex external assignment 4. Translating feedback received into proactive personal development steps 5. Formulating research questions, identifying research contributions, and describe them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: 6. Identifying, selecting and deploying the most adequate game technologies for the given serious game domain and constraints 7. Deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small game studio, according to a tight planning. Also a few plenary sessions and/or lectures.	
Assessment	The final grade of the course consists of the following components: - Product grade (weighting 50%). Unique for the whole group, based on both the game itself and the required documentation. - Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. - Final presentation (weighting 5%) The commissioner will be involved both as advisor and as assessor. The final documentation includes writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation = 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Final presentation: resit opportunity There is no repair possibility for the other components (product and process), since their assessment necessarily requires an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

DSAIT4325	Socio-Cognitive Engineering	5

Education Period
1

Start Education
1

Exam Period
1
2

Course Language
English

Expected prior knowledge
Basic prior knowledge on human-computer interaction is helpful, but not required.

Course Contents
PLEASE NOTE: THIS COURSE WILL BE OFFERED FROM ACADEMIC YEAR 2025-2026

Whether you are playing a game in virtual reality, driving a semi-autonomous car, educating yourself with a mobile app, or harmonizing your health and lifestyle with a robot assistant; nowadays intelligent networked information and communication technology is omnipresent. This course focuses on the design of human-aware intelligence into such environments, to support joint human-technology performances that bring about positive human experiences. The focus will be on social robots in general, with practical design and test assignments on robots that support people with dementia, http://rejam.tudelft.nl).

In the Socio-Cognitive Engineering (SCE) course (MSc level), you will become acquainted with the application of a coherent set of methods for the design and evaluation of human--agent collaboration. Based on the SCE-method, we will elaborate on the state of the art of intelligent user interfaces (ePartners), such as virtual assistants, robotic teammates, eCoaches, and companion agents.
The main topics of study are:
- Design methods: Cognitive Engineering, Value Sensitive Design, Scenario-based Design, Claims Analysis, Design Rationale, Design Patterns.
- Design for collective intelligence: Knowledge Representation, Ontology Engineering, Mental Models, Theory of Mind, ePartners, Adaptive Automation, Social Robots.
- Design Evaluation: Prototyping, Test Methods, Measures, Questionnaires, Ethics.
- Human Factors Theories and Models: Human Cognition & Learning, Memory, Emotion, Task Load, Human-Agent Teamwork, Behavior Change and Persuasive Technology.

Study Goals
At the end of this course, students will be able to:
1. Explain the essential concepts of the design methods addressed in the course.
2. Explain the (dis)advantages of various design methods and their complementarity.
3. Apply the design methods addressed in the course in their research and design projects.
4. Explain what a design rationale is.
5. Construct a design rationale.
6. Create design specifications that are grounded in a design rationale.
7. Evaluate the strengths and weaknesses of a design rationale, e.g. using human-centered evaluations that test the design rationale.
8. Explain some of the state of the art human factors theories, models, and methods relevant to intelligent user interfaces, human-robot collaboration, and ePartner technology.
9. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research.
10. Present work on a design project to an academic audience.
11. Work in a group on collaborative assignments.

Education Method
LECTURES
During the lectures, the teachers will present a range of theories, models, and methods relevant to socio-cognitive engineering. Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see ‘Project’).

PROJECT
In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. This year (like the past years), the design problem is a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve human’s physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/).

Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype.

Literature and Study Materials
Papers from scientific journals on Brightspace.
Lecture notes on Brightspace.

Assessment
The final grade of the course consists of the following components:
- Presentations (weighting 15%)
- Prototype (weighting 15%)
- Project report according to the prescribed format (weighting 60%)
- Individual reflection (weighting 10%)

Final grade calculation = 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection
A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8.

In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for:
- Presentations: resit opportunity
- Prototype: re-submit deliverable
- Project report according to the prescribed format: re-submit deliverable
- Individual reflection: re-submit deliverable

Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).

TPM027A	Cyber Risk Management	5
Module Manager	Prof.dr.ir. P.H.A.J.M. van Gelder	
Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Co-responsible for assignments	Dr. A.P. Afghari	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	MOTIVATION: The challenge of selecting the optimal technical, organisational, legal, and other preventive and repressive measures to reduce cyber risks to acceptable levels can only be understood in the context of the application of Cyber Risk Management. Risk Management is about analysing the relationships between threats, incidents and risks (here in the complex world of cyberspace), based on which an adequate set of countermeasures can be designed. SYNOPSIS: Risk (= the potential to losing something of value) can manifest itself in cyberspace in all kinds of ways: values at stake are financial wealth, health, physical condition (of people, materials, goods, infrastructures, etc.), well-being, reputation, privacy, trust, etc. Based on a conceptualisation of cyberspace and its various sub-domains (discussed in the project week of year 1), we introduce risk assessment approaches, both of qualitative and quantitative manner, illustrated with case studies. CONTENT: Cyberspace Risk concepts Probabilistic and adversarial risk System and attacker models Requirements and policies Risk assessment methods Qualitative and quantitative methods Measuring security and effectiveness	
Study Goals	After this course, students are able to: Understand and apply risk concepts to (possible) incidents in cyberspace Understand the theoretical principles of cyber risk management Explain the role of adversarial risk models, system models, and attacker profiles in cyber risk assessment Analyze and apply state-of-the-art cyber risk assessment methodologies to (complex, multi-step) cyber incidents Analyze the (expected) effect of measures that help to prevent the occurrence of cyber incidents Justify investments in cyber security	
Education Method	LECTURES: There are two lecture slots per week EXAM: There is a written exam at the end of the course. LANGUAGE: The course is taught in English. LECTURERS: Prof. dr. ir. Pieter v. Gelder (TUD/TPM) and guest lecturers.	
Assessment	#### Assessment Grading will be based on the individual written exam at the end of the course. Final mark of this course is determined by the grade of the individual exam (100%). The minimum grade for the individual exam to pass this course is 6.0. The grade for the individual exam is 2 years valid. The individual exam is summative. Detailed feedback on the individual exam can be reviewed by students by contacting the teaching staff of this course. #### Resit The students are allowed one re-sit per examination. It is not allowed to re-sit an examination or assignment for which they have received a pass (6,0 or higher). It is allowed to re-sit an examination which they haven't done during the first occasion. The re-sit format needs to be discussed with the teacher of the course in line with examination regulations. In case the student is granted an extra re-sit by the Board of Examiners, this re-sit has to take place within the study year 2023-2024.	
Elective	Yes	
Targetgroup	MSc	
Category	MSc level	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Broadening electives	
Introduction 1	It is possible to choose electives from other engineering study programmes during your Q5 elective space. By choosing this you are relating your Computer Science knowledge to other disciplines or entrepreneurial ambitions you might have.

TPM401A	Technology Entrepreneurship and Innovation	5
Module Manager	H. Khodaei	
Instructor	P.B.M. Vandekerckhove	
Instructor	Dr.ing. V.E. Scholten	
Responsible for assignments	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	2/0/2/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	This course offers theoretical fundamentals of entrepreneurship and technology innovation. You create a case study of an existing start-up analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an high-tech start-up to academic research on the topics of technology entrepreneurship and innovation.	
Course Contents	Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, dynamics and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Describe and apply the high-tech start-up growth process. LO 4 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 5 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 6 Analyse an existing high-tech start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Enrolment / Application	Enrollment via Brightspace	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft is needed.	
Category	MSc level	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	J. Schmutzler	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10.	
Elective	Yes	

TPM409A	Digital technology entrepreneurship and management	5
Module Manager	J. Gartner	
Instructor	J. Gartner	
Co-responsible for assignments	H. Khodaei	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Parts	The expected workload is 5 ECTS.	
Summary	This course offers the fundaments of digital entrepreneurship and the management of emerging (digital) technologies. In this course, participants choose an upcoming digital technology of interest and assess it with different methods from multiple angles to evaluate its ability to create a competitive advantage or disrupt an industry or market. This course helps participants prepare for various career paths, from launching their product/service as an entrepreneur to making informed technology adoption and rejection decisions as an intrapreneur.	
Course Contents	Throughout the course, participants will learn to distinguish between emerging and disruptive technologies, understand influential dynamics such as innovation diffusion, hypes and emotions, and develop the methodical competencies to assess and evaluate an emerging technology in an environment of high information asymmetry. The goal is to learn to leverage the potential of emerging and digital innovations within a specific industry.	
Study Goals	Upon finishing the course, participants will be able to: LO 1 Describe the difference between emerging and disruptive technologies. LO 2 Understand current social and market dynamics that influence the diffusion and success of emerging technologies. LO 3 Develop a critical perspective on conflicting stakeholder interests. LO 4 Analyse an emerging technology's potential for a specific market/industry/company. LO 5 Adapt a business plan to the realities of digital technology. LO 6 Make informed decisions in the context of high information asymmetries.	
Education Method	The education includes: - Lectures - Self-learning study material - Formative Group assignments and hands-on exercises to apply the understanding of relevant material (contributing to 20% of the final grade). - One individual summative assignment to learn how to analyse and create an informed recommendation for action (contributing to 80% of the final grade).	
Assessment	Group work assessments Students will participate in formative group activities throughout the lectures contributing to 20% of the final grade. These activities may include minor in-class and home assignments, presentations, and collaborative projects. Depending on enrollment, group sizes will range from 2 to 6 members. To be considered successful, each assignment requires a grade of 5.0 or above. Individual assessment Students are expected to submit an individual research report at the course's conclusion, comprising 80% of the final grade. This report will be assessed based on the following criteria: Depth and quality of analysis Proper utilisation of theoretical concepts Scope and quality of research and data collection Strength and coherence of arguments and conclusions Accuracy and comprehensiveness of the bibliography To pass, the final report must be submitted by November 10th, 2023 latest and achieve a grade of 6.0 or above. It is not possible to repeat or resubmit the report.	
Enrolment / Application	This course is part of the MOT specialisation ICT Design and Management and is also open for students from across the Delft University of Technology faculties. You can directly enrol into Brightspace. Brightspace enrollment will be activated by education support a couple of weeks before the course starts. No other means of registration, such as the app MyTuDelft, are needed.	
Special Information	Students are asked to register/enroll on Brightspace. Students are also asked to be present and active in the first lecture session, to facilitate the proper planning of the course.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Targetgroup	Master students from all master programs can apply.	

TPM411A	Idea to Start-up IT & AI	5
Module Manager	E. Çelik	
Instructor	E. Çelik	
Instructor	Dr. A. Giga	
Responsible for assignments	E. Çelik	
Co-responsible for assignments	Dr. A. Giga	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Expected prior knowledge	No prior knowledge is required.	
Summary	In this course, you do not study entrepreneurship, you become an entrepreneur. Working in a team, you identify a business opportunity using emerging digital technologies. Applications can range from mobility, cyber security to predictive maintenance etc. You will test its validity, and build a viable business model a roadmap from an idea to an innovation-driven company. On this journey, you will be coached by lecturers and learn practical insights from guest lecturers from industry and the startup ecosystem. After this course, you will be able to proactively assess the merit of a technology startup, whether you want to build one or join one.	
Course Contents	You will focus on information and computing technologies (e.g AI, big data, blockchain) that are transforming traditional industries and societies at large. To analyse the opportunities for new startups we focus on entrepreneurial tools such as the opportunity canvas; value proposition development and customer empathy mapping, market segmentation and value chain analyses and the initial operations to develop a minimum viable product.	
Study Goals	Develop a business Idea by identifying a relevant economic problem or unmet need that is addressed by digital solutions, and advanced information and computing technologies. Conduct primary and secondary market research in order to rigorously test the value of your business idea and to identify an appropriate market segment Analyse the competitive landscape and devise your products competitive advantage, given specific challenges of protecting IP in the digital world. Create a blueprint of your product. Identify relevant stakeholders in your startups ecosystem, outline the go-to-market strategy, and assess profitability. Understand how these strategies adjust when trying to implement your solution in well-established systems of traditional industries. Integrate and convincingly present all the aspects of your business idea into a coherent business model supported by collected evidence. Emulate effective entrepreneurial behaviour by rapidly gathering new information and acting on it by improving the business model. Evaluate your entrepreneurial skills and motivate constructive feedback through peer reviewing	
Education Method	You will work in teams which are formed in the first sessions of the course. One aspect of a startup will be discussed each week and you will apply into your own projece immediately. You will present your work, provide peer-feedback as well and receive advise from coaches. The course ends with a jury pitch and a 4-pager of the business proposition. Team coaching will help you validate the idea and we provide individual coaching for personal entrepreneurial development.	
Books	The primary textbook for this course is an organic handbook by Bill Aulet: Disciplined Entrepreneurship: 24 Steps to a Successful Startup. Wiley, 2023.	
Assessment	The assessment is based on your project. It includes class assignments and participation (40%), final business model report (30%), online pitch presentation (15%), and team evaluation (15%). Participation and Peer Evaluation grades pertain to each student. Class assignments, final business report, and pitch presentations are evaluated on a group level.	
Enrolment / Application	All students across the various faculties of the Delft University of Technology can take this course by directly enrolling into Brightspace. No other means of registration, such as the app MyTuDelft are needed.	
Remarks	For more information about Entrepreneurship courses: THINK - ACT - START entrepreneurial, please check https://www.tudelft.nl/dce/education.	
Elective	Yes	
Targetgroup	Master students from all master programs can apply.	
Category	MSc level	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Thesis project 2024

CS5000	Thesis Project	45
Responsible Instructor	Dr.ir. A.R. Bidarra	
Responsible Instructor	Dr. K.A. Hildebrandt	
Responsible Instructor	Dr. J.E.A.P. Decouchant	
Responsible Instructor	Dr. E.A. Markatou	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4 Summer Holidays	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	The Computer Science Thesis Project is the final part of the Master's degree programme. During this project, the student will be required to demonstrate their ability to conduct a Computer Science research project, design a solution, communicate and reflect about it. When student and thesis advisor agree on a thesis project topic, they plan a kick-off meeting (start date of the project), here they discuss the planning, milestones of the project and the resources and facilities offered by the faculty. The student and thesis advisor/supervisor plan two formative assessment moment, where they receive feedback from their thesis advisor/supervisor using a standardized rubric. At the end of the project, the student submits their Masters thesis, written in English, and gives an oral presentation of their work to the thesis committee, followed by a question-and-answer session.	
Study Goals	At the end of this course, the student is able to: 1. Conduct a Computer Science research project. 2. Design solutions of Computer Science problems. 3. Communicate about their research. 4. Reflect on their research and the field of Computer Science in the temporal and societal context	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme as shown by a Declaration of Obtained Credits (DCO). To be able to get this DCO, the individual exam program (IEP) should have been approved by the Board of Examiners. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.	
Assessment	The thesis committee assesses the dissertation and the defense based on the degree to which the study goals have been reached. The voting members of the thesis committee determine the final grade. The grade should reflect a weighted average of the four scores above, but need not to be an exact arithmetical mean. The final mark starts from 5 up to and 10. Marks ending in .5 may also be used. If the student shows excellence (is nominated for a 10) the chair of the thesis committee should consult the chair of the Board of Examiners, at least five working days in advance of the defense. The chair may advice to add an extra member to the thesis committee. The justification for the marks for each of the four study goals is summarized on a form and signed by the chair of the doctoral committee. The candidate receives a short report on the assessment, either in private or in front of an audience. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

Year	**2024/2025**
Organization	**Electrical Engineering, Mathematics and Computer Science**
Education	**Master Computer Science**

Additional courses for cohort 2023 and earlier

AP3132	Advanced Digital Image Processing	6
Responsible Instructor	Prof.dr. B. Rieger	
Instructor	Dr. F.M. Vos	
Contact Hours / Week x/x/x/x	0/0/4/2	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Basics of signal processing, image processing, linear algebra, elementary statistics.	
Course Contents	The course Advanced Digital Image Processing covers the principles of several state-of-art image processing techniques. Particularly, students will study the theory of sophisticated algorithms for: 1. Multi-resolution Image Processing: gaussian scale space, windowed Fourier transform, Gabor filters, multi-resolution systems (pyramids, subband coding and Haar transform), multi-resolution expansions (scaling functions and wavelet functions), wavelet Transforms (Wave series expansion, Discrete Wavelet Transform (DWT), Continuous Wavelet Transform (CWT), Fast Wavelet Transform (FWT)); 2. Morpological Image Processing: advanced operations for binary morphology; definitions of gray-scale morphology regarding erosion, dilation, opening, closing; application of gray-scale morphology including smoothing, gradient, second derivatives (top hat) and morphological sieves (granulometry); 3. Image Feature Representation and Description: measurement principles: accuracy vs. precision ; size measurements: area and length (perimeter); shape descriptors of the object outline: form factor, sphericity, eccentricity, curvature signature, bending energy, Fourier descriptors, convex hull, topology; shape descriptors of the gray-scale object: moments, PCA, intensity and density; structure tensor in 2D and 3D: Harris Stephens corner detector, isophote curvature. 4. Motion and optic flow: Taylor expansion method; dual and multi-frame image registration, optic flow; 5. Image Restoration: Noise filtering, Wiener filtering, inverse filtering, geometric transformation, grey value interpolation; 6. Image Segmentation: thresholding, edge and contour detection, data-driven segmentation (boundary detection, region-based segmentation, watersheds, graph-cut, meean shift), model-driven image segmentation (Hough transform, template matching, deformable templates, active contours, ASM/AAM, level sets).	
Study Goals	General learning objectives of the course are: 1. Student has knowledge of can explain the function of state-of-the-art image processing algorithms; 2. Student can solve elementary problems in image processing using Python/MATLAB? programming; 3. Student can solve more advanced problems without implementation, but sketching steps towards a solution; 4. Student can independently acquire new knowledge about image processing from the current literature and present and report about it.	
Education Method	Lectures, practicals and group assignment with plenary presentation and discussion.	
Computer Use	Matlab including the dipimage toolbox and/or other image processing toolboxes.	
Literature and Study Materials	Book 'Digital Image Processing', van R.C. Gonzalez en R.E. Woods, third edition, 2002, ISBN 9780131687288. (Online) Book 'Computer Vision, Algorithms and Applications', R. Szeliski, (http://szeliski.org/Book/). The online version is available for free. We have used the Book Introductory Techniques for 3-D Computer Vision, E. Trucco and A. Verri, ISBN 0-13-261108-2 in the past. Lecture notes Fundamentals of Image Processing (http://homepage.tudelft.nl/e3q6n/education/et4085/sheets/ppt/FIP2.2.pdf) PDF-files of the lecture slides (see Brightspace).	
Assessment	Closed book written exam and assignment. Both parts should be graded 5.8 or higher. A bonus point of 1.5 (to the exam) can be obtained by attending the practicals with 7 out of 8 passed. The final grade is the average of the two parts. The formula for the final grade is: ((0.85*EX + 0.15)+ AS)/ 2 or without the bonus point from the practicals: (EX+AS)/2 With EX the exam grade and AS the grade for the assignment. If you have not passed the exam or the resit, you will need to redo the assignment again next year!	
Permitted Materials during Tests	Closed book exam; books, print-out of pdf files of the lecture slides and lecture notes are not permitted during the written examination.	
Elective	Yes	
Tags	Image processing Matlab Physics	

CESE4025	Real-time Systems	5
Responsible Instructor	Dr. G. Iosifidis	
Instructor	Prof.dr. K.G. Langendoen	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	The course covers the following topics: - basic concepts of RTS - worst case execution time estimation - scheduling policies - response-time analysis - jitter analysis - handling overloads - multiprocessor scheduling - reservation-based scheduling	
Study Goals	At the end of this course, the student is able to: 1. Explain the fundamental concepts and terminology of real-time systems 2. Construct task schedules using different scheduling policies under a given set of realistic system constraints 3. Analyze the timing behavior of a system for a given system model and scheduling policy 4. Discuss advantages and disadvantages of different scheduling policies for a given platform or system 5. Discuss the effect of hardware and software interferences on the timing behavior of a given system 6. Identify (reverse engineer) parameters of a scheduling scheme or a task set from output traces of the system 7. Derive (reverse engineer) the system specification from a given implementation (in the lab) 8. Evaluate the scheduling overheads of a given implementation (in the lab) 9. Implement event-based scheduling policies on a given microcontroller (in the lab)	
Education Method	lectures with exercises (32 hrs); self study (78 hrs); lab assignments (30 hrs)	
Literature and Study Materials	Hard Real-Time Computing Systems by G.C. Buttazzo, Springer 2011	
Prerequisites	Basic software engineering, C system programming, basic Linux operating system knowledge	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 60%) - Individual Assignment: lab work (weighting 40%) Final grade calculation = 0.6 * Written Exam + 0.4 * Individual Assignment A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 6 (5.8 and higher). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit opportunity - Individual Assignment: Repair opportunity Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4030	Embedded Systems Laboratory	5
Responsible Instructor	Prof.dr. K.G. Langendoen	
Instructor	Dr. G. Lan	
Instructor	Ir. J.B. Dönszelmann	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This highly multi-disciplinary course comes with a lab project where teams of 4 students each will have to develop an embedded control unit for a tethered electrical model quad rotor aerial vehicle (the Quadrupel drone), in order to provide stabilization such that it can hover and (ideally!) fly, with only limited user control (one joystick). The control algorithm (which is given) must be mapped onto a home-brew PCB holding a modern RF SoC interfacing a sensor module and the motor controllers. The students will be exposed to simple physics, signal processing, sensors (gyros, accelerometers), actuators (motors, servos), and basic control principles. The embedded software running on the flight controller board (FCB) driving the drone must be written in the Rust programming language. The project work (including written report) covers the entire duration of the course period, and will take approximately 128 hours, of which 32 hours are spent at the lab facilities.	
Study Goals	Student is acquainted with real-time programming in an embedded context, along with a basic understanding of embedded systems, real-time communication, sensor data processing, actuator control, control theory, and simulation. Moreover, the student has had exposure to integrating the various multidisciplinary aspects at the system level. At the end of this course, the student is able to: 1. Explain and analyse real-time communication, sensor data processing, actuator control, control theory in the context of an embedded system 2. Consider various multidisciplinary aspects at the system level 3. Apply real-time programming concepts in an embedded context	
Education Method	Lectures (6*2hrs), lab work (8*4hrs), coding at home (8*12hrs), report (8hrs), so on average 2 days per week	
Prerequisites	Literacy in Rust (CESE4000 or equivalent, and CSE4015)	
Assessment	The final grade of the course consists of the following components: - Lab Project (weighting 75%) - Written Report (weighting 25%) Final grade calculation = 0.75 * Lab Project + 0.25 * Written Report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Lab Project: Additional assignment - Written Report: Additional report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The capacity is limited and -as this is a compulsory course for CESE students- they get preference over other MSc students.	

CESE4045	High-performance data networking	5
Responsible Instructor	Prof.dr.ir. F.A. Kuipers	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Strongly advised: Basic understanding of networking and programming (ideally Python).	
Course Contents	The Internet plays a critical role in modern society, but the complexity of large-scale networks and the diversity of protocols make network management increasingly challenging. As networks become more programmable, the distinction between compute systems and network elements is blurring, leading to a paradigm shift in network operations. The high-performance data networking course is an advanced networking course that will introduce you to the concept of network operations, programmability and management. The course will examine different networking protocols and techniques used across various organisational domains, from edge computing to Internet-scale systems. The course will cover essential topics such as Quality of Service, Internet architecture/organisation, and network measurements.	
Study Goals	At the end of the course, the student is able to: 1. Understand advanced networking concepts, such as Software-Defined Networking, Inter/Intra-domain routing, etc. 2. Understand network management and operation techniques applied in internet, edge, and cellular (5G) domains 3. Evaluate network performance via a network emulator (Mininet) and understand real-world Internet architecture via measurements. 4. Create network programs in the domain-specific language P4.	
Education Method	Approximately 50% of the course will consist of lectures and selfstudy and 50% focuses on (homework) exercises and instruction classes.	
Literature and Study Materials	Slides and a reader containing the exercise material.	
Assessment	The final grade of the course consists of the following components: - Written Exam (weighting 100%) Final grade calculation = 1 * Written Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Exam: Resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. N. Mohan	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	Dr. R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ------------------------ Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. --- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) o Definition of ad-hoc networks o Comparison with infrastructure based systems o Typical applications o Advantages and challenges o Radio technologies for ad-hoc networks o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks o Propagation models o Topology models based on graph theory o Degree and hopcount o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) o Introduction to MAC protocols o Issues and design goals o Classification o Directional, muli-channel MAC protocols o Energy efficiency in MAC protocols o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) o Flooding o Node discovery, neighbour discovery o Route establishment o Topology maintenance, localisation o Proactive, reactive and hybrid routing o Typical protocols o Energy efficiency in routing o Broadcast and multicast o Effects of mobility on connectivity and capacity o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks o Wireless sensor networks (Chapter 12 of the textbook and papers) o Cooperation (Reference papers) o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consist of slides presented at the lectures (Slides are only teaching aids and are not substitutes for textbooks,	

	research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar, C Puttamadappa, and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. === Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. === There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. == (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CESE4060	Wireless IoT and Local Area Networks	5
Responsible Instructor	Dr. P. Pawelczak	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Students are advised to follow the course Wireless Communications (ET4358) before taking this Wireless Networking course. An advantage is to have entry-level programming skills (Matlab, Python, C/C++). Nonetheless, students with little knowledge of programming will be helped.	
Course Contents	The following modules will be discussed during the lectures: Introduction (example topics): - What is wireless networking - Where to search for (academic) wireless network literature and resources Medium Access Control (example topics): - WiFi: hidden/exposed terminal problem, Carrier Sense Multiple Access - Bluetooth standard: in-depth look into the channel hopping, protocol specifications WiFi (example topics): - Review of IEEE 802.11 standards - Protocol format - ISM band regulation - Adaptive Modulation and Coding - WiFi Matlab class (assignment) IoT networking standards (example topics): - LoRa: protocol specifications, energy consumption, modulation format, network design Review of wireless tools (example topics): - Introduction to wireless packet sniffing and analysis using Wireshark (assignment) - Simple simulations of WiFi network with NS3 RFID networking (example topics): - Principles of backscatter - Protocol formats: EPC C1G2 - RFID hackathon (assignment) Cognitive radio (example topics): - Basics of spectrum management - White Space Databases - Theory of spectrum sensing	
Study Goals	At the end of the course students will be able to: (i) to understand how practical wireless systems work and get a deeper understanding of how the theoretical concepts of wireless communications apply to practice; (ii) employ their own analysis methodology to assess new wireless network systems (especially at the physical layer); (iii) understand rapid prototyping of new wireless systems (for instance, with software defined radio).	
Education Method	Lecture presentations, mini-project assignments, assigned paper reading and its critical analysis and presentation.	
Computer Use	Each student should have its own laptop (preferably with a Linux distribution, where Linux must not be installed on a virtual machine). We will be using Matlab, and/or NS3 and/or GNURadio and/or Wireshark for the assignments.	
Literature and Study Materials	WiFi Matlab WLAN toolbox: https://nl.mathworks.com/help/wlan/; Wireshark learn page: https://www.wireshark.org/#learnWS; tutorial on NS3 network simulator: https://www.nsnam.org/documentation/; specific chapters from books provided at the beginning of each lecture.	
Prerequisites	Background in programming (Matlab, Python, Bash)	
Assessment	Points from the mini-project assignments. A research paper analysis from conferences such as IEEE INFOCOM, ACM MobiCom, ACM SIGCOMM will be required to pass the course. Repair option: Individual Assignment = Re-submit deliverable Individual Report = Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

CESE4120	Smart Phone Sensing	5
Responsible Instructor	M.A. Zuñiga Zamalloa	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Requirement 1: Students MUST either (1.1) have passed a JAVA programming course, or (1.2) have passed a C/C++ programming course and be familiar with JAVA, or (1.3) know Objective C (programming language for MACs). This requirement is equivalent to having passed the course CSE1100 in our first year Bachelor curriculum "Object-Oriented Programming" Requirement 2: Students MUST (2.1) have passed a basic course on Probability Theory. This requirement is equivalent to having passed the course CSE1210 in our first year Bachelor curriculum "Probability and Statistics". We will be refreshing some concepts on Probability, but we will not be refreshing concepts on Object Oriented Programming.	
Course Contents	The course provides an introduction to the current research trends in the area of smartphones. The course will be based on a programming project, where students will form groups of two to develop a smartphone application. This is not a programming course; students are expected to have already programming experience. To develop a smartphone application, a user needs to be familiar with (1) the signals and data that smartphones can gather, and (2) the mathematical tools necessary to process this data. This course will provide a solid background for the above two points. During the lectures we will analyze the latest research papers on this emerging field. We will dissect these papers to understand how techniques from algorithms, signal processing and machine learning are used to develop some exciting applications. The students will then use these basic technical tools to develop their own apps.	
Study Goals	The goals of this course are twofold. First, to expose students to the increasingly important area of mobile computing. Students will learn how mobile phones can be used to solve problems in areas ranging from health care and indoor localization to song recognition and traffic management. Second, to provide students with a basic set of tools to develop their own applications. For students aiming for industry, the course should enhance their ability to use theoretical tools to solve practical problems. For students involved in research activities, the course will provide them with the necessary background to use smartphones as a distributed sensing and processing unit that could be used to solve particular problems in their areas. At the end of this course, the student will be able to: (1) Explain the current applications, methods, and research trends in the area of indoor positioning with smartphones. (2) Apply key mathematical tools in the development of smartphone applications. (3) Analyze how a sensing and computing problem can be solved via the use of smartphones, and identify the steps required to design a solution. (4) Create a non--trivial and innovative smartphone application.	
Education Method	Lectures + Lab The project work, including the written report, covers the entire duration of the course period, and will take approximately 120 hours, of which 14 hours are spent on lectures, 10 hours preparing reports, 10 hours reading research papers, and the remaining part programming the App (the time spent in the Lab belong to this latter part).	
Literature and Study Materials	Research Papers and web tutorials	
Assessment	The final grade of the course consists of the following components: - Written Report 1: two intermediate reports (10% each, 2 pages each) (weighting 20%) - Written Report 2: final report (4 pages) (weighting 10%) - Oral Exam: final project demonstration (weighting 70%) The first two reports are due on the third and sixth week; and the final report, project, and oral exam are due on the ninth week. Final grade calculation = 0.2 * Written Report 1 + 0.1 * Written Report + 0.7 * Oral Exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: - Written Report 1: Resit about a new related project - Written Report 2: Resit about a new related project - Oral Exam: Resit about a new related project Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	1. You need to enrol in Brightspace 2. The first lecture will be compulsory 3. This course can only accommodate 60 students, with CESE students having a preference when demand exceeds capacity. If your program marks this course as required, you are guaranteed a spot.	

CS4090	Quantum Communication and Cryptography	5
Responsible Instructor	Dr. S.D.C. Wehner	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Linear Algebra, Probability & Statistics, Q101 (AP3421 Fundamentals of quantum information) or CSE3130 (Introduction to Quantum Computer Science)	
Course Contents	This class will introduce you to the fascinating field of quantum communication! We will look at the state of the art of quantum networks, and explore techniques for building quantum repeaters that promise to deliver qubits over long distances. We also briefly look at one of the most famous application of quantum cryptography, quantum key distribution.	
Study Goals	The student will acquire: A good understanding of the fundamental concepts of quantum communication The ability to understand, design and analyze quantum communication protocols Insight into the differences between classical and quantum communication and cryptography	
Education Method	- Flipped classroom (Online videos and materials, prepared for flip classroom teaching) - Weekly Quiz and Discussion Session (In person) - Weekly Tutorials on Homework Problems (In person)	
Literature and Study Materials	Book on tools and quantum cryptography: Introduction to Quantum Cryptography, Cambridge University Press, 2023 Lecture Notes Slides Review Articles Auxilliary: Nielsen and Chuang Quantum computation and information, Cambridge University Press.	
Assessment	The final grade of the course consists of the following components: Homework (weighting 60%) Final Project (weighting 30%) Weekly Quiz (weighting 10%) Final grade calculation: 0.6 * Homework + 0.3 * Final Project + 0.1 * Quiz A passing final grade for the course can only be earned when for the homework and final project at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance to TER Implementation Regulations Art 5, sub 5., for: Homework: re-submit deliverable Final Project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Tags	Abstract Adventurous Algorithmics Challenging Group Dynamics/Project Organisation Information & Communication Integrated Intensive Involved Lineair Algebra Mathematics Physics Quantum Signals Technology Telecommunication	

CS4195	Modeling and Data Analysis in Complex Networks	5
Responsible Instructor	H. Wang	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	The assignment and final project require basic programming skill.	
Course Contents	Big Data is mostly obtained from features of components and the interactions between components in large complex systems. Examples are (1) end user features and interactions in both online and real-world social networks like Twitter, LinkedIn (2) data from content sharing platforms such as YouTube (3) physiological data of the brain and (4) stock prices etc. in economic systems. Such a dataset is networked in nature i.e. the data of the system components or interactions are (cor)related to each other. This course introduces the basic methodologies to analyze, model, interpret and predict such Networked Data that enable us to further intervene or optimise the system, combining advances from network science, modeling of dynamic processes and statistical physics, beyond machine learning algorithms. These methods will be applied to diverse real-world datasets obtained from e.g. Facebook, LinkedIn, YouTube, the brain etc.	
Study Goals	After this course, students could construct a network based on the dataset, characterize and model the network in order to e.g. detect patterns and anomalies, model the data via dynamic processes (e.g. viral spreading) on networks to decode the underlying governing mechanisms of e.g. information/error/behavior contagion and to predict e.g. the popularity of a product, news, disease, computer virus, control the contagion process such as maximize the information prevalence and market share. Students could obtain an overview of the Msc/Phd projects on the frontiers of networked data analysis.	
Education Method	In total, there will be about 7 lectures. Students will also learn via an assignment and a final project (each group gets individual supervision).	
Assessment	The final grade of the course consists of the following components: Assignments (weighting 20%) Final project (weighting 80%) Final grade calculation: 0.2 * Assignments + 0.8 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Assignments: re-submit deliverable Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4). In case the assignment or project is not submitted on time, which means it won't be evaluated, there won't be any repair opportunity.	

CS4225	Educational Technologies	5
Responsible Instructor	Prof. M.M. Specht	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	* Theories of Human Information Processing and Learning * Learning Analytics * Personalisation and Adaptive Educational Systems * Multimodal Feedback Systems * Artificial Intelligence in Education * Realtime Learning Technologies	
Study Goals	The course will enable you to classify, understand, design and implement the core functionalities and systems for supporting human learning processes. As well current practices implemented as also approaches for technology enhanced learning currently researched will be presented. You will learn how educational technologies provide human learning process support, implement guidance and recommendation, create personalised learning support, as also give real-time feedback and support reflection of learners. In the final project you will identify a problem, design a solution based on the presented approaches and implement your own educational technology solution.	
Education Method	Lectures, Peer Reviews, Presentation and Final Report.	
Assessment	The final grade of the course consists of the following components: * Peer Reviews and midterm presentation (weighting 30%) * Final project (weighting 70%) Final grade calculation: 0.3 * weekly assignments + 0.7 * Final project A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: * Weekly assignments: re-submit deliverable * Final project: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	M.A. Migut	

CS4235	Socio-Cognitive Engineering	5

Responsible Instructor	Prof.dr. M.A. Neerincx
Instructor	Dr. B.J.W. Dudzik
Contact Hours / Week x/x/x/x	0/0/4/0
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Expected prior knowledge	Basic prior knowledge on human-computer interaction is helpful, but not required.
Course Contents	Artificial intelligence (AI) is increasingly integrated into various aspects of our daily lives, from work and leisure to care settings. Hybrid Intelligence (HI) aims to create a beneficial integration that leverages the strengths of human and artificial intelligence, while compensating for their respective weaknesses. By aligning AI with human values and cognitive processes, it can collaborate with humans to enhance hybrid decision-making, problem-solving and creativity. This collaborative AI, embodied as virtual or physical agents, fosters the development of mutualistic, symbiotic human-AI relationships, promoting partnership behaviors and collaborative cognitions. This MSc-course centers on theories, models and methods for developing such behaviors and cognitions, including value alignment, goal sharing, workload harmonization, co-learning and collaborative memories. Examples will be provided from diverse application domains, including childrens diabetes management, people with dementias activities of daily living (ADL) and first responders disaster management. Next to the lectures, students work in small groups on a specific Hybrid Intelligence design challenge, focusing on social robots that support people with dementia (see http://rejam.tudelft.nl). This way, you will become acquainted with the application of a coherent set of Socio-Cognitive Engineering (SCE) methods: - Value Alignment; Scenario-based Design; Claims Analysis; Design Patterns. - Social, Cognitive and Affective Modeling. - Prototyping; Evaluation; Research Ethics.
Study Goals	At the end of this course, students will be able to: 1. Explain the core models and methods of Socio-Cognitive Engineering (SCE), including their strengths, weaknesses and complementarity. 2. Apply these models and methods in research and development projects, constructing a theoretically and empirically grounded design rationale and prototype. 3. Systematically build on state-of-the-art developments in an iterative process, explicating and testing the strengths and weaknesses of the design choices. 4. Explain which new insights are acquired by SCE for a specific Hybrid Intelligence (HI) development. 5. Write a structured report about a design-test cycle, with sufficient detail for a new group of researchers to continue the research. 6. Collaborate in a group to develop a HI-system, presenting intermediate and final results.
Education Method	LECTURES During the lectures, the teachers will present a range of theories, models, and methods relevant to Socio-Cognitive Engineering (SCE). Students are required to read a number of scientific papers which are made available on Brightspace, along with the sheets/slides of the lectures. Together, the sheets/slides and the papers provide the students with the required theoretical knowledge to work on the practical project, and to learn about relevant design and evaluation methods, human factors theories, conceptual solutions, and design principles. Most of the lectures include practical assignments and discussions stimulating the students to apply the contents of the lecture to their project (also see Project). PROJECT In the project, students work in groups to apply the knowledge acquired during the lectures. Students are required to plan, execute, present, and report on a complete design cycle (i.e. design, prototype, and evaluation) for a given design problem. The SCE-assignment centers on a social robot for older adults with dementia, and their social environment (https://rejam.tudelft.nl). The objective of the social robot is to improve humans physical, social, cognitive, and emotional well-being. The students will use the Wiki Socio-Cognitive Engineering (WiSCE) tool to specify the design rationale and its evaluation, step-by-step (see also https://xwiki.ewi.tudelft.nl/). Throughout the course, students will give presentations about their progress, on the design and evaluation of their prototype. The instructors provide feedback on the WiSCE content and the presentations.
Literature and Study Materials	Papers from scientific journals on Brightspace. Lecture notes on Brightspace.
Assessment	The module assessment concerns the processing and application of the theory and methods; the construction of the design (rationale) and the evaluation; and the provision of the resulting concise and coherent report (including the lessons learned): The final grade of the course consists of the following components: Presentations (weighting 15%) Prototype (weighting 15%) Project report according to the prescribed format (weighting 60%) Individual reflection (weighting 10%) Final grade calculation: 0.15 * Presentations + 0.15 * Prototype + 0.6 * Project report + 0.1 * Individual reflection A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Presentations: resit Prototype: re-submit deliverable Project report according to the prescribed format: re-submit deliverable Individual reflection: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).
Exam Hours	There is no exam. The assessment is based on a paper, presentation and report. During the course, students will receive feedback on interim work. There is no resit after the end of the course.

CS4345	Seminar Formal Methods for Learned Systems	5
Responsible Instructor	Dr. A. Lukina	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course will cover several topics at the intersection of formal methods and machine learning, for instance, abstraction refinement for analyzing deep neural networks, algorithms for reliable learning-based control, interpretable model design, and formal methods for verification and monitoring of learned systems. The course should be useful for students of both formal methods and machine learning, and lies at the intersection of these areas. Since the area is relatively new, the course material will be primarily based on research papers.	
Study Goals	After this course students will acquire formal methods knowledge and skills such as: - modeling complex systems, - formally defining critical properties, - applying verification tools to learned systems, - synthesizing controllers for learned systems, - providing interpretability for learned models.	
Education Method	The course will consist of one lecture and one discussion session per week. The students will be assigned projects in groups of two, where they will need to apply the material from the lectures, e.g., train a learned system, define a property and verify it using one of the open-source tools from state-of-the-art research papers. Discussion sessions will be used for questions about the projects.	
Assessment	The final grade of the course consists of the following components: - Participation (weighting 10%) - Presenting Papers (weighting 30%) - Assignment (weighting 20%) - Final Report (weighting 40%) Final grade calculation: 0.1 * Participation + 0.3 * Presenting Papers + 0.2 * Assignment + 0.4 * Final report A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Presenting Papers: resit - Assignment #1: re-submit deliverable - Assignment #1: re-submit deliverable - Final report: re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Co-Instructor	Prof.dr. M.T.J. Spaan	

CS4350	Machine Learning for Graph Data	5
Responsible Instructor	Dr. E. Isufi	
Responsible Instructor	M. Khosla	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Basics Machine Learning: data division in train-validation-test, overfitting, kernels, regularization; Basics Deep Learning: back-propagation, stochastic gradient descent (SGD), multi-layer perceptron. Basics in programming: Python or equivalents, minimal experience with Pytorch / TensorFlow or equivalents is helpful but not strongly expected.	
Course Contents	Graph data are present in a myriad of modern computer sciences systems and applications. Examples include data generated over social, brain, financial, power, water and sensor networks. Because these data have a complicated structure they require different tools conventionally used in machine and deep learning to develop end to end solutions. These solutions falls generally under the umbrella of graph-based machine learning and can be used to perform recommendations, detect anomalies in the brain, predict financial crisis, estimate the sate of a power or water network, and coordinate group of autonomous moving sensor, to name a few. This course deals with the foundations and principles of machine and deep learning for network data. Topics include: unsupervised and semi-supervised learning on graphs; graph representation learning; graph signal processing; graph convolutions; graph neural networks; spatiotemporal learning on graphs; scalable algorithms; explainability.	
Study Goals	By the end of this course, you should be able to: Lo1: Recognise key challenges in learning from graph-structured data. You are able to recognise challenges in real applications involving graph-based data, explain them with a machine learning language, and discuss strategies to solve them. You are able to identify the issues of lack of interpretability in graph-based machine learning models and describe some possible solutions. You are also able to explain a few techniques to improve scalability of existing techniques for large graphs. Lo2: Explain and compare representation learning techniques on graphs. You are able to explain, and compare different graph-based machine learning techniques, identifying advantages and limitations in each of them, and discuss the latter with respect to a specific application. Lo3: Implement and analyse machine and deep learning techniques for graph data. You are able to apply different graph-based machine learning techniques including graph kernels, graph neural networks, graph-time learning solutions to real data. You are also able to analyse the different trade-off of these solutions, compare them with each other and inspects their behaviour. LO4: Design machine learning solutions for real world graph data tasks. You are able to create a novel solution for a practical graph-based challenge, discusses different graph-machine learning strategies for solving the task, test one or more of them, and contrast the performance of the developed approach with alternative baselines.	
Education Method	The course consists of: - 12 lectures in which the main concepts are explained and discussed; - 3 hands-on homework labs, in which practical assignments need to be solved individually; - At least one feedback sessions about the final project;	
Assessment	The final grade of the course consists of the following components: - Group assignment (weighting 60%) - Oral Exam (weighting 40%) Final grade calculation: 0.6 * group assignments + 0.4 * Oral exam A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Group assignment: repair by re-submitting deliverable - Oral exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	

CS4410	Category Theory for Programmers	5
Responsible Instructor	Dr. B.P. Ahrens	
Responsible Instructor	Dr. J.G.H. Cockx	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Categorical structures occur in programming languages on different levels: (1) within programming languages, providing design principles and guidance on how to write modular and correct-by-design programmes (as demonstrated in the practical programming language Haskell) and (2) in the design and study of programming languages, as a guiding meta-theory. In particular, category theory provides a mathematical justification for recursion schemes for inductive datatypes. This course aims to provide solid foundations on both (1) and (2).	
Study Goals	At the end of the course, the student is able to: 1. Use categorical constructions in the design and structuring of computer programmes 2. Specify datatypes as initial algebras 3. Prove properties of computer programmes, guided by categorical intuition 4. Use categorical fusion laws to optimize functions	
Education Method	Lectures and problem sessions	
Assessment	The final grade of the course consists of the following components: Individual project (weighting 100%) Final grade calculation: 1 * Individual project A passing final grade for this course is obtained when the final grade for the project is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Individual project: re-submit deliverable	

EE4715	Array Processing	5
Responsible Instructor	Dr.ir. R.C. Hendriks	
Responsible Instructor	Prof.dr.ir. G.J.T. Leus	
Instructor	Prof.dr.ir. A.J. van der Veen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Linear algebra, signal processing, Fourier transform, stochastic processes and preferably statistical signal processing and some experience with matlab	
Summary	In this course we discuss array processing techniques for signal separation and parameter estimation, using arrays of sensors. After a review/introduction of the necessary linear algebra tools we will start with deriving the signal processing model for narrowband applications, followed by the wideband extension, and apply these to several applications among which array processing for wireless communication, audio and speech processing, biomedical signal processing and astronomy.	
Course Contents	Signal processing models for narrowband and wideband array processing, elementary beamforming concepts (spatial filtering), tools from linear algebra: QR, SVD, eigenvalue decompositions, projections and GEVD. Elementary beamformers/receivers: the matched filter, the Wiener filter, MVDR, LCMV, etc. Estimation of angles and delays using ESPRIT, adaptive space-time filters, the LMS algorithm and factor anaylsis.	
Study Goals	To be able to explain some key problems regarding data models, estimation and detection that occur in array processing applications. - To be able to explain the major signal processing tools required to solve array processing problems. - To be able to implement these signal processing techniques in Matlab. - To be able to apply these techniques to new array processing problems.	
Education Method	Lectures + mini project	
Literature and Study Materials	References from literature and notes	
Assessment	Oral exam: 2 Take-home assignments with oral discussion to test how well the corresponding theory and assignment has been understood. Final grade calculation: 0.5 * Oral discussion 1 + 0.5 * Oral discussion 2. Both partial grades must be sufficient, i.e., .a 6 or higher. Repair possibility: In case one or both partial exams have an insufficient score (i.e., <6), the corresponding exam can be redone by making an appointment. Notice: The partial results only remain valid in the academic year in which they are obtained. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l0 norm is shown to be NP-hard, and several classical approximate algorithms like l1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (25%), a mini project (15%), and homework submission (10%). The course is closed by a project report, presentation, and Q&A (50%). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

EE4C06	Networking	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	PART 1: Basics, concepts and computations of networks 1. Basics of networking & introduction to Network Science - what is a network? Representation of a graph, basics of graph theory, overview of the relatively new theory of complex networks, called Network Science. - important characterizers of a network (network metrics) - basic network/graph models - examples of real-world networks (airline transportation, the web and Internet, social networks, brain networks, etc.) and applications of network science 2. Concepts of networking - routing - Quality of Service (QoS) - traffic management and scheduling - network robustness (failure, cascading effects,...) - overlay networking and new aspects of networking such as interdependent networks PART 2: Applications and examples of networks (as listed below) will be taught (some of those by a guest lecturer). Ranging from year to year, a selection among the following will be covered: 1. Electrical networks (smart grids) 2. Networks on Chip (NoC) 3. Optical networks 4. Computer Networks (the Internet) 5. Mobile communication networks 6. Sensor networks 7. Biological networks 8. Social networks	
Study Goals	The course on Networking aims to provide a general and basic introduction to the art of networking, that tries to unravel the operation and behavior of networks, both man-made (infrastructures such as the Internet and power grids) as well as networks appearing in nature (such as the human brain, biological networks and social human interactions). The course on Networking will introduce concepts of the new Network Science, that basically studies the interplay between, on the one hand, the processes (also called functions or services) on the network and on the other hand, the underlying topology, that is mostly changing over time as an evolving organism, rather than as given or fixed object. Network Science combines many disciplines such as graph and network theory, probability theory, physical processes, control theory and algorithms. After this course, students are expected to represent/abstract real-world infrastructural network (e.g. a communication system) as a complex network, understand the basic methods to analyze properties of networks and dynamic processes on networks. Students will also understand why processes on networks and design of networks are so complex. Finally, students may appreciate the fascinatingly rich structure and behavior of networks and may realize that much in the theory of networks still lies open to be discovered.	
Education Method	Lectures, slides & homework. On the slides, references to articles (i.e. numbered paragraphs) or to pages in the book "Graph Spectra for Complex Networks" (see Literature and Study Material) are made. The information in the book thus complements the slides.	
Literature and Study Materials	Graph Spectra For Complex Networks, second edition (2023) Cambridge University Press ISBN: 978-009-36680-9 See: https://www.nas.ewi.tudelft.nl/people/Piet/bookGraphSpectraCN.html	
Assessment	The examination consists of multiple choice questions. The examination is executed on a computer. Each student logs in with his own student ID. The kind of questions is the same for each student, but the parameters and the order of multiple choice questions is individual. All information can be used (except consulting humans), i.e. the exam is "open book". Further details are given in Brightspace. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4030	Error Correcting Codes	4
Responsible Instructor	Dr.ir. J.H. Weber	
Contact Hours / Week x/x/x/x	0/0/0/3	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Basic linear algebra and probability theory, as being treated in B.Sc. programmes like Electrical Engineering, Computer Science, and Mathematics	
Course Contents	Introduction into error-correcting codes; mathematical basics; block codes fundamentals; cyclic codes; co-operating codes; soft-decision decoding; convolutional codes; iterative decoding (turbo codes, LDPC codes); applications.	
Study Goals	The general overall objectives of this course are that the students can - reproduce the error control concepts as presented in the lecture notes; - explain the meaning and usefulness of these concepts; - apply the concepts to transmission and storage systems, including basic calculations, at the level of the exercises in the lecture notes and previous exams. More specifically, the students are able to - discuss the basic trade-offs between efficiency, reliability, and complexity in designing coding schemes for transmission and storage systems; - apply finite fields theory in order to construct (cyclic) codes; - encode data into code sequences that are resistant to a certain number of random and/or burst errors; - decode received sequences using (iterative) hard or soft decision techniques.	
Education Method	Lectures; expected workload is 20 hours attending lectures, 60 hours preparing for the lectures, studying the lecture notes, and making suggested exercises, and 32 hours for preparing and making the exam.	
Literature and Study Materials	Lecture notes "Error-Correcting Codes" by J.H. Weber, available on Brightspace. Also previous exams as well as demos are available via Brightspace.	
Assessment	The final grade will be fully determined by a scheduled written exam, which will be held at the end of Q4. The resit will take place as an oral exam in Q5, on appointment with the lecturer. The exam format is closed-book in any case. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Remarks	Actual course information available on Brightspace.	
Co-Instructor	Dr. J.A.M. de Groot	

IFEEMCS4070	Multivariate Data Analysis	5
Responsible Instructor	Dr. J. Söhl	
Responsible Instructor	Dr. rer. nat. F. Mies	
Contact Hours / Week x/x/x/x	(8/0/0/0)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Required: * Introduction Probability Theory and Statistics: see for instance A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover ISBN: 1-85233-896-2 * Basic calculus * Linear Algebra: matrix multiplication, the inverse of a matrix, the transpose of a matrix, least square solution. see: David C. Lay: Linear Algebra and Its Applications ISBN-10: 0321385179 ISBN-13: 9780321385178 ©2012 Pearson)	
Course Contents	PART I: 1. Refresher probability theory and basic statistics 2. Markov Chains 3. Introduction to stochastic processes: Brownian motion, Poisson process, autocorrelation function, stationarity, estimation of correlation function, seasonality and trend, 4. Stochastic processes in time domain: ARMA processes, linear prediction (theory and computer exercise) 5. Stochastic processes in frequency domain: spectral density function, white noise, periodogram PART II: A course in advanced statistics about linear models, Bayesian inference, classification problems, Gaussian processes and Markov Chain Monte Carlo.	
Study Goals	PART I: By the end of the course, you should be able to: - Compute probabilities and moments in a mathematical probability model. - Analyze a stochastic process in terms of its moments and distribution. - Formalize problems in electrical engineering or computer science in the form of a statistical model using stochastic processes. - Fit a time series model to temporal data using the R programming language. PART II: By the end of the course, you should be able to: - compute properties of frequentist and Bayesian estimators in the linear model - derive posterior distributions and their properties - apply hyperparameter optimization in Bayesian models - apply computational methods for posterior approximation - derive properties of frequentist and Bayesian classification methods - derive properties of MCMC methods for sampling from the posterior distributions - derive properties of Gaussian process regression	
Education Method	Classes and exercises	
Literature and Study Materials	PART I: Matthew A. Carlton and Jay L. Devore, "Probability with Applications in Engineering, Science, and Technology" Roy D. Yates and David J. Goodman "Probability and stochastic processes: a friendly introduction for electrical and computer engineers" Robert H. Shumway and David S. Stoffer "Time Series Analysis and Its Applications: With R Examples" PART II: Simon Rogers and Mark Girolami "A first course in machine learning, 2nd edition" Chapman & Hall	
Assessment	The assessment of the course consists of the following components: Lab assignments (pass/fail) Written exam (weighting 100%) Resit/ Repair opportunities: Written exam: Resit (written or oral exam, depending on the number of students) In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2024-2025, for: Lab assignments: Re-submit deliverable Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Exam Hours	Exam of 3 hours	
Permitted Materials during Tests	None	
Co-Instructor	Dr. J. Söhl	
Co-Instructor	Dr. rer. nat. F. Mies	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties (incl. LDE universities). In this project we put together a team of students with varying talents, backgrounds, and perspectives to achieve what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: o identifying and valuing the soft skills necessary to work effectively in an interdisciplinary team, integrating its members' talents in a variety of team roles o adapting with flexibility to the dynamic requirements of a complex external assignment o proactively translating feedback received into personal development steps o formulating research questions, identifying research contributions, and describing them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: o identifying and deploying the most adequate game technologies for the given serious game domain and constraints o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small indie game studio, according to a tight planning. Also a few plenary sessions and/or lectures are held.	
Assessment	The final grade of the course consists of the following components: 1. Product grade (for the whole group) (weighting 50%). Based on both the game itself and the required documentation. 2. Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. 3. Final presentation (for the whole group) (weighting 5%). The course deliverables include writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation: 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: Final presentation: has no repair, as it is done in group. There is no repair possibility for the other components (product and process), since their assessment necessarily requires intensive teamwork throughout an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Marroquim	

IN4306	Literature Survey	10
Responsible Instructor	Dr. K.A. Hildebrandt	
Responsible Instructor	L. Miranda da Cruz	
Contact Hours / Week x/x/x/x	n/a	
Education Period	None (Self Study)	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Literature Survey is an individual assignment carried out under the supervision of a Computer Science staff member, i.e. an assistant, associate or full professor. For this assignment the student reads a broad range of papers in the chosen specialisation field and writes a report in which the ideas found in the papers are discussed and compared. It is not allowed to merge this assignment with the thesis project.	
Study Goals	After the course, the student will be able to: LO1 - Formulate a research question that is relevant, meaningful, and scientifically sound. LO2 - Find high-quality contemporary scientific literature in the chosen research field. LO3 - Explain the scientific contribution of the reviewed papers within the context of the research question.	
Education Method	Individual assignment and individual guidance by a scientific staff member.	
Assessment	Writing a scientific report, individually and under supervision of a staff member. This staff member will also mark the report. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Enrolment / Application	The Literature Survey may be part of an individual exam programme of a student, which has to be approved by the Board of Examiners (BoE). To apply for a literature study the student should contact a staff member of the research group of her/his chosen specialisation after having received approval of her/his individual exam programme. The staff member and the student make arrangements regarding content and scope of the survey. The aforementioned staff member will supervise the student during this course.	

IN4310	Seminar Computer Graphics	5
Responsible Instructor	Dr. P. Kellnhofer	
Instructor	Dr. M. Skrodzki	
Instructor	Dr. M. Weinmann	
Instructor	Dr.ir. A.R. Bidarra	
Instructor	Prof.dr. E. Eisemann	
Instructor	Dr. K.A. Hildebrandt	
Instructor	T. Höllt	
Instructor	Dr. R. Marroquim	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	Strongly advised: One of the CS core courses (IN4089 Data Visualization, and IN4152 3D Computer Graphics and Animation), and at least one of the Computer Graphics specialization courses (IN4255 Geometric Modeling, IN4302TU Building Serious Games, CS4365 Applied Image Processing, and IN4307 Medical Visualization) are expected as prior knowledge.	
Course Contents	In this seminar you work on a selection of recent topics and results in one of the areas of Computer Graphics.	
Study Goals	To obtain in-depth knowledge about an advanced topic within Computer Graphics, in particular in rendering, game technology, visualization, appearance capture, animation or geometric modeling. The seminar may be used as a preparation for an MSc thesis topic. The student will acquire practical skills in reading, presenting, explaining, and discussing scientific papers, as well as writing scientific papers.	
Education Method	This course has the format of a student seminar. You will prepare a scientific presentation of a recent research paper. The presentation goes in-depth and covers the strengths and weaknesses of the described technique. Depending on number of participants, we may organize the paper presentations individually or in groups. After each presentation in the plenary colloquium session, a research discussion takes place. All students participate in a scientific discussion and ask relevant critical questions. Finally, you will implement a tool that further explores the presented method. Depending on the number of participants and scheduling constraints, you will then be asked to present your work either in a form of a short public demonstration or in a pre-recorded video presentation. Additionally, you will prepare a short report summarizing your solution.	
Literature and Study Materials	Recent research papers about the selected topic.	
Assessment	The final grade of the course consists of the following components: - Presentation of a recent scientific paper (50%) - Implementation component reproducing an aspect of the paper (40%, including short report and presentation) - Preparation of questions for the presented research papers (10%) Final grade calculation: (0.5 * Presentation + 0.4 * Implementation + 0.1 * Questions) The course is passed if all components have at least a grade of 5.0 and the weighted average rounded on 0.5 is at least a 6. Repair possibilities (under conditions described in TER Implementation Regulations Art 5, sub 5): - Presentation of a recent scientific paper can be repaired by an individual video presentation of a state-of-the-art report + an oral examination. The content of each presented paper must be understood by the student. - Implementation component can be repaired by a new implementation following a new approved plan. - Preparation of each question for the assigned paper can be repaired by a short video presentation of the same paper. The maximum obtainable result for each repair option is a 6.0 (see: TER Art 5, sub 5). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

IN4341	Performance Analysis	5
Responsible Instructor	Prof.dr.ir. P.F.A. Van Mieghem	
Instructor	Dr. C. De Bacco	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course applies probability theory and the theory of stochastic processes to the design and performance evaluation of complex networks such as man-made networks as telecommunication, computer and embedded networks and biological networks. The computation with random variables is reviewed. Markov processes and queuing theory will be introduced to the current important concept of "Quality of Service (QoS)" provisioning and to the computation of the blocking probabilities in telephony (both fixed as mobile). Several applications (e.g. the robustness of networks, epidemics in networks, the Internet shortest path routing) are also included. More details are found on brightspace.	
Study Goals	The course intends to provide students with mathematical techniques, in particular probabilistic methods and graph theory, to compare the performance of different network designs and protocols.	
Education Method	Lectures and homework after each class The content of the lectures follows chapters in the book "Performance Analysis of Complex Networks and Systems" (see Literature and Study Material).	
Literature and Study Materials	Performance Analysis of Complex Networks and Systems. ISBN: 978-1-107-05860-6 See http://www.nas.ewi.tudelft.nl/people/Piet/bookPA.html	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 100%). Written and closed book. A formularium is provided that can be consulted at the examination. Final grade calculation: 1 * Written exam A passing final grade for this course is obtained when the final grade for the exam is at least a 5.8. Resit/ Repair opportunities: Written exam: resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

IN5000	Final Project	45
Responsible Instructor	Dr.ir. A.R. Bidarra	
Responsible Instructor	Dr. K.A. Hildebrandt	
Responsible Instructor	C.B. Poulsen	
Responsible Instructor	Dr. E.A. Markatou	
Responsible Instructor	L. Miranda da Cruz	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	IN5000 Final Project is the final part of the Master's degree programme. During this project, you will be required to demonstrate your ability to solve a research or engineering problem. The project must be carried out using the techniques of project management. You will begin by making a project plan in cooperation with your Master’s thesis advisor. Several aspects of the project are defined within the plan, including the assignment, the frequency of interaction with the advisors, the milestones of the project and the resources and facilities offered by the faculty. You will be required to adhere to your plan throughout the project. It is obviously possible to adjust your plan under certain circumstances and after discussion with your daily supervisor. At the end of the project, you will submit your Master’s thesis, which must be written in English, and make an oral presentation of your work to the Thesis Committee. The Thesis Committee will announce the final mark, which is based on the project performance, the thesis, the presentation and the subsequent discussion. More information about the graduation process: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/ More information on the thesis project on Brightspace: MSc CS Thesis project EEMCS https://brightspace.tudelft.nl/d2l/home/492868	
Study Goals	1. The student is able to design a research project: - The student is able to use, explain and justify adequate research and design methodologies; - The student is able to apply theory to the performed project; - The student is able to use techniques for interpretation and verification and bases his/her conclusions on results; - The student is able to do reliable work with scientific significance. 2. The student is able to execute a research project: - The student has a critical attitude towards his/her own results, literature and specialists; - The student makes an original contribution to the project; - The student takes initiative (together with the supervisor) to give his/her own input within the research project; - The student interacts sufficiently with peers and superiors; - The student is able to make and execute a project plan. 3. The student is able to write a research report: - The student is able to write a research report that shows sufficient coherence of content; - The students is able to structure the research report and sufficiently present the content (text and figures); - The student expresses argumentation using correct spelling and grammar; 4. The student is able to present and defend the research project: - The student is able to present the content using sufficient detail to support conclusions; - The student is able to logical structure the presentation and use visual aids; - The student is able to adequately formulate and express himself/herself as well as sufficiently address the audience; - The student is able to argument and answer the questions asked by the committee.	
Education Method	Project	
Prerequisites	Before starting the project, students must have completed at least 60 EC of the Master's degree programme as shown by a Declaration of Obtained Credits (DCO). To be able to get this DCO, the individual exam program (IEP) should have been approved by the Board of Examiners. Note: In some cases, the thesis supervisor may impose additional conditions for starting this project.	
Assessment	The thesis committee assesses the thesis and the defense on the following criteria: - quality of work: novelty, volume, grasp, methodology, publishable; - personal performance: autonomy, planning, creativity, attitude; - quality of thesis report: clarity, organisation, argumentation; - oral presentation and defense: clarity, focus, relevance, discussion. More information on thesis grading: https://www.tudelft.nl/en/student/faculties/eemcs-student-portal/education/graduation-policy-msc/assessment/ The voting members of the thesis committee determine the final grade. The grade should reflect a weighted average of the four scores above, but need not to be an exact arithmetical mean. The final mark starts from 5 up to and 10. Marks ending in .5 may also be used. If the student shows excellence (is nominated for a 10) the chair of the thesis committee should consult the chair of the Board of Examiners, at least five working days in advance of the defense. The chair may advice to add an extra member to the thesis committee. The motivation for the grade at each of the four criteria as listed above is summarized on a form and signed by the chairman of the thesis committee. The candidate is given a short account of the assessment, either in private or in front of the audience. In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

QIST4310	Fundamentals of Quantum Information	4
Responsible Instructor	Dr. M. Blaauboer	
Responsible Instructor	Dr. L. di DiCarlo	
Contact Hours / Week x/x/x/x	0/4/0/0/	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Required for	AP3432, CS4090	
Expected prior knowledge	Knowledge of linear algebra, probability and statistics (at undergraduate level).	
Course Contents	Approximate syllabus: - quantum states, unitary operations, and measurements; - universal gate sets; - entanglement, Bell test; - basic quantum communication protocols; - basic quantum algorithmic techniques; - principles of quantum error correction and basic quantum error correction protocols; - simple physical implementations of qubits.	
Study Goals	The aim of this course is for you to learn and to be able to apply the fundamental principles of quantum information and the fundamental concepts underlying quantum computation and communication systems. Specifically: -You will learn essential concepts that distinguish quantum from classical devices and how these can be harnessed in quantum information applications. -You will learn about quantum bits and the quantum operations and measurements that can be performed on them and how to apply these to concrete problems. -You will learn basic techniques underlying quantum communication protocols and be able to apply these to concrete problems. -You will learn the basic techniques used in quantum algorithms (this is studied in greater depth in the course Quantum Computing and Quantum Control, QIST4320) and quantum error correction -You will take the first steps in understanding how a quantum bit can be physically implemented, as introduction to the more advanced courses Quantum Hardware I and II (AP3432 and AP3442)	
Education Method	Lectures and tutorials	
Literature and Study Materials	The main reference textbooks for the course are: - N.D. Mermin, "Quantum Computer Science", Cambridge University Press. - M.A. Nielsen and I.L. Chuang, Quantum Computation and Information, Cambridge University Press.	
Assessment	The final grade for the final exam is composed of the homework assignment grades (total of 30%), online quiz grades (total of 10%) and the grade obtained for the final exam (60%). A minimum grade of 5.0 (unrounded) for the final exam is required to pass the course. The final grade for the resit exam consists of the resit exam grade only (100%). In other words grades for homework and quizzes do not count towards the final grade of the resit exam.	

SEN1141	Managing Multi-actor Decision-making	5
Module Manager	Prof.dr. T.A.P. Metze	
Instructor	B. Wagner	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. M. Leijten	
Instructor	F. Hirschhorn Zonana	
Instructor	Dr. F.S. Gürses	
Co-responsible for assignments	Prof.dr. W.W. Veeneman	
Co-responsible for assignments	Dipl.-Math. B. Kasparek	
Co-responsible for assignments	B. Wagner	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Previous courses have focused on, first, diagnosing the complexity of systems and problems and, then, designing for systems and their institutions in the midst of this complexity. This course on multi-actor decision-making goes one step further: (reflecting on) realizing change. We focus on the actors in the system. How do they behave, individually and collectively? What does the system look like from an actor perspective? Why does intentional change often appear so hard? What limits the designability of systems? By studying characteristics of complex engineering systems (e.g. the increased importance of IT for design, the increased interdependencies between systems) as well as the specific actor characteristics (e.g. the global character of the manufacturers and supply chains, issues of plurality and inclusiveness we identify key factors that affect intentional change (i.e. the wickedness of problems, public values, strategic behaviour of actors and dynamics of the system). How do these factors affect decision-making and design and change of complex engineering systems? How can we assess the consequences of this decision-making complexity and how does this affect the potential for interventions for (re)design in these systems in deliberate and responsible ways? In this course, we identify key characteristics which affect multi-actor decision making in complex engineering designs. Furthermore, we assess how and under which conditions process managerial tactics may help to manage multi-actor decision-making with its plural rationalities and dynamics of negotiation and design.	
Study Goals	At the end of this course, students will be able to explain why actors in networks behave as they do, and how their behaviour evolves describe different network structures and their practical implications assess the designability of institutions, systems, policies and technologies argue why certain process managerial tactics might work under which conditions create case-specific process managerial tactics to anticipate complex decision-making. assess the quality of process designs reflect on multi-actor decision-making in light of broader scientific and public policy trends and debates	
Education Method	Lectures Assignments Readings	
Assessment	Assessments group assignment analysing a multi-actor decision making process (1/2) individual assignments which enable students to reflect on prescribed literature (1/2)	

SEN1611	I&C Architecture Design	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Dr. B.D. Rukanova	
Instructor	Dr. F. Kleiman	
Responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Co-responsible for assignments	G.A. De Reuver	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	· Basic knowledge of databases, including Entity-Relationship Diagram (ERD) · Basic programming skills (such as JavaScript) · Basic knowledge of Unified Modelling Language (UML) and software engineering (agile development) · Basic understanding of multi-actor system and stakeholder analyses methods This prior knowledge are part of the linkage program.	
Course Contents	More and more data is available using various sources like databases, social media or the Internet of Things (IoT). Information sharing requires the integration of these systems. The ability to execute and process data is a key capability required by public and private organizations. The large variety of heterogeneous data, applications and business processes results in a complex and fragmented landscape that need to be dealt with when architecting and engineering new systems. Path dependencies and legacy block the easy integration, whereas at the same time new technologies appear that need to be integrated in the complex systems landscape. The purpose of this course is to teach the architectural design of innovative and large-scale ICT infrastructures and services in the light of the challenges imposed by the requirements from the systems physical, economic and social environment. Emphasis will be put on the concepts and role of ICT-architecture and modelling in order to properly design ICT solutions within a multi-actor context. For this purpose application and data integration technologies and information and system quality theories will be addressed.	
Study Goals	-The student is familiar with the state-of-the-art knowledge of ICT-architecting, design and governance within the field of large-scale ICT-systems within a multi-actor context -The student is able to describe basic concepts related to architecting and designing large and complex ICT-infrastructures and service systems within a multi-actor context. -The student should master architecture theories, methods and tools with the ability to combine and switch between them when dealing with complex ICT-problems. -The student is able to structure and analyze problems and identify dilemmas arising during the design process with regard to designing large ICT-systems within a multi-actor context. -The student is able to apply system engineering and architecture-based approaches, methods and tools to deal with problems with regard to designing large ICT-systems. -The student is able to report about the use of architectural concepts, methods and tools for translating business needs into ICT-designs within constellation of public and private actors.	
Education Method	· Lectures. The lectures are aimed at providing an overview of the knowledge of this course. During the lectures exercises are given to practice to internalize the knowledge. · Guest lectures (obliged). The guest lectures are aimed at giving an overview of the state-of-the art in practice and provide insight into practical challenges when using ICT in a multi-actor domain. · Exercises During some of the lectures assignments are given to practice what is learned. · Assignments. Various assignments are given which together create the final report. This includes developing various architectural models. · Presentation. Students should present their results during a lectures. · Report. The results should be reported in a concise report including the argumentation of design choices, the resulting architecture and an evaluation	
Computer Use	Architecture modelling in Archimate tool (http://www.archimatetool.com/) BPMN modelling using Eunomia process builder (http://www.eunomia-process.com/), Bizagi (http://www.bizagi.com/) or LucidChart (https://www.lucidchart.com/)	
Literature and Study Materials	slides book (open access e-book) Nitesh Bharosa, Remco van Wijk, Niels de Winne, Marijn F.W.H.A. Janssen (2015). Challenging the chain. Governing the Automated Exchange and Processing of Business Information. ISBN: 978-1-61499-497-8 (open access). This book covers a range from governance to technical issues and integrates them. Open access book: http://ebooks.iospress.nl/book/challenging-the-chain-governing-the-automated-exchange-and-processing-of-business-information Reader providing an outline of the area and focus on some aspects which is complemented by papers which goes in more depth.	
Assessment	Group assignment is 60% of the grade. The students work together in groups of 4-6 persons. The assignment is assessed by a report which should adhere to the APA reference standards. The report cannot be updated after handing it in. Groups that will not pass will be given the opportunity to write a new report. The exam is 40% of the grade. The exam will be digital and held in a computer room at the university. Each grade should be 5.75 or higher If one part is not passed and the other part is then the other part will remain valid for the next year.	
Elective	Yes	
Tags	Design Information & Communication Modelling Prototyping Technology	
Targetgroup	SEN I&C track Information Architecture (IA) students Electives for MOT, EPA, Computer science students, and non-TBM students having the required knowledge in this field are welcome	

SEN1622	I&C Service Design	5
Module Manager	Dr. Y. Ding	
Instructor	Prof.dr.ir. G.A. de Reuver	
Instructor	Dr. Y. Ding	
Responsible for assignments	Dr. Y. Ding	
Co-responsible for assignments	Prof.dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Mobile apps, AI, Internet-of-Things, Cloud Edge computing are enabling a range of new ICT services. Since these services are typically offered in ecosystems of interdependent actors, designing services that add value for users as well as ecosystem stakeholders is challenging. In this course, you learn how to design practical ICT services and develop a valorization plan. The course contains two main parts. In the first part, it is about designing innovative services that meet the needs of users. You will design service mockups and interview users to test your ideas. In the second part, it is about bringing that service idea to reality. You will design a valorization plan that specifies how to deal with external technologies, stakeholders and revenues. The course is practical and hands-on: you will create, test and plan your service ideas. At the same time, you will learn how to support design choices using relevant kernel and design theories.	
Study Goals	After the course, you are able to: - Analyze practical ICT services by applying concepts of design science research, action design research, agile development and service engineering - Evaluate practical requirements and make informed choices on supporting technologies and platforms, with specific attention to autonomous driving, healthcare IoT, AI, XR, and edge computing technologies - Evaluate an ICT-enabled service concept through semi-structured interviews - Design and illustrate a value-added service concept driven by ICT - Design a valorization plan that explicitly covers how and when to involve external stakeholders in a design process and integrate the ICT service with value network	
Education Method	Lectures, design project, peer review, presentations, guest lecture.	
Assessment	Group essay = 3 EC; Group mockup = 1.25 EC, Individual reflection document = 0.75 EC All three marks should be >5.75. Participation in peer reviewing, guest lecture and presentations is mandatory.	

TPM010A	Cyber Crime Science	5
Module Manager	Dr. R.S. van Wegberg	
Instructor	F.E.G. Miedema	
Instructor	Dr. R.S. van Wegberg	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Course Contents	- Introduction to crime science (i.e., how do scientists study crime), - Socio-technical research skills (i.e., research design, data collection, data analysis), - Studying cybercrime from a socio-technical perspective (e.g., evaluate the effectiveness of phishing detection).	
Study Goals	The student will acquire: A good understanding of the theoretical principles of crime science, An appreciation of the whole spectrum of different cybercrimes, Skills necessary to empirically research cybercrimes.	
Assessment	1. Research proposal (groups of 3 students) pass/no-pass; 2. Final conference grade - Research paper - Presentation	

TPM025B	User-Centred Security	5
Module Manager	S.E. Parkin	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	Cybersecurity attacks on organizations and services increasingly target the people who are involved, such as employees in companies or home computer users. Solutions to address this problem need to be feasible for individuals, alongside other user priorities (such as completing paid work). Without consideration of user skills or needs, solutions such as security training or browser warning pop-ups add effort and burden to the user, and encourage less secure behaviours as workarounds (such as writing down difficult-to-remember passwords to ensure system access). In this course, students will learn about the user perspective of security technologies. This will leverage key human factors concepts, around security usability and its connections to decisions in policy, planning, and technology investment. This will include how to assess a security solution from the perspective of different kinds of technology users and their tasks. By assessing the strengths and weaknesses of particular security mechanisms for users in practice (policies, training, monitoring, etc.), security implementation and management decisions can be made which better fit the context in which mechanisms are used. This can ensure long-term security which better matches the requirements of a particular user organisation or community.	
Study Goals	The student will: -Gain a sound understanding of security usability as a discipline, from assessing the context of use alongside primary tasks, to identifying the time and effort costs to users. -Obtain foundational skills in matching security technologies and processes to user abilities, motivations, and perceptions of security-related technologies. This will include examination of authentication and identity technologies, employee security training, trust in online contexts, privacy-related evaluations of personal data disclosure, etc. -Gain insights into the design of effective security technologies and their deployment from a human-centred perspective, to inform interface and policy design and ensure compliance with security expectations. This will include management of security behaviour change activities, and identifying how to position expert support to aid users, e.g., effective communication of policy, use of persuasive design in security.	
Education Method	Structured lectures, background reading, problem-driven group discussions	
Assessment	Group assignment on assigned reading material (20%); a final individual assignment is an essay with a feedback moment (80%). The group assignment can be repaired (only after a group report was first submitted) but cannot get more than 6.0 for a repair; the resit option for the individual assignment is an oral exam. To students that need to retake parts of TPM025A: you can still pass the course TPM025A with accompanying individual assignments in its old form of: Individual assignment on assigned reading material (20%); a final individual assignment is an essay with feedback moment (80%). Or, you can choose to partake in the new course TPM025B, with all new assessment; Written group assignment 20% minimum grade of 5.0 + Individual assignment 80% minimum grade of 5.0.	
Elective	Yes	
Category	MSc level	

TPM030A	Introduction to Cloud as Infrastructure: The effects of the new business of computing on practice	5
Module Manager	Dr. F.S. Gürses	
Instructor	Dr. F.S. Gürses	
Co-responsible for assignments	Prof.dr. M.J.G. van Eeten	
Co-responsible for assignments	Dipl.-Math. B. Kasparek	
Contact Hours / Week x/x/x/x	0/0/0/2x2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Some understanding of software engineering or computer science is desirable.	
Course Contents	Why is there a rush to AI? What is the role of cloud computing in the multi-trillion dollar valuation of tech companies? Are mobile phones really personal devices? How has the way software is produced today changed the theory and practice of computing? And, what are the implications of the technical breakthroughs pushed by big tech companies (Google, Apple, Facebook, Amazon and Microsoft) for our societies? These are some of the questions we will ask during this course. The thesis of this course is that contemporary computational infrastructures (CI [0]) present an environment for software production that has become a key driver of political and economic developments globally. With computational infrastructures we refer to cloud + mobile devices as their accessories and by software production, we refer to the creation of economic value through the engineering of software based services. In this course, we will study how CI, developed and controlled by a handful of corporations in the business of computing, is vying to become the default environment for a movement that will transform relationships of production across the globe. To make things concrete, we will look at how digital policy on privacy and data protection attempts to but falls short of responding to the societal impact of computatinal infrastructures. The pivotal role that these computational infrastructures aim to play in production has great societal consequences, for example, in Europe. In an opinion piece published in the Financial Times on May 12, 2023, the president of France, Emmanual Macron underlined the importance of re-industrialization of Europe to ensure its "strategic autonomy" while transitioning to a green and digital economy [1]. By that he refers to "taking back control of our supply chains, energy and innovation". However, a commitment to bringing back industrial production to mainland Europe, when digitalization requires outsourcing key aspects of production to US or Chinese based technology companies poses a fundamental challenge to this industrial strategy. The ambition to create a "European cloud", or replicating the cloud model in any other region of the world, lacks an understanding of the historical and economic underpinnings of current day computational infrastructures. In order to equip you with the ability to assess their global impact, the course will give you insights into how these computational infrastructures came to be and provide you with knowledge to better understand where their future political, economic and technical challenges may lie. [0] A mainframe, server rack, or even a personal computer can serve as any entitys computational infrastructure. In this course, however, we will use the term CI to refer to the dominant and corporate form of materially organizing computing for the development and global delivery of digital services, for example AWS by Amazon, Azure by Microsoft, or mobile devices offered by Apple or Google. Using popular terms, CI is cloud infrastructures plus end- or mobile devices as their accessories. [1] https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-industrial-strategy_en Strategic dependency: https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/european-industrial-stra	
Study Goals	In this course you will discover how the way we produce software impacts how society is organized by reaching the following learning-objectives: Demonstrate a basic understanding of the history of computing through exemplary shifts in software production Develop an understanding of how corporate computational infrastructures serve as a powerful environment for software production Analyze how privacy or data protection by design are weakened in their protective role due to the rise of computational infrastructures Express strengths and weaknesses of current EU digital policy vis a vis computational infrastructures Speculate and predict possible futures of software production in CI and its ambition to transform global economic production	
Education Method	Lectures, work groups, case studies, presentations, self-study, writing.	
Assessment	10% engagement 40% 5 assignments 50% oral exam The total of the assignments, engagement as well as the oral exam must be graded with at least 5.0. The final grade consists of the average of the three parts (engagement, assignments and exam), and the rounded grade must be at least 6.0 (5.75 rounded up).	
Elective	Yes	
Tags	Artificial intelligence Business Computer Engineering Economics Energy & Industry Finance Group work Industry Information & Communication Software Software Engineering Sustainability Technology Telecommunication Transport & Logistics Water management	
Category	MSc level	

WI4049TU	Introduction to High Performance Computing	6
Responsible Instructor	Prof.dr.ir. H.X. Lin	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Required for	Required: Linear algebra (matrix and vector operations), numerical analysis (solution of a system of linear equations Strongly advised: some experience with a programming language (e.g., C) is preferred.	
Course Contents	This course is intended for students who are interested in computing-intensive research. In the course, a number of algorithms that are being used within a diversity of research areas are considered. The scaling behavior of these algorithms in case of increasing problem size and/or an increasing number of processors is analyzed. Attention is paid to those aspects of computer architectures that are important to understanding the resulting performance, such as the memory hierarchy and the interconnection network. By analyzing their computing-intensive characters, determine the possible bottlenecks of several case studies (applications). Based on performance analysis, it will be indicated how the effect of those bottlenecks can be reduced. The goal is to learn how to achieve high performance with the available hardware architecture. The lab exercises will be done on the supercomputer at TU Delft. The emphasis will be on designing efficient parallel algorithms and optimizing for high performance. During the lab exercises, the following types of problems will be elaborated on: a parallel Poisson solver, a parallel finite element simulation, a parallel N-body simulation, and matrix computation on GPUs. More information, such as handouts and slides, can be found on Brightspace. Note: This course is the same as the previous ocurse IN4049TU. Starting from 2024/2025, lectures and exams will be given under the new course code WI4049TU (which is equivalent to IN4049TU)	
Study Goals	1. Can categorize high-performance computer systems 2. Can explain and classify parallel and distributed architectures, and programming models; 3. Can explain the concepts of data decomposition and parallel algorithms; 4. Can apply various parallelization methods to benefit from the potential parallel computer systems; 5. Capable to implement parallel programs (using MPI) on parallel computers and GPU (using Cuda); 6. Capable to evaluate and analyze the performance of parallel programs.	
Study Goals continuation	High Performance Computing, parallel programming, parallel algorithm	
Education Method	Lectures, computer lab exercise using MPI. As an option, answers to the bi-weekly quizzes can be handed in, and a maximum of one bonus point to the exam grade can be obtained.	
Computer Use	Lab exercises (mandatory): Given the sequential numerical simulation programs, make parallel implementations with C, MPI, and/or Cuda on the Delftblue supercomputer.	
Literature and Study Materials	An online book will be used for parts of the course. Lecture notes and lab exercises Will be made available throughout the course and can be downloaded from the Brightspace.	
Assessment	The final grade of the course consists of the following components: Written exam (weighting 50%) Lab work (weighting 50%) Quizzes, a bonus of max 1 point can be obtained (only valid on the first examination occasion after the lectures) Final grade calculation: 0.5 * Written exam + 0.5 * lab work You can only pass the course if both exam and lab work scores are at least a 5.0, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: Written exam: Resit In case of an insufficient final result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: Lab work: re-submit the lab report, the repair grade will be capped at 6.0 Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
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